

Structured bilevel optimization with many follower variables: Global responses, accuracy, and structural boundaries

1 Problem setting and foundations

Many leader–follower models contain a large number of local follower choices but only a few shared resources, aggregate performance measures, and leader decisions. A tariff may influence the operation of many units, while their nonlinear interaction depends only on total load or emissions. The relevant question is whether the number of shared quantities can control the complexity of global bilevel optimization even as the number of follower variables grows. The distinction between the dimension of the follower and the dimension needed to describe its responses is central.

A tractable follower problem alone does not answer this question. The leader optimizes over the responses induced by all its possible decisions; this optimization can be nonconvex even if every follower has a unique convex minimizer. Conversely, nonconvexity confined to a fixed number of aggregate coordinates need not prevent an exact description of all global responses. The development below separates three tasks: describing global follower responses, optimizing over those responses under a specified selection convention, and representing the requested output at a stated accuracy.

1.1 Related work and the structural question

Classical positive results for optimistic linear bilevel optimization fix the number of *follower* variables. Liu and Spencer [32] give a polynomial algorithm under that restriction; Deng [13] discusses the resulting complexity theory. This restriction differs from fixing the number of leader variables while allowing the follower dimension to grow. Indeed, Sugishita and Carvalho [41] prove NP-completeness with one leader variable, unit-bounded variables, and no additional upper constraints for a linear bilevel class with coupled follower constraints. Their polynomial algorithm for finding a local optimum does not give global tractability.

Ketkov and Prokopyev [28] distinguish fixed follower variable and constraint counts, optimistic and pessimistic responses, and linear and quadratic models. Their optimistic convex-quadratic positive result fixes the follower dimension and imposes their stated convex quadratic upper structure. Neither that result nor a restriction on the total follower constraint count encompasses a model with arbitrarily many local follower variables and local bounds, few shared constraints, and a possibly nonconvex polynomial aggregate cost. In particular, counting shared resource rows separately from local rows is essential. Their pessimistic convention also illustrates why upper feasibility must be specified before complexity statements are compared.

The local elimination tools have direct antecedents. Megiddo and Tamir [34] use low-dimensional multiplier arrangements for separable convex quadratic optimization and for fixed-dimensional quadratic blocks with shared constraints. The local quadratic formulas and multiplier partitions used here build on this principle. Exact sign determination and quantifier elimination in fixed dimension are classical real algebraic algorithms [4]. The structural question is how these ingredients

provide a description of *globally* optimal responses when the aggregate cost is nonconvex, and how that description supports the upper optimization. Stationarity alone cannot supply this last step.

For polynomial local costs, earlier work already supplies relevant approximation methods. Hochbaum and Shanthikumar [25] give logarithmic accuracy dependence for continuous separable resource allocation under their function-oracle and constraint-matrix assumptions. Vigneron [42] gives fully polynomial approximation schemes for global optimization of sums of nonnegative algebraic functions of constant description complexity, with polynomial dependence on inverse relative error. A high-accuracy follower evaluation at a fixed leader is different from a global approximation algorithm for the induced, possibly signed upper objective. Both the numerical degree and the accuracy dependence must be tracked in making that comparison.

Cost-near-optimal follower behavior is an established form of decision uncertainty. Besançon, Anjos, and Brotcorne [9] study robustness of upper feasibility against such responses; Beck, Ljubić, and Schmidt [5] place this model in the broader uncertainty literature. The robust model below also allows a worst-case upper objective. The structural contribution concerns exact compression of its adversarial problems, including nonconvex nominal followers, rather than the introduction of near-optimality robustness.

Finally, polynomial-size optimality certificates need not produce polynomial-time optimization. Buchheim [11] proves NP membership for linear bilevel optimization and polynomial encoding bounds for valid big- M values in optimistic KKT reformulations. Practical reformulations, branching, cuts, and decomposition are surveyed by Kleinert et al. [30]. They provide the computational context for the specializations studied here. Polynomial time for fixed structural dimensions is a worst-case statement; it is not by itself a claim of favorable numerical performance.

1.2 The quadratic-block model

Every polynomial datum in this model is supplied explicitly with rational coefficients.

The leader chooses $x \in C \subset \mathbb{R}^r$, where C is a compact set specified by a rational quantifier-free polynomial formula. The follower vector is $z = (z_1, \dots, z_B)$, with $z_b \in \mathbb{R}^{d_b}$, $d_b \leq d$, and total dimension $N = \sum_{b=1}^B d_b$. Both B and N may grow. For the primary model, let

$$P_b(x) = \{y \in \mathbb{R}^{d_b} : E_b y = e_b(x), G_b y \leq h_b(x)\}, \quad (1)$$

$$P(x) = \{z : z_b \in P_b(x) \ (b = 1, \dots, B), Az = b(x), Hz \leq a(x)\}. \quad (2)$$

The matrices E_b, G_b, A, H are rational and independent of x . There are k_+ shared equality rows and k_- shared inequality rows, and $k = k_+ + k_-$. The number of local rows is unrestricted. All right-hand sides are polynomial in x . Among the local inequalities are coordinate bounds $\ell_{bi}(x) \leq z_{bi} \leq u_{bi}(x)$, with polynomial endpoints satisfying $\ell_{bi}(x) \leq u_{bi}(x)$ on C . Thus the union of the follower feasible sets is bounded. A set $P_b(x)$ or $P(x)$ may be empty; follower feasibility is not a promise at every leader.

Let $U(x) \in \mathbb{Q}[x]^{s \times N}$, and partition its columns into blocks $U_b(x)$. For rational polynomial vectors $c_b(x) \in \mathbb{Q}[x]^{d_b}$, the follower objective is

$$f(x, z) = \sum_{b=1}^B \left(\frac{1}{2} z_b^\top Q_b(x) z_b + c_b(x)^\top z_b \right) + \phi(x, U(x)z). \quad (3)$$

Each $Q_b(x)$ is a symmetric polynomial matrix positive definite for every $x \in C$. The polynomial $\phi(x, w)$ can be nonconvex in w . Its argument $w = U(x)z \in \mathbb{R}^s$ collects the shared aggregate quantities. Positive definiteness is imposed on the local blocks, not on the entire follower Hessian. In particular, the follower need not have a unique minimizer.

The upper objective $F(x, z)$ and upper rows $g_j(x, z) \leq 0$ ($j = 1, \dots, m$) are explicitly encoded rational polynomials. Their monomials may involve many different follower coordinates; neither m nor their support sizes are structural parameters. Unless a theorem says otherwise, the fixed parameters of the exact quadratic model are

$$(r, d, s, k_+, k_-). \quad (4)$$

The scalar local model has $d = 1$ and only interval constraints within a block. Extensions with polynomially varying constraint normals will be stated separately. The approximation results use strictly convex univariate polynomial local costs and a convex aggregate cost, with their own leader-domain and data assumptions; those assumptions are not implicit extensions of (3).

1.3 Follower selection and upper feasibility

For $P(x) \neq \emptyset$, compactness gives the value and solution set

$$v(x) = \min_{z \in P(x)} f(x, z), \quad S(x) = \{z \in P(x) : f(x, z) = v(x)\}. \quad (5)$$

Put $S(x) = \emptyset$ if $P(x) = \emptyset$, and use $v(x)$ only on feasible leaders. The follower always optimizes over $P(x)$ without upper constraints. Define the optimistic feasible set and value by

$$\mathcal{O} = \{(x, z) : x \in C, z \in S(x), g_j(x, z) \leq 0 \ (j = 1, \dots, m)\}, \quad (6)$$

$$\text{val}_{\text{opt}} = \inf_{(x, z) \in \mathcal{O}} F(x, z). \quad (7)$$

Thus the leader may select a favorable global response that satisfies its upper rows. A minimizer is asserted only when attainment is established.

For pessimistic optimization, upper feasibility is universal:

$$D_0 = \{x \in C : S(x) \neq \emptyset, g_j(x, z) \leq 0 \text{ for all } z \in S(x), j = 1, \dots, m\}, \quad (8)$$

$$W_0(x) = \max_{z \in S(x)} F(x, z), \quad \text{val}_{\text{pes}} = \inf_{x \in D_0} W_0(x). \quad (9)$$

The maximum at a fixed feasible leader is attained. The infimum over leaders need not be. An upper constraint cannot discard an inconvenient follower optimum before W_0 is computed.

Given a nonnegative polynomial budget $\rho(x)$ on C , the set of cost-near-optimal responses is

$$S_\rho(x) = \{z \in P(x) : f(x, z) \leq v(x) + \rho(x)\} \quad (10)$$

at feasible leaders, and is empty otherwise. Replacing $S(x)$ by $S_\rho(x)$ in (8)–(9) defines D_ρ , W_ρ , and val_ρ . A bounded budget may itself be a leader coordinate when the leader model permits it. Near-optimal points need not satisfy follower stationarity. The robust compression theorem will restrict the number of affine measurements of z used by each upper criterion; this is an additional hypothesis, distinct from the unrestricted polynomial upper data in the exact optimistic model.

For revenue maximization, the corresponding optimistic objective is a supremum and the pessimistic objective maximizes the minimum revenue over all responses. Equivalently, negate revenue in the minimization conventions above. The exact algorithms report infeasibility separately and state whether the finite optimum value is attained. Approximation algorithms have the guarantees stated in their respective theorems: an original-infeasibility certificate, output satisfying relaxed upper rows, and an empty inner surrogate have different meanings. In particular, neither relaxed output nor an empty inner surrogate decides original feasibility in general.

1.4 Input length, degree, and output

The bit model counts arithmetic on binary-encoded rational data. Let L be the total rational input encoding length, including dimensions, row indices, monomial lists, exponents, and the Boolean formula for C . Let $\delta \geq 2$ bound the actual total degree of every input polynomial. The exact algorithms have running time polynomial in (L, δ) for fixed (4). A claim of polynomial time in L alone assumes that δ is polynomially bounded by L , as for dense or unary-exponent encodings. A sparse monomial representation with unrestricted binary exponents does not have that property. Polynomial local costs in the approximation results are supplied densely; a power-specific result will state its dependence on the numerical power.

A common-field algebraic output consists of a square-free polynomial $p \in \mathbb{Z}[T]$, a rational isolating interval selecting a real root α , and rational polynomials p_i, q_i such that each returned coordinate equals $p_i(\alpha)/q_i(\alpha)$ with $q_i(\alpha) \neq 0$. The guarantee includes the degree of p , all coefficient bit lengths, and the combined encoding of all coordinates. They are bounded polynomially in (L, δ) in the stated fixed dimensions, and hence polynomially in L under the degree-encoding condition above. The polynomial p need not be a minimal polynomial. It suffices that its chosen root specifies the same real extension for every recovered coordinate. If several adversarial problems are requested independently, a common-field guarantee for each leader-witness pair does not imply a polynomial-degree field containing all their unrelated witnesses at once.

An accuracy-bit guarantee takes a positive rational tolerance ε , whose encoding contributes to the running time. Writing $\varepsilon = 2^{-\beta}$ illustrates the distinction between a bound polynomial in β and one polynomial in $1/\varepsilon$. In the base approximation model, the output is a rational leader that is exactly feasible for the base leader domain and admits a feasible follower, with an additive guarantee for the true induced upper objective. An approximate follower vector does not replace the true response in that guarantee. Exact satisfaction of response-dependent upper rows requires additional conditions; without them, a theorem must specify its allowed upper violation. Exact algebraic output, rational near-optimal output, and numerical evaluation are different computational tasks.

The exponents of the polynomial bounds depend on the structural parameters. For the decision versions this gives XP-type bounds, $L^{h(r,d,s,k)}$, under the degree convention. It does not establish a fixed-parameter bound $h(r,d,s,k)L^c$ with an absolute exponent c .

1.5 Algebraic and polyhedral tools

We use the following standard consequences of fixed-dimensional real algebraic computation [4, Theorem 1.3.1 and Section 3.1.3]. For a rational polynomial family in a fixed total number of real variables, realizable sign conditions, including zero signs, can be enumerated in time polynomial in the number of polynomials, their degrees, and coefficient bit lengths. A first-order formula with a fixed total number of free and bound variables can be replaced by an equivalent quantifier-free formula within the same kind of bound. Nonempty semialgebraic sets admit algebraic samples of polynomial degree and encoding length. Applied simultaneously to all free coordinates, sampling gives the common-field representation described above. The total number of variable occurrences introduced by quantified copies must be controlled; fixing only the number of free variables is insufficient.

One elementary encoding fact will be used repeatedly.

Lemma 1.1 (Substitution in fixed dimension). *Let $v \in \mathbb{R}^q$, with q fixed, and suppose that $Z_1(v), \dots, Z_N(v)$ and $D(v)$ have explicit polynomial encodings of polynomial size and degree. On a region where $D > 0$, put $z_i = Z_i/D$. Let $p(x, z)$ be an explicitly listed rational polynomial of total degree at most δ , with the coordinates of x among those of v . Then*

$$D(v)^\delta p(x, Z(v)/D(v))$$

is a polynomial that can be constructed in time polynomial in the input encoding lengths, the degrees of Z_i, D , and δ . Its sign agrees with that of $p(x, z)$ on the region.

Proof. An input monomial $ax^\nu z^\gamma$ becomes $ax^\nu \prod_i Z_i^{\gamma_i} D^{-|\gamma|}$. Its degree is at most $\delta(1+T)$ if the degrees of Z_i, D are at most T . A polynomial of degree at most $\delta(1+T)$ in q variables has at most $\binom{\delta(1+T)+q}{q}$ monomials. This number is polynomial for fixed q . Expanding and adding the explicit input monomials therefore takes polynomially many rational operations; coefficient bit lengths grow at most polynomially under these products and sums. Positivity of D^δ proves sign preservation. $\square$

For polyhedral limits we use Hoffman's bound [26]. Given fixed matrices M, T , there is a constant $H_{M,T}$ such that, whenever $K(a, b) = \{y : My \leq a, Ty = b\}$ is nonempty,

$$\text{dist}(y, K(a, b)) \leq H_{M,T}(\|(My - a)_+\| + \|Ty - b\|). \quad (11)$$

Any fixed choices of finite-dimensional norms can be used, with a changed constant. The constant is uniform in the right-hand sides. Equalities can be encoded as opposite inequalities. In the exact attainment proof, only existence of this constant for each instance is needed. The approximation proofs require quantitative encoding bounds and will establish them explicitly. Neither form of the argument grants a uniform constant when the constraint normals vary with the leader.

2 Exact global response compression

The positive quadratic curvature within each local block permits its elimination after fixing the aggregate and shared multipliers. The full follower can still have many stationary points and several global optima. We first encode all feasible stationary candidates, then compare their actual follower values. This comparison is what makes the method valid for a nonconvex aggregate cost.

The underlying geometry can be seen before introducing multipliers. Fix a leader x and an attainable aggregate w . On the fiber $\{z \in P(x) : U(x)z = w\}$ the aggregate cost is constant, and the remaining cost is a strictly convex quadratic. It has exactly one minimizer. A global follower optimum must be this minimizer on its own fiber. Thus aggregate coordinates determine a global response uniquely when that response exists. Constructing its formula still requires handling the many local rows and shared constraints; the multipliers below provide that construction without quantifying all follower variables.

Theorem 2.1 (Exact quadratic-block optimization). *For the model of section 1.2, fix (r, d, s, k_+, k_-) . There is an algorithm, polynomial in (L, δ) , that decides optimistic bilevel feasibility. If feasible, the upper minimum is attained and the algorithm returns an optimizer (x, z) and its value in one common real-algebraic field, with polynomial total encoding length. The number of blocks, their local constraint counts, and the number and support sizes of upper polynomials may grow. Consequently the algorithm is polynomial in L under the degree-encoding condition of section 1.4.*

The theorem includes scalar local quadratics as a special case. The aggregate polynomial need not be convex, and no linear independence or strict feasibility of the follower constraints is required.

Positive definiteness of each $Q_b(x)$ on C , compactness, and the stated coordinate bounds are hypotheses of the problem class. The following construction proves the theorem and also provides the response formula used in subsequent extensions.

2.1 Local quadratic responses

Write A_b, H_b for the block columns of the shared matrices. Introduce

$$\zeta = (x, w, \lambda, \mu) \in \mathbb{R}^{r+s+k_+=k_-}, \quad w = U(x)z, \quad (12)$$

where λ is unrestricted and $\mu \geq 0$. Define

$$q_b(\zeta) = c_b(x) + U_b(x)^\top \nabla_w \phi(x, w) + A_b^\top \lambda + H_b^\top \mu. \quad (13)$$

At a follower local minimizer, the negative gradient belongs to the normal cone of its polyhedron. To recall why no constraint qualification is needed, every feasible tangent direction has nonnegative objective derivative at a local minimum. The polar of the polyhedral tangent cone, by the linear alternative, is the span of the equality normals plus the cone generated by the active inequality normals. This gives multipliers for all rows, with nonnegative inequality multipliers and complementary slackness. Separating shared and local rows shows that each block satisfies the optimality conditions of

$$\min_{y \in P_b(x)} \frac{1}{2} y^\top Q_b(x) y + q_b(\zeta)^\top y. \quad (14)$$

For fixed ζ with $x \in C$, this is a strictly convex quadratic program on a bounded polyhedron. If feasible it has a unique minimizer. Its gradient agrees, at the proposed block response, with the original follower gradient after the shared multipliers have been added. This statement does not assert convexity of the original follower.

The scalar case makes the elimination transparent. If $Q_i(x) = a_i(x) > 0$ and the local set is $[\ell_i(x), u_i(x)]$, the response in (14) is

$$z_i = \text{clip}_{[\ell_i(x), u_i(x)]} \left(-\frac{q_i(\zeta)}{a_i(x)} \right). \quad (15)$$

The signs of $q_i + a_i \ell_i$ and $q_i + a_i u_i$ determine the branch. Choose the lower bound if the first expression is nonnegative, otherwise the upper bound if the second is nonpositive, and otherwise the interior formula $-q_i/a_i$. Boundary formulas agree, including when the interval is a singleton. Joint regimes are determined by realizable signs in (12), rather than by an independent choice among three states for every follower variable.

Lemma 2.2 (A polynomial list for each block). *For fixed d , one can construct polynomially many rational formulas for the solution of (14), each with a polynomial validity predicate in ζ . On its validity region in $x \in C$, each formula has a positive polynomial denominator. Every feasible local optimum is represented, and all valid formulas at the same ζ agree. Their degrees are $O(d\delta)$ and their coefficient bit lengths are polynomial in (L, δ) .*

Proof. Choose an independent row basis $\bar{E}_b$ of the fixed matrix E_b ; let its size be t_b . Retain all original equality rows for feasibility tests. Enumerate inequality subsets J for which $V_J = [\bar{E}_b; (G_b)_J]$ has independent rows. There are at most $\sum_{j=0}^{d_b-t_b} \binom{m_b}{j}$ such subsets, where m_b is the number of local inequalities. With $r_J(x)$ denoting the corresponding right-hand sides, solve

$$M_J(x) \begin{pmatrix} y \\ \theta \end{pmatrix} = \begin{pmatrix} -q_b(\zeta) \\ r_J(x) \end{pmatrix}, \quad M_J(x) = \begin{pmatrix} Q_b(x) & V_J^\top \\ V_J & 0 \end{pmatrix}. \quad (16)$$

This matrix is nonsingular on C . Indeed, a null vector obeys $Q_b y + V_J^\top \theta = 0$ and $V_J y = 0$, so $y^\top Q_b y = 0$. Positive definiteness gives $y = 0$, and row independence gives $\theta = 0$.

Let $\Delta_J = \det M_J$ and let P_J be the vector of Cramer numerators for (y, θ) . Use

$$D_J = \Delta_J^2, \quad Y_J = \Delta_J P_J, \quad (y, \theta) = Y_J / D_J.$$

Thus $D_J > 0$ on C . A branch is valid if its y satisfies every original local row and the entries of θ belonging to J are nonnegative. Multiplication by D_J turns these conditions into polynomial equalities and inequalities. Omitted inequality multipliers are zero. The conditions imply sufficient KKT conditions for the strictly convex local problem, proving soundness and agreement of valid branches.

For completeness, represent the negative gradient of the local minimizer by equality normals and a nonnegative combination of active inequality normals. Project onto the quotient by the equality row space. If the positive-support projected generators are linearly dependent, let a be a nonzero dependence, change its sign so some $a_i > 0$, and replace their positive coefficients θ_i by $\theta_i - ta_i$, where $t = \min_{a_i > 0} \theta_i / a_i$. All coefficients remain nonnegative, at least one vanishes, and the represented vector is unchanged in the quotient. Absorb the change in the equality multipliers and repeat. Eventually the surviving projected normals are independent. Together with $\bar{E}_b$ they form one of the enumerated V_J , and their stationarity system is (16).

The matrices have order at most $2d$. Their entries have degree $O(\delta)$ in a fixed number of variables, so determinants, adjugates, and cleared tests have degree $O(d\delta)$. Fixed matrix order and the explicit input encoding bound their expanded sizes and coefficient lengths polynomially. Rational linear algebra constructs $\bar{E}_b$ and tests independence in polynomial bit time. $\square$

2.2 One common description of the candidate graph

Collect all local branch-validity polynomials. Their total number is polynomial for fixed d , even if the number of rows in an individual block grows. Enumerate their realizable sign conditions in ζ , including zero signs, using the tools of section 1.5. A sign condition fixes which branches are valid in every block. Discard it if some block has no valid branch. For each remaining condition σ , choose the first valid branch of each block in a fixed ordering. The branches agree wherever they overlap by lemma 2.2, so this choice loses no response. A sign condition may have disconnected realization; its decisions still depend only on signs.

Let $D_{b\sigma}$ be the chosen denominator for block b , and write

$$D_\sigma = \prod_{b=1}^B D_{b\sigma}, \quad z_i = Z_{i\sigma}(\zeta) / D_\sigma(\zeta). \quad (17)$$

The polynomial numerator $Z_{i\sigma}$ is the chosen local numerator multiplied by the other block denominators. Empty products are one; empty blocks can be removed after checking their rows. In particular the construction also applies if there are no follower coordinates. Every D_σ is strictly positive on its declared region.

Define $K_\sigma(\zeta)$ to be its sign condition together with $x \in C$ and the following polynomial conditions:

$$\begin{aligned} U(x)Z_\sigma &= D_\sigma w, & AZ_\sigma &= D_\sigma b(x), \\ HZ_\sigma &\leq D_\sigma a(x), & \mu &\geq 0, \\ \mu_j((HZ_\sigma)_j - D_\sigma a_j(x)) &= 0 & (j &= 1, \dots, k_\leq). \end{aligned} \quad (18)$$

The local constraints and local multiplier signs already hold through the chosen valid branches. Thus K_σ describes exactly the follower-feasible KKT candidates encoded in that regime.

The numerator of the follower value is

$$T_\sigma = \sum_b \left(\frac{1}{2} Z_{b\sigma}^\top Q_b(x) Z_{b\sigma} + D_\sigma c_b(x)^\top Z_{b\sigma} \right) + D_\sigma^2 \phi(x, w). \quad (19)$$

Set $\xi = (w, \lambda, \mu, t)$ and define

$$\mathcal{E}(x, \xi) = \bigvee_\sigma \left[K_\sigma(x, w, \lambda, \mu) \wedge t D_\sigma^2 = T_\sigma \right]. \quad (20)$$

This is a quantifier-free formula in $r + s + k + 1$ variables. Its represented pairs consist only of feasible follower points with their actual values; every global follower minimizer has such an encoding by polyhedral first-order necessity. There may be several multiplier encodings of the same response and several nonglobal candidates.

Lemma 2.3 (Polynomial construction size). *The regime list, the polynomial response functions (17), and (20) can be constructed in polynomial bit time in (L, δ) for the fixed structural dimensions. Substituting these response functions in all upper data preserves this bound.*

Proof. The number of local subsets and tests is polynomial for fixed d . The ambient dimension of their sign conditions is $r + s + k$, fixed, and their degrees are $O(d\delta)$. Hence the realizable sign list and its explicit conjunctions have polynomial size and construction cost. No Cartesian product of local lists is taken.

Products of the selected denominators have degree $O(Bd\delta)$. The numerators $Z_{i\sigma}$ have the same order of degree, with an additional $O(d\delta)$ term when $B = 0$ is included in the bound. Equations (18) and (19) have degree $O((B + 1)d\delta)$. Each such polynomial has a fixed number of variables, so its dense monomial count is polynomial in that degree. Repeated multiplication adds coefficient bit lengths and logarithmic convolution counts, giving polynomial coefficient lengths as well. For example, after clearing rational coefficient denominators, a product of B factors with at most M monomials and coefficient magnitudes at most 2^H has coefficients bounded by $M^B 2^{BH}$, whose bit lengths are $O(B(H + \log M))$. Rational denominator clearing has the same polynomial bound.

Finally lemma 1.1 applies separately to every explicitly listed upper polynomial, giving degree $O((B + 1)d\delta^2)$ and polynomial total size. Positive denominators preserve all weak and strict inequality signs. The objective equation for a new value coordinate η is treated as $\eta D_\sigma^\delta - D_\sigma^\delta F(x, Z_\sigma/D_\sigma) = 0$. The added coordinate is a fixed increase in dimension. $\square$

2.3 Globality, optimization, and algebraic recovery

The exact global-response predicate is

$$\mathcal{R}(x, \xi) = \mathcal{E}(x, \xi) \wedge \forall \xi' [\mathcal{E}(x, \xi') \implies t \leq t']. \quad (21)$$

There are just two fixed-dimensional candidate copies. If $P(x) \neq \emptyset$, a global minimum exists and has an encoding in $\mathcal{E}$. Therefore any candidate passing the comparison has value at most this global minimum, and, since it is itself feasible, has equal value. Conversely, a global optimum cannot exceed the value of any feasible candidate. Thus (21) represents *every and only* global responses, including all ties.

An explicit upper predicate $\mathcal{U}(x, \xi, \eta)$ is obtained by a single disjunction over regimes: in regime σ impose the substituted upper rows and $\eta = F(x, Z_\sigma/D_\sigma)$, clearing positive denominators. Use

the regime conditions of K_σ , so its decoded vector is the one encoded by $\mathcal{E}$. The optimistic feasible value relation is

$$\Psi(x, \xi, \eta) = \mathcal{R}(x, \xi) \wedge \mathcal{U}(x, \xi, \eta). \quad (22)$$

There is one copy of the quantified global comparison, outside the upper regime disjunction. One may eliminate its quantifier once to obtain a polynomial-size quantifier-free $\mathcal{R}_0$ before any further construction. In particular, a growing number of regimes or upper rows does not introduce a growing number of real variables.

Quantifier elimination decides whether (22) has a solution. To recover an attained optimum jointly, use

$$\Psi(x, \xi, \eta) \wedge \neg \exists \bar{x}, \bar{\xi}, \bar{\eta} [\Psi(\bar{x}, \bar{\xi}, \bar{\eta}) \wedge \bar{\eta} < \eta]. \quad (23)$$

After replacing $\mathcal{R}$ by $\mathcal{R}_0$, this still uses only a fixed number of variables and has polynomial length, degrees, and coefficient bit lengths. Algebraic sampling after elimination provides (x, ξ, η) in one field of polynomial degree and encoding length. Testing its regime signs selects a valid rational decoding formula; evaluating every $Z_{i\sigma}/D_\sigma$ in that same field recovers z . Polynomial degree and coefficient bounds ensure polynomial bit cost for these evaluations and the total output. No product of N unrelated algebraic extensions is formed.

Lemma 2.4 (Closed response graph with fixed normals). *For the fixed-normal model of section 1.2, the graph $\{(x, z) : x \in C, z \in S(x)\}$ is compact. Consequently $\mathcal{O}$ is compact and an optimistic optimum is attained whenever $\mathcal{O} \neq \emptyset$.*

Proof. Continuity of the coordinate bounds on compact C gives uniform boundedness. Consider a convergent sequence (x_n, z_n) in the graph, with limit (x, z) . Its limit is follower-feasible by continuity of all rows, so $P(x)$ is nonempty. Fix $y \in P(x)$. All local, shared, and box normals form one matrix independent of the leader. The systems $P(x_n)$ are nonempty because they contain z_n . Hoffman's bound (11), applied to y in each such system, gives $y_n \in P(x_n)$ with $y_n \rightarrow y$: the residual at y tends to zero as the polynomial right-hand sides converge. Since $f(x_n, z_n) \leq f(x_n, y_n)$, continuity implies $f(x, z) \leq f(x, y)$. This holds for every feasible y , so $z \in S(x)$. The graph is closed and bounded over compact C , hence compact. Upper weak polynomial rows define a closed subset, and F is continuous. $\square$

Lemma 2.4 makes (23) nonempty whenever the optimistic problem is feasible. Together with lemmas 2.2 and 2.3 and the exact global comparison, this completes the proof of theorem 2.1. The response formula uses a fixed number of shared quantities rather than a fixed follower dimension. The local multiplier-cell principle is classical [34]; global comparison and upper optimization are the additional steps needed here.

2.4 Moving normals and changing local ranks

Now allow $E_b(x), G_b(x), A(x), H(x)$ to be polynomial matrices, preserving the same row-count restrictions, positive-definite local quadratics, compact C , and the polynomial coordinate bounds. Every follower feasible set is still a compact polyhedron at a fixed leader, but its geometry can change discontinuously with x .

Lemma 2.5 (Rank changes preserve compression). *The moving-normal model has a polynomial-size exact global-response formula with the same fixed total dimension and common-field decoding as (21).*

Proof. Replace A_b, H_b by $A_b(x), H_b(x)$ in (13). For a local block enumerate all pairs (I, J) of equality and inequality row subsets with $|I| + |J| \leq d_b$, including the empty pair. Put $V_{I,J}(x) = [(E_b(x))_I; (G_b(x))_J]$ and form the KKT matrix as in (16). Use the branch only where $\Delta_{I,J}(x)^2 > 0$, and again multiply the Cramer numerators by $\Delta_{I,J}$ to obtain this positive denominator. Validity also tests all original local rows and the nonnegative multipliers for J .

A valid branch satisfies local KKT conditions with zero multipliers on omitted rows, so it is the unique local minimizer. Soundness does not require that I span all equality rows. For completeness, at any fixed leader select I to be a basis of its equality row space, whose rank may depend on that leader. Reduce the conic support of active inequality normals modulo this row space exactly as in lemma 2.2. This supplies an enumerated J for which $V_{I,J}(x)$ is independent. Positive definiteness makes its KKT matrix nonsingular, and its branch returns the optimum.

There are polynomially many pairs for fixed d . Their determinant nonvanishing tests are included among the sign polynomials, so all clearing operations occur only where the chosen denominators are positive. The sign-list selection, product denominators, resource consistency and global comparison proceed as before. Row counts and rank changes do not add compressed variables. Pointwise polyhedral KKT necessity still covers every global minimum. The degree and encoding bounds in lemma 2.3 remain valid, since multiplying by moving polynomial normals changes them by only $O(\delta)$. $\square$

Example 2.6 (Optimistic nonattainment with a moving row). Let $x, z \in [0, 1]$, require $xz = 0$, and let the follower minimize $(z - 1)^2$. The response is $z = 0$ for $x > 0$ and $z = 1$ for $x = 0$. For upper objective $F = x + z$, the infimum is zero and is not attained. This is a moving shared equality, a one-dimensional positive local quadratic, and compact uniform boxes. Interpreting the same equality as a local row also illustrates lemma 2.5: for $x \neq 0$ its equality-active KKT determinant is nonzero; at $x = 0$ that determinant vanishes and the empty equality basis supplies the response $z = 1$. Dropping the determinant guard or fixing the equality basis would respectively introduce an invalid branch or lose a response.

2.5 Pessimistic values and attainment decisions

Theorem 2.7 (Exact infima and pessimistic optimization). *For either the fixed-normal or moving-normal quadratic-block model, with the same fixed structural dimensions and encoding, optimistic feasibility, its exact infimum, and attainment are decidable in polynomial bit time in (L, δ) . The same holds for pessimistic feasibility and value under (8)–(9). If the infimum is attained, the algorithm returns an optimal leader and a corresponding response, worst for the upper objective in the pessimistic case, in one common field of polynomial total encoding size. Every feasible problem has finite value. Fixed-normal optimistic attainment is guaranteed by lemma 2.4; no unconditional pessimistic attainment assertion is made.*

Proof. Use a quantifier-free exact response predicate $\mathcal{R}_0(x, \xi)$ from (21) or lemma 2.5. Let $\mathcal{F}(x, \xi, \eta)$ denote the regime-wise, denominator-cleared equation for the upper value, and let $\mathcal{B}(x, \xi)$ denote the disjunction of violated upper rows in their respective regimes. Both use a single regime disjunction and have polynomial size. Define

$$\begin{aligned} T(x, \eta) &\iff \exists \xi [\mathcal{R}_0(x, \xi) \wedge \mathcal{F}(x, \xi, \eta)], \\ \text{Bad}(x) &\iff \exists \xi [\mathcal{R}_0(x, \xi) \wedge \mathcal{B}(x, \xi)], \\ D(x) &\iff x \in C \wedge (\exists \xi \mathcal{R}_0(x, \xi)) \wedge \neg \text{Bad}(x). \end{aligned} \tag{24}$$

These are exact: the existence conjunct excludes empty follower sets, and Bad detects any globally optimal response violating any upper row. It never filters the follower set by those rows. At a fixed feasible leader $S(x)$ is compact, so the maximum of F exists. Its graph on the pessimistic domain is

$$\text{Worst}(x, \eta) \iff D(x) \wedge T(x, \eta) \wedge \forall \eta' [T(x, \eta') \implies \eta' \leq \eta]. \quad (25)$$

One may eliminate T, D before using this expression, or expand a fixed number of copies with fresh internal variables. Thus the number of variables stays fixed even if there are many upper constraints.

Elimination describes the attainable worst values $V = \{\eta : \exists x \text{ Worst}(x, \eta)\}$ as a univariate semialgebraic set. It also decides whether D is empty. Uniform boundedness of z and compactness of C bound F on all feasible pairs, even if the response graph is not closed. Hence nonempty V has a finite infimum. If $V_0(\eta)$ is its quantifier-free description, that infimum η_0 is characterized exactly by

$$\begin{aligned} I(\eta_0) \iff & \forall \eta [V_0(\eta) \implies \eta_0 \leq \eta] \quad \wedge \\ & \forall \varepsilon [\varepsilon > 0 \implies \exists \eta (V_0(\eta) \wedge \eta < \eta_0 + \varepsilon)]. \end{aligned} \quad (26)$$

It has a unique solution when V is nonempty and uses a fixed number of variables. Univariate algebra or elimination and sampling computes η_0 exactly. Membership $V_0(\eta_0)$ decides attainment. When attained, sample the joint formula

$$I(\eta_0) \wedge \text{Worst}(x, \eta_0) \wedge \mathcal{R}_0(x, \xi) \wedge \mathcal{F}(x, \xi, \eta_0).$$

This returns the leader, its value, and a worst global response in one field, after the rational regime decoding. Multiple multiplier encodings have no effect on the represented upper values or feasibility. For the optimistic variant replace V by the attainable upper values of (22), and use its relation in the joint sample. All formulas satisfy the polynomial bounds already proved. If the endpoint is not in V , it is reported as an unattained infimum, not as an optimizer. $\square$

Example 2.8 (Fixed normals do not ensure pessimistic attainment). Let $x, z \in [0, 1]$ and

$$f(x, z) = z^2(1 - z)^2 + xz, \quad F(x, z) = x + z.$$

This is the model (3) with local term $z^2/2$, $w = z$, and aggregate $\phi(x, w) = w^2(1 - w)^2 - w^2/2 + xw$. For $x > 0$ the unique optimum is $z = 0$; for $x = 0$ the global optima are exactly $z = 0, 1$. Thus $W_0(x) = x$ for $x > 0$, while $W_0(0) = 1$. The pessimistic infimum is zero and unattained. Adding the upper row $z \leq 1/2$ makes precisely the leaders $x > 0$ pessimistically admissible, whereas the optimistic model can use $(x, z) = (0, 0)$. At $x = 0$ the stationary point $z = 1/2$ has cost $1/16$ and is rejected by (21); the two zero-cost endpoints survive. This example separates nonglobal stationarity, true ties, and universal upper feasibility within the same bounded instance.

Corollary 2.9 (Bounded algebraic degree with constant local Hessians). *In the fixed-normal model, suppose additionally that every Q_b is independent of x . If the structural dimensions and input degree δ are fixed, the algebraic degree of a common-field optimizer and value in theorem 2.1 is bounded by a constant independent of the number of follower variables, local or upper rows, and input coefficient bit lengths. The same degree bound holds for the infimum in theorem 2.7, and jointly for an optimal leader and worst response when the pessimistic infimum is attained. Total output encoding length remains polynomial in L .*

Proof. Each selected KKT matrix in (16) is a constant rational matrix. Its inverse has constant rational entries, so every local response is a polynomial in ζ of degree $O(\delta)$; only its coefficient lengths

and the number of regimes depend on the number of blocks or rows. Summing block contributions does not raise degree. Thus the candidate equations and substituted upper polynomials have degree $O(\delta^2)$, independent of B and N . There is no need to use a growing-degree product denominator: all denominators are rational constants.

The degree bounds for quantifier elimination in [4, Theorem 1.3.1] depend on the input degrees and variable-block dimensions, not on the number of polynomials. Its simultaneous sample-point construction likewise bounds the degree of each univariate representation in terms of polynomial degree and ambient dimension [4, Section 3.1.3]. Apply these bounds to the fixed-number-of-variables formulas (23), (25), and (26). The required number of elimination and sampling steps is fixed, so the resulting degree depends only on δ and the structural dimensions. Polynomial evaluation of the responses stays in the sampled field. The bit bounds already proved control the growing coefficient and output lengths. $\square$

2.6 A rational LP specialization for supplied low-rank coupling

When both the response formulas and the upper data are affine, linear programming replaces quantifier elimination.

Corollary 2.10 (Supplied diagonal-plus-fixed-rank Hessian). *Fix r and s . Let $X \subset [0, 1]^r$ be a rational polytope and suppose the constant rational follower Hessian is supplied in the form*

$$Q = \text{diag}(d_1, \dots, d_N) + VMV^\top \succ 0, \quad d_i > 0, \quad V \in \mathbb{Q}^{N \times s}, \quad M = M^\top \in \mathbb{Q}^{s \times s}.$$

For leader $x \in X$ the follower minimizes $\frac{1}{2}z^\top Qz + (c + Cx)^\top z$ on $[0, 1]^N$. With any explicit rational affine upper rows and objective, feasibility and an attained global optimum, when feasible, are computable in polynomial bit time, with a rational optimizer and value. No bound on the condition number or magnitude of M is needed, and M need not be positive semidefinite. The decomposition is part of the input; this is not an algorithm for finding one.

Proof. Let $w = V^\top z$. The unique follower optimum satisfies

$$z_i = \text{clip}_{[0,1]} \left(-\frac{c_i + C_i x + V_i M w}{d_i} \right).$$

The two clipping thresholds per coordinate are affine hyperplanes in $(x, w) \in \mathbb{R}^{r+s}$. Enumerate their realizable cells, including all faces and constant threshold cases. On each cell a clipped response is a specified affine function; it extends with the same values to the cell closure because adjacent formulas agree at a threshold. The closure is given by the corresponding weak inequalities and the cell equalities. Impose $w = V^\top z$ after affine substitution, along with the upper rows, $x \in X$, and bounds

$$\sum_i \min(V_{i\ell}, 0) \leq w_\ell \leq \sum_i \max(V_{i\ell}, 0) \quad (\ell = 1, \dots, s).$$

This is a bounded rational LP in fixed dimension. Every feasible point satisfies the complete box KKT conditions for the SPD follower and thus is its unique global response. Conversely every leader–response pair is included in a cell LP. There are polynomially many cells in fixed $r + s$, and rational LP optimization and response reconstruction have polynomial bit cost. Taking the best of the finitely many attained cell optima proves the claim. Uniqueness makes optimistic and pessimistic selection identical in this specialization. $\square$

Positive local quadratics are what keep all decoded follower coordinates in one field generated by fixed-dimensional data. General polynomial local costs need a different output and accuracy analysis. Even independent cubic costs $z_i^3/3 - a_i z_i$ on $[1, a_i + 1]$, for integers $a_i \geq 1$, have unique responses $\sqrt{a_i}$. An upper inequality on their sum therefore embeds exact square-root-sum comparison. This observation motivates the approximation results; it does not label that arithmetic problem NP-hard. Detailed output and structural boundaries are treated separately. A related support-function elimination theorem for polyhedral blocks is proved in appendix A; it clarifies why many local variables can be recovered without giving each one a quantified coordinate.

3 Robust responses and certified dense recovery

The response description admits two different extensions. First, the leader can protect itself against followers whose cost is close to the true optimum, without requiring those followers to be stationary. Second, a structured surrogate can identify enough coordinate statuses to recover the exact response of a dense convex quadratic follower. The first extension changes the set of responses considered by the leader; the second preserves the original follower and certifies its solution.

3.1 Robustness through criterion-specific measurements

Use the moving-normal quadratic-block model of section 2.4, including its compact domain C , polynomial coordinate bounds, and positive-definite local quadratic matrices. The fixed-normal model is included. The follower value $v(x)$ and cost-near-optimal set $S_\rho(x)$ are those of (5) and (10), with a supplied polynomial budget $\rho(x) \geq 0$ on C . Let each upper criterion, including the objective indexed by zero, have the form

$$G_j(x, z) = p_j(x, T_j(x)z), \quad j = 0, \dots, m, \quad (27)$$

where $T_j(x)$ has $h_j \leq h$ rows and p_j, T_j are explicitly encoded rational polynomials. Only h is fixed; the number of criteria and the rank of their combined measurement matrices may grow. An affine upper row needs only one measurement, regardless of its follower support. Leader-only terms belong directly to p_j . Affine measurements with an additional offset can be written in this form by absorbing the offset into p_j .

The robust feasible leaders and objective are

$$D_\rho = \{x \in C : P(x) \neq \emptyset, G_j(x, z) \leq 0 \text{ for every } z \in S_\rho(x), j = 1, \dots, m\}, \quad (28)$$

$$W_\rho(x) = \max_{z \in S_\rho(x)} G_0(x, z), \quad \text{val}_\rho = \inf_{x \in D_\rho} W_\rho(x). \quad (29)$$

The budget is measured from the global follower minimum. Upper rows never remove responses from $S_\rho(x)$ before an adversarial choice. The near-optimality model and separate upper-row adversaries are established in [9]; related complexity-class membership results and an objective-robust variant appear in [8]. The following theorem concerns exact global optimization for the stated structural dimensions.

Theorem 3.1 (Exact near-optimal robustness). *Fix $(r, d, s, k_=: k_≤, h)$. For (28)–(29), feasibility, the finite infimum when feasible, and attainment are computable in polynomial bit time in (L, δ) . If attained, an optimal leader and a worst near-optimal response can be returned jointly in one real-algebraic field with polynomial total encoding length. At any supplied rational leader with $P(x) \neq \emptyset$, the exact worst value and a witnessing near-optimal response for any specified criterion can also be computed in polynomial bit time. For many separately requested criteria, the polynomial encoding guarantee applies separately to each witness; a common field containing all unrelated witnesses is not promised.*

Lemma 3.2 (Measurement-fiber threshold). *Fix one measurement matrix $T(x)$ with at most h rows. Let*

$$P_T(x, \tau) = \{z \in P(x) : T(x)z = \tau\}, \quad m_T(x, \tau) = \min_{z \in P_T(x, \tau)} f(x, z)$$

when the fiber is nonempty. At a follower-feasible leader,

$$\tau \in T(x)S_\rho(x) \iff P_T(x, \tau) \neq \emptyset \text{ and } m_T(x, \tau) \leq v(x) + \rho(x). \quad (30)$$

Moreover the right side is equivalent to existence of a feasible KKT candidate of the measurement-fiber problem whose cost is at most $v(x) + \rho(x)$. That candidate need not be a stationary point of the unrestricted follower problem.

Proof. If a near-optimal response has measurement τ , minimizing on its compact fiber cannot increase cost. Conversely, a fiber minimizer whose cost meets the budget is itself a near-optimal response with that measurement. A global fiber minimizer satisfies polyhedral KKT conditions, so it supplies the asserted candidate whenever the right side holds. Conversely every feasible fiber candidate meeting the budget is already a near-optimal point. Its global optimality on the fiber is unnecessary for this converse. Global comparison remains essential in computing the nominal value $v(x)$. $\square$

Proof of theorem 3.1. If C is empty, report infeasibility. Otherwise first obtain the nominal value graph $\mathcal{V}(x, v)$ by eliminating the response variables from the global predicate (21). It is quantifier-free of polynomial size, and it has exactly one value $v = v(x)$ at a feasible leader and none otherwise.

For one criterion j , add $T_j(x)z = \tau$ as h_j shared equalities. The parameters are now (x, τ) , of fixed dimension $r + h_j$. All possible measurements lie in a rational box with polynomial encoding length: maximize the absolute values of the polynomial entries of T_j and the coordinate bounds on compact C , using fixed-dimensional real algebra, and round the resulting bounds outwards to integers. Summing the products of these bounds gives a bound for each measurement. Univariate root bounds applied to the polynomial-size algebraic outputs ensure the integer bounds have polynomial bit length. Thus the enlarged parameter domain can be taken compact. The positive-definite local quadratics are unchanged, and the added normals are polynomial, so lemma 2.5 applies.

Let $\mathcal{E}_j^{\text{fib}}(x, \tau, \xi_j)$ be the quantifier-free candidate predicate constructed as in (20), including the new equality multipliers. Its coordinate t_j is the candidate's actual follower cost. Define

$$\mathcal{A}_j(x, v, q) \iff \exists \tau, \xi_j [\mathcal{E}_j^{\text{fib}}(x, \tau, \xi_j) \wedge t_j \leq v + \rho(x) \wedge q = p_j(x, \tau)]. \quad (31)$$

By lemma 3.2, at $v = v(x)$ this describes exactly all attainable criterion values on $S_\rho(x)$. A representative may be a minimizer on its measurement fiber even when the response it replaces is nonstationary for the unrestricted follower.

Eliminate the variables in (31) *separately for every criterion*, producing quantifier-free $\mathcal{A}_j$. Then eliminate q in

$$\mathcal{B}_j(x, v) \iff \exists q [\mathcal{A}_j(x, v, q) \wedge q > 0].$$

Each elimination has a fixed total number of variables. There are polynomially many such computations, whose output formulas have polynomial total length and degree. After this elimination, the formula

$$\mathcal{D}(x) \iff x \in C \wedge \exists v [\mathcal{V}(x, v) \wedge \bigwedge_{j=1}^m \neg \mathcal{B}_j(x, v)] \quad (32)$$

uses only one shared nominal-value variable. The empty-follower case is excluded by $\mathcal{V}$. Introducing simultaneous quantified adversaries for all rows would not give the claimed bound.

Since $S_\rho(x)$ is nonempty and compact at every feasible leader, its continuous criterion attains a maximum. The worst-objective graph is

$$\mathcal{W}(x, q) \iff \mathcal{D}(x) \wedge \exists v [\mathcal{V}(x, v) \wedge \mathcal{A}_0(x, v, q) \wedge \neg \exists q' (\mathcal{A}_0(x, v, q') \wedge q' > q)]. \quad (33)$$

Eliminate $\mathcal{D}$ first if desired; every formula still has a fixed number of variables. The attained worst-value set is bounded by uniform follower bounds and compact C . Its emptiness, exact infimum, and endpoint membership follow by the construction (26). Endpoint membership decides attainment.

When attained, sample the leader, nominal value and worst value together with *one* measurement and a thresholded fiber candidate realizing that worst value, using the uneliminated relation (31) for $j = 0$. This adds only a fixed number of variables. Simultaneous sampling returns a common field of polynomial degree and encoding length. The rational block formulas of the fiber candidate recover all follower coordinates in that field. Their cost meets the true budget and their criterion equals the exact worst value. The same procedure at a supplied rational follower-feasible leader computes one criterion maximum and witness, including a violating witness whenever that maximum is positive. Repeating it for different criteria gives separate encodings and does not require a joint field for all those witnesses. $\square$

A bounded budget can be an additional leader coordinate. For example, choosing $G_0(x, z) = -\rho$ maximizes tolerated follower cost loss under all robust upper rows. This adds one fixed leader dimension. Exact rational adversarial output is generally impossible: minimizing $z^2/2$ on $[0, 1]$ with budget $1/4$ gives near-optimal interval $[0, 1/\sqrt{2}]$, whose worst response for criterion z is irrational.

Corollary 3.3 (Constant local matrices and algebraic degree). *If the local Q_b, E_b, G_b are independent of the leader, the common-field outputs of theorem 3.1 have degree bounded solely by input degree and the fixed structural dimensions. This includes polynomially moving shared normals and measurement matrices $T_j(x)$. For fixed input degree, the bound is independent of the follower and criterion counts and coefficient bit lengths. It applies to one jointly recovered leader and adverse witness, or to each separately requested criterion witness. It does not assert attainment.*

Proof. Shared normals and added measurement rows enter the effective linear cost and shared consistency equations; they do not enter the local KKT matrix (16). That matrix is constant under the stated assumptions, so each local response is polynomial in the compressed variables of degree $O(\delta)$ with rational coefficients. All candidate, threshold and criterion formulas have degree bounded in terms of δ , independent of the numbers of rows and blocks. The degree-independent-of-polynomial-count elimination and sampling bounds used in corollary 2.9 apply to each separate criterion computation and then to (33) and one witness. The number of successive elimination and sampling steps along each dependency chain is fixed. Rational decoding preserves the resulting field. $\square$

3.2 When the robust optimum is attained

Proposition 3.4 (Convex fixed-normal attainment). *Under the hypotheses of theorem 3.1, suppose all follower constraint normals are fixed and $\phi(x, \cdot)$ is convex for each $x \in C$. Then every feasible robust problem attains its minimum, including when the nonnegative budget vanishes at some leaders.*

Proof. Let $C_F = \{x \in C : P(x) \neq \emptyset\}$. It is compact: any sequence of feasible leaders has, along a convergent subsequence, bounded feasible followers whose limit remains feasible. Fixed normals and (11) let every $z \in P(x)$ be approximated by feasible $z_n \in P(x_n)$ whenever $x_n \rightarrow x$ in C_F . Closed constraints and uniform bounds give the converse limit property.

The full follower objective is strictly convex in z , so it has one nominal minimizer $z^*(x)$. The comparison-and-repair argument in lemma 2.4, together with uniqueness, implies continuity of z^* on C_F , and hence of $v(x) = f(x, z^*(x))$. The graph of S_ρ is closed by continuity of f, v, ρ and the constraints, and its values are uniformly bounded.

To prove the needed approximation property for S_ρ , fix $x_n \rightarrow x$ in C_F and $z \in S_\rho(x)$. If $\rho(x) = 0$, then $z = z^*(x)$ and the nearby nominal minimizers supply near-optimal approximations. If $\rho(x) > 0$, for any $\theta \in (0, 1)$ the point $z_\theta = (1 - \theta)z + \theta z^*(x)$ is feasible and convexity gives

$$f(x, z_\theta) \leq (1 - \theta)f(x, z) + \theta v(x) \leq v(x) + (1 - \theta)\rho(x) < v(x) + \rho(x).$$

Repair z_θ into $P(x_n)$ by Hoffman. Continuity of the cost, nominal value and budget makes the repaired point near-optimal for all sufficiently large n . Taking θ successively to zero and using a diagonal sequence gives $z_n \in S_\rho(x_n)$ with $z_n \rightarrow z$.

For any continuous criterion, the upper limit of its near-optimal maximum is bounded by its maximum at x : choose maximizing responses and use a convergent subsequence and the closed graph. The reverse inequality follows by approximating a maximizing response at x using the preceding construction. Thus each criterion maximum is continuous on C_F . The robust upper rows define a closed subset of compact C_F , and the continuous worst objective attains its minimum there. $\square$

Example 3.5 (A positive budget does not ensure attainment). Set $\rho = 1/16$, $h(z) = 3z^2 - 2z^3$, and, for $c \in [\rho, 1/8]$, let

$$f_c(z) = z^2(1 - z)^2 + ch(z), \quad z \in [0, 1].$$

Both summands are nonnegative and $h(z) > 0$ for $z > 0$, so the unique nominal optimum is zero with value zero. Separating off $z^2/2$ puts this model in the positive-local-quadratic class with a scalar polynomial aggregate and fixed box normals. Direct factorization gives

$$f_\rho(z) - \rho = (z - 1)^2(z^2 - z/8 - 1/16).$$

Consequently $S_\rho(\rho) = [0, a] \cup \{1\}$, where $a = (1 + \sqrt{17})/16 < 1/3$. For $c > \rho$, the near-optimal set is $[0, a_c]$, with $0 < a_c < a$: the derivative $f'_c(z) = 2z(1 - z)(1 - 2z + 3c)$ changes sign once in $(0, 1)$, and $f_c(1) = c > \rho$. Its first crossing of the budget is therefore its only sublevel endpoint, and $a_c \rightarrow a$ as $c \downarrow \rho$.

If the leader minimizes c subject to $z \leq 1/2$ for every near-optimal response, its feasible set is $(\rho, 1/8]$ and its infimum ρ is unattained. Even without upper rows, put $c = \rho + x$ for $x \in [0, \rho]$ and minimize the worst value of $4x + z$. Its value is one at $x = 0$ and $4x + a_{\rho+x}$ for $x > 0$. On $0 < z \leq a < 1/3$,

$$f'_\rho(z) = z(4z^2 - 51z/8 + 19/8) > z/4, \quad h(z) < z.$$

Writing $c = \rho + x$, the identity $xh(a_c) = f_\rho(a) - f_\rho(a_c)$ implies $xa_c > (a - a_c)a_c/4$, and thus $4x + a_c > a$. The objective tends to a as $x \downarrow 0$, so this unconstrained robust infimum is also

unattained. Unique nominal optima and a strictly positive budget therefore do not replace the convexity assumption in proposition 3.4.

Proposition 3.6 (An unrestricted quadratic criterion is hard). *Without a bound on the measurement dimension, exact worst-criterion computation is NP-hard even for a leader-free diagonal quadratic follower with unit-box constraints and no aggregate or resource coupling. Universal upper-threshold feasibility is coNP-complete on the following restricted family.*

Proof. For a graph on N vertices, let $f(z) = \sum_i z_i^2/2$ on $[0, 1]^N$ and take budget $\rho = N/2$. The nominal value is zero, so every point of the cube is near-optimal. Set

$$G(z) = \sum_{ij \in E} (z_i - z_j)^2.$$

This is convex. Every cube point is a convex combination of binary vertices, so its criterion is no greater than the maximum criterion of a binary vertex. At such a vertex, G is the cut size. Unweighted Max-Cut threshold decision is NP-complete [20]. Thus computing the worst value is NP-hard, and the robust row $G(z) \leq K - 1$ is satisfied for all responses exactly when there is no cut of size at least K . A violating binary vertex is a polynomial certificate for the complement on this family, proving coNP-completeness. This is an elementary transfer of the classical cut problem. A growing collection of separately checked affine rows is not an arbitrary quadratic criterion in a growing number of joint measurements. $\square$

3.3 A structured surrogate for a dense quadratic follower

We now consider a different model. Let $X \subset \mathbb{R}^r$ be a nonempty compact rational polytope. The true follower response is

$$z(x) = \arg \min_{0 \leq z \leq 1} \left\{ \frac{1}{2} z^\top Q z + (c + Cx)^\top z \right\}, \quad Q = Q^\top \succ 0. \quad (34)$$

All data are rational and the upper objective and rows are affine in $(x, z(x))$. Equalities can be represented by two rows. Supply a surrogate

$$\widehat{Q} = \text{diag}(d) + V M V^\top \succ 0, \quad d_i > 0, \quad V \in \mathbb{Q}^{N \times s_0}, \quad M = M^\top \in \mathbb{Q}^{s_0 \times s_0},$$

with the same box and linear cost. Write $y(x)$ for its exact response and $E = Q - \widehat{Q}$. No norm or rank bound on E is needed for correctness. The fixed-rank construction of corollary 2.10 provides a finite closed rational polyhedral cover of the surrogate response graph in $v = (x, w)$, $w = V^\top y(x)$. Each nonempty cover polytope P is compact, projects into X , and has an affine response formula $y_P(v)$. All of its points are consistent surrogate responses, and every leader has a graph point in the cover. The method accepts any supplied cover with these properties, including subdivisions. Its size and description length are part of the complexity bound.

Surrogate-based safe elimination has a direct precedent in the structured dictionary screening of Dantas and Gribonval [12]. Variational-inequality enclosures precede it in [31], and complete parameter-region analysis of QP algorithms appears in [2]. Here the target is an exact global upper optimum for the true dense follower, with the exponential enumeration confined to statuses not certified on a whole surrogate cell.

Lemma 3.7 (Directional response enclosure). *At a fixed leader put $a = c + Cx$, $e = z - y$, $p = Ey$, and $\widehat{g} = \widehat{Q}y + a$. Then*

$$e^\top Q e + p^\top e \leq 0, \quad (35)$$

and for any signed vector $b \in \mathbb{R}^N$,

$$\left| b^\top e + \frac{1}{2} b^\top Q^{-1} p \right| \leq \frac{1}{2} \sqrt{(b^\top Q^{-1} b)(p^\top Q^{-1} p)}. \quad (36)$$

Proof. The two box variational inequalities are $(Qz + a)^\top (y - z) \geq 0$ and $(\hat{Q}y + a)^\top (z - y) \geq 0$. Combining the two inequalities gives (35). Completing the square yields

$$(e + \frac{1}{2} Q^{-1} p)^\top Q (e + \frac{1}{2} Q^{-1} p) \leq \frac{1}{4} p^\top Q^{-1} p.$$

Cauchy–Schwarz in the Q norm gives (36). This is an enclosure: points inside the ellipsoid need not be realizable follower responses. It uses the exact surrogate response, not an unchecked numerical approximation. $\square$

3.4 Certificates on a whole response cell

On a nonempty cover polytope P , the functions $y(v)$ and $p(v) = Ey(v)$ are affine. Compute the rational number

$$\eta_P = \max_{v \in \text{vert}(P)} p(v)^\top Q^{-1} p(v). \quad (37)$$

It is also the maximum over P , since the quadratic is convex and P is the convex hull of its vertices. In fixed ambient dimension $r + s_0$, all vertices can be enumerated by rational row-basis tests, including lower-dimensional cells and singletons, as in appendix A.1. Define

$$\begin{aligned} s_i(v) &= y_i(v) - \frac{1}{2} (Q^{-1} p(v))_i, & R_i &= \frac{1}{2} \sqrt{(Q^{-1})_{ii} \eta_P}, \\ t_i(v) &= \hat{g}_i(v) + \frac{1}{2} p_i(v), & S_i &= \frac{1}{2} \sqrt{Q_{ii} \eta_P}. \end{aligned} \quad (38)$$

Applying (36) to a coordinate vector and to its image under Q gives, throughout P ,

$$s_i - R_i \leq z_i(x) \leq s_i + R_i, \quad t_i - S_i \leq (Qz(x) + a(x))_i \leq t_i + S_i. \quad (39)$$

We use labels L, U, F to certify, respectively, $z_i = 0$, $z_i = 1$, and zero true gradient in coordinate i throughout the cell. The label F is a stationarity equation; it allows a zero-gradient bound point. The following tests are sufficient:

Label	Whole-cell test
L	$\min_P t_i > S_i$ or $\max_P s_i + R_i \leq 0$
U	$\max_P t_i < -S_i$ or $\min_P s_i - R_i \geq 1$
F	$\min_P s_i > R_i$ and $\max_P s_i < 1 - R_i$

A positive gradient in a box optimum forces the lower bound, a negative gradient forces the upper bound, and strict interiority forces zero gradient. Zero gradient alone does not force either bound; hence the strict signs in the first tests for L and U . All affine extrema are attained at the already enumerated vertices. Square-root comparisons are exact rational tests: for rational A and $B \geq 0$, for instance, $A > \sqrt{B}/2$ means $A > 0$ and $4A^2 > B$.

The following boundary refinement avoids unnecessary ambiguity on closed cells. The weaker conditions

$$\min_P s_i \geq R_i, \quad \max_P s_i \leq 1 - R_i, \quad \max_P s_i > R_i, \quad \min_P s_i < 1 - R_i \quad (40)$$

also certify F . Indeed the affine functions $s_i - R_i$ and $1 - R_i - s_i$ are nonnegative on P and neither is identically zero. Each is positive throughout its relative interior: otherwise a small

relative-neighborhood segment through a zero interior point would force an affine nonnegative function to vanish on the affine hull. Thus the response is strictly interior there, giving zero gradient. The true box-QP response is continuous in x . More explicitly, adding its two variational inequalities at x, x' gives

$$\lambda_{\min}(Q) \|z(x) - z(x')\|_2 \leq \|C(x - x')\|_2$$

when the responses differ. Zero gradient therefore extends to all of P by continuity and density of its relative interior. Similarly, $\min_P t_i \geq S_i$ and $\max_P t_i > S_i$ certify L ; $\max_P t_i \leq -S_i$ and $\min_P t_i < -S_i$ certify U . These arguments apply in the affine hull of a lower-dimensional cell. The strict-somewhere conditions cannot be omitted.

Choose any sound available label for a certified coordinate. Call the others ambiguous, let t_P be their number on P , and put $t = \max_P t_P$. Stronger sound certificates may reduce this number. In particular, it measures the chosen certificates rather than an intrinsic complexity of the response.

Theorem 3.8 (Exact dense recovery after screening). *Suppose a valid surrogate cover has M nonempty rational polytopes in fixed dimension $r + s_0$, and whole-cell sound certificates leave at most t ambiguous coordinates on each. Exact bilevel feasibility and a rational global optimum, when feasible, can be obtained with at most $M3^t$ rational recovery LPs. Each LP and its preparation have polynomial bit complexity in the data and cover descriptions. This count excludes construction and screening of the cover. The displayed tests provide their certificates by rational matrix computations and vertex enumeration in fixed dimension, without additional LP calls for affine extrema. Starting with the supplied fixed-rank surrogate, the total bound is $3^t L^{O(r+s_0+1)}$, with L including all rational input and upper rows. No bound on the number of certified F coordinates is required.*

Proof. On a cover polytope P , keep all certified labels and enumerate the three labels for its ambiguous coordinates. A completed assignment partitions the coordinates into L, F, U . Set $z_L = 0$, $z_U = \mathbf{1}$, and solve the free equations with the true Hessian:

$$z_F(x) = -Q_{FF}^{-1}(c_F + C_F x + Q_{FU}\mathbf{1}). \quad (41)$$

Every principal submatrix of an SPD matrix is SPD, so it is invertible; the empty free set requires no inversion. Rational elimination has polynomial bit cost even when F is large and densely coupled. Impose

$$\begin{aligned} v \in P, \quad 0 \leq z_F(x) \leq 1, \\ (Qz(x) + c + Cx)_L \geq 0, \quad (Qz(x) + c + Cx)_U \leq 0, \end{aligned} \quad (42)$$

together with all original affine upper rows, and minimize the original upper objective after substitution. This is a rational LP in at most $r + s_0$ variables, bounded by compact P . Its feasible points satisfy all true box KKT conditions, so they give the unique response to the original dense follower.

Conversely, any true feasible leader–response pair has a surrogate graph point in at least one cell. Every certified label holds there. Assign an ambiguous coordinate label F when its gradient is zero, L when its gradient is positive, and U when its gradient is negative. Box optimality makes this a compatible assignment. Thus the pair is included in one recovery LP. Zero-gradient bound coordinates may appear under more than one assignment, which loses no solutions. The best nonempty LP yields an exact rational optimum; if all are empty, the original problem is infeasible. Summing 3^{t_P} over the cells gives the LP count. Fixed-dimensional cover construction and screening and polynomial rational inversion/substitution/LP give the stated bit bound. $\square$

An inexpensive additional sufficient test is often useful. Guess a full status assignment on P , form (41), and check all true KKT signs and bounds at every vertex of P . All tested functions are

affine, so successful checks certify the response on the entire cell without status enumeration. Failure merely means that the assignment is not certified on the *whole* cell. It must remain available in the recovery enumeration on a smaller feasible part; a failed guess is not an infeasibility certificate.

3.5 A computable perturbation neighborhood

The ambiguity bound can be certified from the geometry of the supplied surrogate. Use its original closed clipping cells, retaining their nominal labels L, F, U . At a vertex v , call coordinate i a transition coordinate when

$$y_i(v) \in \{0, 1\}, \quad \hat{g}_i(v) = 0.$$

Let q be the largest number of such coordinates at a vertex of any nonempty cell. Vertices created by the leader polytope and consistency equalities are included; no general-position assumption is made.

Every such cell has affine dimension at most r . Its projection to x is injective, since a leader has a unique surrogate response and aggregate. An injective linear projection on a polytope is injective on its affine hull: a nonzero kernel direction in that affine hull would produce two nearby points with the same projection in a relative-interior neighborhood. Hence its dimension is at most the leader dimension.

Cover each cell by the convex hulls of all its affinely independent vertex subsets of size at most $r + 1$. These are rational simplices and they cover the cell by the finite-dimensional convex-combination reduction proved in lemma A.3. A triangulation is unnecessary: overlapping simplices are allowed. There are at most $\sum_{j=1}^{r+1} \binom{V_P}{j}$ such simplices for a cell with V_P vertices, polynomial in fixed dimension.

For a simplex T , let J_T be the union of its vertices' transition coordinates. Then $|J_T| \leq (r + 1)q$. If $i \notin J_T$, the containing cell's nominal label gives strictly positive vertex margins

$$L : \hat{g}_i(v), \quad U : -\hat{g}_i(v), \quad F : y_i(v) \text{ and } 1 - y_i(v).$$

For a nominal F coordinate its surrogate gradient vanishes throughout the closed cell, so any zero bound margin would indeed be a transition. Let $\sigma > 0$ be the minimum of all these margins over all simplices and coordinates outside their J_T , setting $\sigma = 1$ if the list is empty. It is rational, and every listed margin is at least σ throughout the corresponding simplex by affine interpolation.

For $N \geq 1$, compute

$$\begin{aligned} m_0 &= \frac{1}{\|\hat{Q}^{-1}\|_\infty}, & L_0 &= \|\hat{Q}\|_\infty, & R &= \lceil \sqrt{N} \rceil, \\ \varepsilon_0 &= \min \left\{ \frac{m_0}{2}, \frac{\sigma}{4R \max\{1/m_0, 1 + L_0/m_0\}} \right\}. \end{aligned} \tag{43}$$

Matrix infinity norms are induced maximum absolute row sums. Symmetry implies $\|A\|_2 \leq \|A\|_\infty$, so $\lambda_{\min}(\hat{Q}) \geq m_0$ and $\|\hat{Q}\|_2 \leq L_0$. The case $N = 0$ is an ordinary upper LP.

Corollary 3.9 (Transition-controlled dense neighborhood). *Every rational symmetric Q satisfying $\|Q - \hat{Q}\|_\infty \leq \varepsilon_0$ is positive definite. On each of the constructed simplices, at most $(r + 1)q$ coordinates require status enumeration in theorem 3.8. All radius and cover data can be computed with polynomial bit complexity for fixed r, s_0 , so exact optimization is polynomial for fixed r, s_0, q . The dense perturbation may have arbitrary rank and signs.*

Proof. Write $\varepsilon = \|E\|_\infty \leq m_0/2$. Then $\lambda_{\min}(Q) \geq m_0/2$. From (35), Cauchy–Schwarz and the unit-box bound $\|y\|_2 \leq R$ give

$$\|z - y\|_2 \leq 2\varepsilon R/m_0. \quad (44)$$

The true gradient minus the surrogate gradient equals $\widehat{Q}(z - y) + Ez$. Since $\|z\|_2 \leq R$,

$$\|Qz + a - \widehat{g}\|_2 \leq 2\varepsilon RL_0/m_0 + \varepsilon R \leq 2\varepsilon R(1 + L_0/m_0). \quad (45)$$

The choice of ε_0 makes both bounds at most $\sigma/2$. Every nontransition nominal L coordinate has strictly positive true gradient and hence is zero; a nominal U coordinate has strictly negative gradient and hence is one. A nontransition nominal F coordinate lies in $[\sigma/2, 1 - \sigma/2]$, so its true gradient is zero. These are sound whole-simplex certificates, independent of whether the ellipsoid tests happened to identify them. Only J_T needs enumeration.

Rational vertex enumeration, fixed-size simplex descriptions, matrix inversion, margin evaluation and the displayed norm calculations have polynomial bit cost. The simplex count above is polynomial for fixed r, s_0 ; it may increase the exponent of the original arrangement bound. Apply theorem 3.8 to this cover. $\square$

The radius is instance-dependent and can be small. Neither low transition multiplicity nor a numerically large margin is automatic. This is a verifiable neighborhood of a supplied surrogate, not a general method for finding a useful surrogate or a uniform neighborhood of every well-conditioned Hessian.

3.6 Value and feasibility bounds without full recovery

Even when many statuses remain ambiguous, the directional enclosure provides computable global bounds. For a signed output vector b , put

$$q_b(v) = b^\top y(v) - \frac{1}{2} b^\top Q^{-1} p(v), \quad r_b = \frac{1}{2} \sqrt{(b^\top Q^{-1} b) \eta_P}.$$

Choose a rational outward upper bound $\bar{r}_b \geq r_b$ by rational square-root enclosure. Its accuracy is an explicit computational choice; any such upper bound is safe. Then $q_b - \bar{r}_b \leq b^\top z \leq q_b + \bar{r}_b$ on P . For a row $a_j^\top x + b_j^\top z \leq h_j$, an inner sufficient condition and an outer necessary condition are, respectively,

$$a_j^\top x + q_{b_j}(v) + \bar{r}_{b_j} \leq h_j, \quad a_j^\top x + q_{b_j}(v) - \bar{r}_{b_j} \leq h_j. \quad (46)$$

Use these conditions with $v \in P$ in rational inner and outer LPs, representing equalities by both signs. If the objective is $a_0^\top x + b_0^\top z + c_0$, minimize $a_0^\top x + q_{b_0} - \bar{r}_{b_0} + c_0$ over the outer LPs. Their best value is a certified lower bound on the true optimum: every true feasible leader has a point in an outer LP and its objective is no smaller than that lower enclosure. Minimize the corresponding upper enclosure over the inner LPs for a certified upper bound and a leader feasible for the true response. Evaluating the true follower with an exact KKT certificate at that leader can improve this upper bound.

Empty outer LPs for every cell certify original infeasibility. Empty inner LPs alone say nothing about original feasibility, and a minimizer of an outer LP need not be truly feasible. The inner construction can miss exactly feasible upper equalities. Signed objective coefficients are handled by the directional formula, without a monotonicity assumption. Only the actual computed finite upper-minus-lower gap gives an objective guarantee; a small perturbation by itself supplies no small global loss under restrictive upper rows.

Example 3.10 (Small residual but complete screening ambiguity). Let $N \geq 2$, $\widehat{Q} = I$, $Q = I + (\varepsilon/N)\mathbf{1}\mathbf{1}^\top$, $c = 0$, $C = -\mathbf{1}$, and $X = [0, 1]$, where $\varepsilon > 0$ is rational and arbitrarily small. The surrogate response is $y = x\mathbf{1}$ and the true response is $z = x\mathbf{1}/(1 + \varepsilon)$, whose gradient is zero throughout the interval. On the whole cell $[0, 1]$, however, the coordinate enclosure radius is positive and $s_i(0) = 0$, so neither strict nor refined F screening passes. The lower-gradient center has minimum zero and positive half-width, the upper-gradient test also fails, and the coordinate tests force neither bound. Thus the displayed ellipsoid tests leave all N coordinates ambiguous. Here $\|E\|_\infty = \varepsilon$ can tend to zero. The all- F guessed assignment in (41) immediately certifies the response throughout the cell and needs no enumeration. This separates conservative screening from intrinsic difficulty.

Small perturbations can also move a switching point. With one follower, $\widehat{Q} = 1$, $Q = 1 + \varepsilon$, linear cost $-2xz$, and $x \in [0, 1]$, the upper clipping point moves from $1/2$ to $(1 + \varepsilon)/2$ for $0 < \varepsilon < 1$. Reusing the surrogate's upper-bound label on its entire cell would therefore be incorrect. Recovery must use the true KKT conditions. Dense rational elimination and construction of many cover cells can dominate cost even when screening is sound. The results do not establish practical superiority, a uniformly effective cell refinement, or automatic discovery of an accurate low-rank surrogate.

4 Global optimization in accuracy bits

Exact response coordinates for higher-degree local costs can generate large algebraic fields. Approximation avoids assembling those fields. The construction in this section approximates each scalar inverse by rational polynomials, optimizes their compositions in a fixed number of leader and dual variables, and recovers a rational leader with controlled error. The requested accuracy enters through its number of bits, rather than its reciprocal.

4.1 Model, output, and approximation tools

Let $X \subseteq [0, 1]^r$ be a rational polytope, and let $C \in \mathbb{Q}^{k \times N}$ and $U \in \mathbb{Q}^{s \times N}$ be independent of the leader. The dimensions r, k, s are fixed. The functions $b(x)$ and $\ell_i(x)$ are rational affine. Each rational polynomial marginal g_i is strictly increasing on $[0, 1]$, with $g_i(0) = 0$ and $G_i = g_i(1) > 0$. Its coefficients may have either sign. Write $P = \max_i \deg g_i$ and $f_i(t) = \int_0^t g_i(u) du$. An arbitrary strictly convex polynomial local cost is included by subtracting its initial derivative from both its marginal and its linear incentive. Additive local constants do not affect responses.

Put $R = \max\{1, \max_a \sum_i |U_{ai}|\}$ and $W = [-R, R]^s$. The rational polynomial $\phi(x, w)$ is convex in $w \in W$ for every follower-feasible leader. Convexity and strict increase are promises; strict increase can alternatively be checked by univariate sign methods. No positive lower bound on either local or aggregate curvature is imposed. The follower minimizes

$$F_x(z) = \sum_{i=1}^N [f_i(z_i) - \ell_i(x)z_i] + \phi(x, Uz), \quad z \in [0, 1]^N, \quad Cz \leq b(x). \quad (47)$$

Equality resources can be represented by pairs of inequalities, with the resulting row count fixed. The follower-feasible leaders form the explicit rational polytope X_F of lemma C.1. At every $x \in X_F$, strict convexity gives a unique response $z(x)$.

Local marginals and the aggregate are densely encoded. Upper polynomials may instead be supplied by explicit monomial lists in (x, z) ; all their terms and numerical total degrees count toward the complexity bound. In particular no expansion in all N follower variables is required. Let

L_{in} denote total rational input bit length and D_{all} the maximum numerical degree of all supplied polynomials.

Theorem 4.1 (Accuracy-bit global optimization). *For fixed r, k, s , follower-feasible leader existence for (47) is decidable in polynomial time. If $X_F \neq \emptyset$, then for any rational polynomial upper objective $H(x, z)$ and requested integer $B \geq 1$, one can return a rational $\hat{x} \in X_F$ and a rational number $\hat{V}$ such that*

$$H(\hat{x}, z(\hat{x})) \leq \min_{x \in X_F} H(x, z(x)) + 2^{-B}, \quad |\hat{V} - \min_{x \in X_F} H(x, z(x))| \leq 2^{-B}.$$

The bit complexity and rational output lengths are polynomial in $(L_{\text{in}}, D_{\text{all}}, B)$. The minima are attained. No exact common-field representation of the full response or optimum is promised. Response-dependent upper constraints require the qualifications in section 4.7.

Scalar inverse resource allocation and logarithmic dependence on inverse accuracy for the convex allocation objective are established [25, 40]. Approximation of algebraic sums in fixed dimension also has an established bit-model treatment [42]. Here the objective being optimized is an independent, possibly signed and nonconvex function of the allocation response. The proof must control the entire response and the feasibility of the returned leader, rather than only the follower's objective gap. General polynomial bilevel relaxation methods provide a complementary global-optimization approach [27].

The clipped inverse h_i of g_i is zero for targets at most zero, one for targets at least G_i , and g_i^{-1} between them. Theorem B.2 supplies rational polynomial branches approximating h_i to any tolerance η , with a complete uniform error and polynomial-bit construction. Its proof in appendix B handles real and complex critical values, interior derivative zeros, rational analytic panels, and exact Taylor reversion. Proposition B.3 retains sharper bounds when coefficients are nonnegative. We first explain how such approximations are used in the diagonal and single-resource cases, before proving the full theorem.

Lemma 4.2 (Rational recovery in a polytope). *In fixed dimension n , let $Q \subseteq [0, 1]^n$ be a nonempty rational polytope and let $v \in Q$ have a polynomial-size common-field algebraic encoding. For any rational $h > 0$, a rational $\hat{v} \in Q$ with $\|\hat{v} - v\|_\infty \leq h$ can be computed in polynomial bit time in these encodings and $\max\{0, \log(1/h)\}$. For a rational polynomial $S(v) = \sum_\alpha a_\alpha v^\alpha$,*

$$L(S) = \max\{1, \sum_\alpha |a_\alpha| |\alpha|_1\} \quad (48)$$

bounds its infinity-norm Lipschitz constant on the cube. Thus recovery can preserve any polynomial-size finite list of polynomial values to specified positive tolerances, while keeping membership in Q exact.

Proof. Enumerate vertices by independent active-row systems and use the convex-combination reduction in lemma A.3 to find an affinely independent subset of at most $n + 1$ vertices with $v = \sum_{j=1}^t \lambda_j v_j$, $\lambda_j \geq 0$, $\sum \lambda_j = 1$. Enumerating these subsets and checking barycentric coordinates takes polynomial time in fixed dimension, in the existing algebraic field. For $j < t$ replace λ_j by $\lfloor M\lambda_j \rfloor / M$ and put the remaining mass on vertex t . All weights stay nonnegative; the point stays in the same simplex and moves by at most $2n/M$. Choose an integer $M \geq \max\{1, 2n/h\}$. The floors require only $O(\log M)$ exact algebraic comparisons each. Lower-dimensional and singleton polytopes are included. Finally sum the absolute first partial derivatives on the cube to obtain (48), and take h no larger than the smallest tolerance divided by its bound. $\square$

4.2 Diagonal followers and dyadic powers

When $k = s = 0$, the response is $z_i(x) = h_i(\ell_i(x))$. For an affine upper objective $d_0 + d^\top x + c^\top z(x)$, let $\varepsilon = 2^{-B}$, $A = \sum_i |c_i|$ and $\eta = \varepsilon/[16 \max\{1, A\}]$. If $A = 0$, solve the leader LP. Otherwise pull back every inverse breakpoint through ℓ_i and construct the realizable affine sign cells in X , including zero signs. There are polynomially many in fixed dimension. Take their closures: weakening the strict affine tests gives the closure, since mixing any point satisfying the weak tests with a point of the original relative cell approaches it through that cell. Choose a valid branch on each closed cell Q . Their polynomial upper objective H_Q satisfies

$$|H_Q(x) - H(x, z(x))| \leq A\eta \leq \varepsilon/16. \quad (49)$$

At a threshold either adjacent branch is valid on the boundary.

Minimize every H_Q exactly on Q by section 1.5, using one additional copy of the r variables to express that no point has a smaller value. Compare the resulting algebraic values pairwise and select the smallest. This requires at most products of two polynomial field degrees, not a field containing all response radicals. If v^* is the winning point, then $H_Q(v^*) \leq \text{OPT} + \varepsilon/16$. Use lemma 4.2 inside Q so that its objective increases by at most $\varepsilon/4$. The true objective of the resulting rational leader is at most $\text{OPT} + 3\varepsilon/8$ by (49). Bisection of each true inverse at that leader to weighted total error $\varepsilon/4$ gives a rational optimum estimate of error less than ε . Rational vertex recovery is essential: coordinatewise rounding of an algebraic minimizer can leave X .

For the important power marginal $g_i(t) = t^{p_i}$, one can use a simpler fully explicit approximation. Put $m = \lceil \log_2(1/\eta) \rceil$, $P = \max_i p_i$, $K_0 = Pm$, and $q = m + 1$. Below 2^{-K_0} use zero; its root error is at most 2^{-m} . On $2^{-(j+1)} \leq t \leq 2^{-j}$, $j = 0, \dots, K_0 - 1$, put $b_j = 3 \cdot 2^{-(j+2)}$ and $u = t/b_j - 1 \in [-1/3, 1/3]$. For $\alpha = 1/p_i$, the rational binomial coefficients have absolute value at most one, by their consecutive ratio. Thus

$$T_q(u) = \sum_{n=0}^q \binom{\alpha}{n} u^n, \quad |T_q(u)| \leq 3/2, \quad |(1+u)^\alpha - T_q(u)| \leq 3^{-q}/2 \leq \eta/4.$$

Compute a rational σ_j within $\eta/4$ of b_j^{1/p_i} by bisection and use $\sigma_j T_q(t/b_j - 1)$. Its error is at most $3\eta/8 + \eta/4 < \eta$. There are Pm dyadic bands and degree $O(m)$. Bisection compares rational p_i th powers of $O(m)$ -bit arguments against $O(Pm)$ -bit dyadic targets. Binomial coefficient lengths are $O(q(\log P + \log q))$, and expansion adds $O(qK_0)$ bits from scaling. These are polynomial in numerical P and accuracy bits. In fixed leader dimension the substituted polynomial has at most $\binom{r+q}{r}$ monomials. This proves the diagonal accuracy-bit guarantee directly, without a critical-value calculation.

Example 4.3 (Sparse exponents can force long rational output). Let even $p \geq 2$ be encoded in binary. Two independent unit-box followers with marginals z^p and $z^{p/2}$ and incentive $x \in [0, 1]$ have responses $x^{1/p}$ and $x^{2/p}$. Their difference has minimum $-1/4$ at $x = 2^{-p}$, since it is $(x^{1/p} - 1/2)^2 - 1/4$. Suboptimality at most $1/64$ implies $3/8 \leq x^{1/p} \leq 5/8$, hence $0 < x \leq (5/8)^p$. For reduced rational $x = a/b > 0$, this requires $b \geq (8/5)^p$, so explicit rational output takes $\Omega(p)$ bits, although the sparse input has $O(\log p)$ bits and accuracy is constant. The example uses two different powers. It is an output obstruction, not a hardness claim for the same-power subclass.

4.3 One equality and an exact signed balance identity

For $s = 0$ with one resource equality $\sum_i w_i z_i = b(x)$, where $w_i \in \mathbb{Q}$ may be negative or zero, there is a sharper argument than the general residual estimate below. Define

$$b_- = \sum_i \min\{w_i, 0\}, \quad b_+ = \sum_i \max\{w_i, 0\}, \quad W_1 = \sum_i |w_i|.$$

The box image is exactly $[b_-, b_+]$: its endpoint vertices and their segment realize that interval. Thus $X_F = X \cap \{b_- \leq b(x) \leq b_+\}$. If $W_1 = 0$, this is simply $b(x) = 0$ and the diagonal construction applies. Otherwise put $w_{\min} = \min_{w_i \neq 0} |w_i|$.

Bound each affine incentive on the leader cube by rational ℓ_i^-, ℓ_i^+ . Among all i with $w_i \neq 0$, take the minimum λ_- and maximum λ_+ of

$$\ell_i^-/w_i, \quad \ell_i^+/w_i, \quad (\ell_i^- - G_i)/w_i, \quad (\ell_i^+ - G_i)/w_i.$$

They have polynomial bit length and $\lambda_- < \lambda_+$. The box-Lagrangian response and its balance are

$$q_i(x, \lambda) = h_i(\ell_i(x) - \lambda w_i), \quad B_x(\lambda) = \sum_i w_i q_i(x, \lambda).$$

The continuous nonincreasing function B_x equals b_+ at λ_- and b_- at λ_+ . Hence every feasible leader has a balancing multiplier in that interval. Its box minimizer is the resource optimum by Lagrangian comparison against any balanced point. This includes endpoint resources and requires no constraint qualification. All balancing multipliers give the same response by strict convexity.

For a balancing λ^* , every nonzero term $w_i(q_i(x, \lambda) - q_i(x, \lambda^*))$ has the same weak sign. Zero-weight responses do not change. Therefore

$$\sum_i |w_i| |q_i(x, \lambda) - z_i(x)| = |B_x(\lambda) - b(x)|. \quad (50)$$

In particular, for $c_{\max} = \max_i |c_i|$,

$$|c^\top(q - z(x))| \leq (c_{\max}/w_{\min}) |B_x(\lambda) - b(x)|.$$

This identity uses monotonicity and the single dual parameter, including signed weights. There is no corresponding no-cancellation identity for several resource rows.

Here is the complete affine-objective accuracy ledger for this case. If $c = 0$ solve the LP over X_F . Otherwise let $A = \sum_i |c_i|$, $C_1 = c_{\max} W_1 / w_{\min}$ and $\eta = \varepsilon / [32(1 + A + C_1)]$. Normalize $\lambda = \lambda_- + (\lambda_+ - \lambda_-)\theta$, $0 \leq \theta \leq 1$, and arrange the affine inverse breakpoints in $X_F \times [0, 1]$. On each closed rational cell Q let p_i be its valid approximants, and put

$$H_Q = d_0 + d^\top x + c^\top p, \quad T_Q = \sum_i w_i p_i - b(x).$$

Minimize H_Q subject to $v = (x, \theta) \in Q$ and $|T_Q(v)| \leq W_1 \eta$. A true optimal leader and a balancing multiplier have such a lift, so the winning algebraic value is at most $\text{OPT} + A\eta$. Optimize in $2(r+1)$ variables and recover a rational point in Q within distance

$$h \leq \min\{1, \varepsilon/(8L(H_Q)), W_1 \eta / L(T_Q)\}.$$

The recovered polynomial residual is at most $2W_1 \eta$ and the true box response residual at most $3W_1 \eta$. Consequently

$$|H(\hat{x}, z(\hat{x})) - H_Q(\hat{v})| \leq (A + 3C_1)\eta,$$

and its suboptimality is at most $\varepsilon/8 + (2A + 3C_1)\eta \leq 7\varepsilon/32$. The rational polynomial value at $\hat{v}$ lies between $\text{OPT} - (A + 3C_1)\eta$ and $\text{OPT} + A\eta + \varepsilon/8$, hence estimates the optimum within $5\varepsilon/32$. The leader is exactly in X_F even though its auxiliary response need not be balanced. Small weights affect logarithmic precision only. Compact balancing multipliers also prove response continuity by subsequence passage and uniqueness, and therefore attainment.

4.4 Resource and aggregate residuals control the response

We now return to (47), assuming $N \geq 1$. Construct X_F and K by lemmas C.1 and C.2. Choose a rational $\bar{G} \geq 1$ bounding $\|\nabla_z F_x(z)\|_\infty$ on the leader and follower cubes. Absolute coefficient sums after normalizing $w = R(2t - \mathbf{1})$ supply such a bound with polynomial encoding. Put

$$\begin{aligned} L_F &= N\bar{G}, & \Lambda &= \max\{1, K\bar{G}\}, \\ S &= \max\{1, \max_j \sum_i |C_{ji}|\}, & A_U &= \max\{1, \max_a \sum_i |U_{ai}|\} = R. \end{aligned}$$

Empty row maxima are zero. The constant L_F bounds the follower objective's infinity-norm Lipschitz constant. Let $L_\phi \geq 0$ bound the infinity-norm Lipschitz constant of $\nabla_w \phi$ on W , uniformly on the leader cube; the maximum row sum of absolute Hessian coefficient bounds suffices. Define

$$D_\phi = L_\phi \sum_i \|U_i\|_1, \quad \mu = \frac{\min_i G_i}{4(4P)^P}.$$

These are effective rational constants of polynomial bit length. With positive marginal coefficients one may use the larger μ from lemma C.3.

For $x \in X_F$, $w \in W$ and $\lambda \in [0, \Lambda]^k$, define the frozen-gradient box response

$$q_i = h_i(\ell_i(x) - U_i^\top \nabla_w \phi(x, w) - C_i^\top \lambda). \quad (51)$$

This minimizes a strictly convex separable box objective with aggregate gradient frozen at w .

Lemma 4.4 (Response certificate). *If the true frozen response satisfies*

$$Cq - b(x) \leq \delta \mathbf{1}, \quad |\lambda^\top (Cq - b(x))| \leq \zeta, \quad \|Uq - w\|_\infty \leq \rho, \quad (52)$$

where $\delta, \zeta, \rho \geq 0$, then

$$\|q - z(x)\|_\infty \leq K\delta + \left(\frac{L_F K\delta + \zeta + D_\phi \rho}{\mu} \right)^{1/(P+1)}. \quad (53)$$

An exact response has a certificate with all three residuals zero and $\lambda \in [0, \Lambda]^k$. The response $z(x)$ is continuous on X_F .

Proof. Put $a = \nabla_w \phi(x, w)$ and $a_q = \nabla_w \phi(x, Uq)$. Testing the frozen box variational inequality at $z^* = z(x)$ and using convexity of the actual follower objective gives

$$F_x(z^*) \geq F_x(q) - \lambda^\top C(z^* - q) + (a_q - a)^\top U(z^* - q) \geq F_x(q) + \lambda^\top (Cq - b(x)) - D_\phi \rho.$$

Here $Cz^* \leq b(x)$, $\lambda \geq 0$, and $\|U(z^* - q)\|_1 \leq \sum_i \|U_i\|_1$. Repair q to feasible y within $K\delta$ by lemma C.2. Then

$$0 \leq F_x(y) - F_x(z^*) \leq L_F K\delta + \zeta + D_\phi \rho.$$

First-order optimality at z^* and lemma C.3 bound $\mu \|y - z^*\|_\infty^{P+1}$ by this gap. Add the repair distance to prove (53).

At an exact optimum the bounded multipliers of lemma C.2 and $w = Uz^*$ give (51) with zero residuals. Conversely zero residuals imply the full convex follower KKT conditions and identify its unique response. Along a sequence in X_F , the aggregate and multiplier boxes are compact; every convergent subsequence of such certificates passes to a zero-residual certificate by continuity of the clipped inverses. Uniqueness identifies every response limit. This proves continuity, including at degenerate feasible faces. $\square$

The aggregate error has a useful sharper exponent. Retaining $\|z^* - q\|_\infty$ in the preceding dot-product bound and writing $t = \|y - z^*\|_\infty$ yields

$$\mu t^{P+1} \leq L_F K \delta + \zeta + D_\phi \rho (K \delta + t).$$

If t exceeded both terms in the following maximum, each right-hand term would be less than half the left side. Therefore

$$\|q - z^*\|_\infty \leq K \delta + \max \left\{ \left(\frac{2(L_F K \delta + \zeta + D_\phi \rho K \delta)}{\mu} \right)^{1/(P+1)}, \left(\frac{2D_\phi \rho}{\mu} \right)^{1/P} \right\}. \quad (54)$$

When $\delta = \zeta = 0$, take $y = q$ and divide by the response distance if it is positive to obtain the sharper $\|q - z^*\|_\infty \leq (D_\phi \rho / \mu)^{1/P}$. The simpler bound (53) suffices below. For $s = 0$ it reduces to the multiple-resource residual/complementarity bound, with no scalar balance identity presumed.

4.5 Polynomial response approximations and rational recovery

Normalize $w = R(2t - \mathbf{1})$, $\lambda = \Lambda \theta$, so the compressed variable $v = (x, t, \theta)$ lies in the rational polytope $Q_0 = X_F \times [0, 1]^{s+k}$. Its dimension $r + s + k$ is fixed. The inverse arguments

$$a_i(v) = \ell_i(x) - U_i^\top \nabla_w \phi(x, R(2t - \mathbf{1})) - \Lambda C_i^\top \theta$$

are rational polynomials. For tolerance $0 < \eta \leq 1/4$, pull back all rational inverse breakpoints by these arguments and compute their realizable sign conditions on Q_0 using section 1.5. At each sign sample choose a valid branch for every inverse. Define Q by Q_0 and the *closed branch-interval conditions* for that chosen branch vector. These compact sets cover Q_0 , and on each,

$$|p_i(v) - q_i(v)| \leq \eta. \quad (55)$$

There are polynomially many sets, since there are polynomially many realizable signs in fixed dimension. They may overlap or be disconnected. The closed validity set may enlarge its originating sign condition, but every added point retains all branch error bounds. This avoids both a Cartesian enumeration of branch choices and an unjustified closure operation on nonlinear cells. When $s = 0$, the arguments are affine and the sets can be the closed rational arrangement cells.

On each Q define the polynomial residuals

$$E = Cp - b(x), \quad J = \lambda^\top E, \quad T = w - Up,$$

and the compact candidate set

$$\mathcal{S}_Q = \{v \in Q : E_j \leq S\eta, \quad |J| \leq k\Lambda S\eta, \quad |T_a| \leq A_U \eta\}. \quad (56)$$

Empty row families are omitted. Every true response has a lift in some candidate set, with p within η of that response: use its true aggregate and a bounded multiplier. The polynomial residual allowances follow from (55). Every candidate in turn has true residual allowances $2S\eta$, $2k\Lambda S\eta$ and $2A_U\eta$. All polynomials have polynomial degree and coefficient length after expansion in the fixed compressed dimension.

Lemma 4.5 (Recovery across nonlinear branch boundaries). *Given $0 < \tau \leq 1/4$, choose*

$$\eta \leq \min \left\{ \frac{1}{4}, \tau, \frac{\tau}{8KS}, \frac{\mu(\tau/2)^{P+1}}{1 + 4L_FKS + 5k\Lambda S + 4D_\phi A_U} \right\}. \quad (57)$$

For any algebraically encoded $v^ \in \mathcal{S}_Q$ and any rational $h_0 > 0$, one can compute a rational $\hat{v} \in Q_0$ with $\|\hat{v} - v^*\|_\infty \leq h_0$ such that*

$$\|q(\hat{v}) - q(v^*)\|_\infty \leq \eta, \quad \|q(\hat{v}) - z(\hat{x})\|_\infty \leq \tau, \quad \|p(v^*) - z(\hat{x})\|_\infty \leq 2\eta + \tau. \quad (58)$$

All additional precision and bit cost are polynomial in the supplied encodings, in $\log(1/\tau)$, and in the positive part of $\log(1/h_0)$. Membership in the nonlinear branch set Q is not promised.

Proof. Set

$$\alpha = \min_i \frac{G_i}{2} \left(\frac{\eta}{2P} \right)^P.$$

By lemma B.1, an argument change less than α changes every clipped inverse by less than η : for actual degree $d \leq P$, $(\eta/(2d))^d \geq (\eta/(2P))^P$. Let $M_a \geq 1$ bound the Lipschitz constants of all a_i on the normalized cube. Let $B_b \geq 1$ bound the affine Lipschitz constant of b and let $\bar{E} = S + \sup_{x \in [0,1]^r} \|b(x)\|_\infty$, using an absolute coefficient upper bound for the supremum. Choose positive rational h strictly smaller than

$$\min \left\{ h_0, \frac{\alpha}{2M_a}, \frac{S\eta}{B_b + 1}, \frac{A_U\eta}{2R + 1}, \frac{S\eta}{\bar{E} + 1} \right\}. \quad (59)$$

Use lemma 4.2 in Q_0 . Then true clipped responses change by at most η , right-hand sides by at most $S\eta$, aggregates by at most $A_U\eta$, and multipliers by at most $\Lambda S\eta/(\bar{E} + 1)$.

At v^* the true residual allowances were twice those in (56). At $\hat{v}$ they are therefore

$$\delta = 4S\eta, \quad \zeta = 5k\Lambda S\eta, \rho = 4A_U\eta. \quad (60)$$

For complementarity write, with the *true* residuals $e^*, \hat{e}$,

$$\hat{\lambda}^\top \hat{e} - (\lambda^*)^\top e^* = \hat{\lambda}^\top (\hat{e} - e^*) + (\hat{\lambda} - \lambda^*)^\top e^*.$$

The first term has magnitude at most $2k\Lambda S\eta$ and the second at most $k\Lambda S\eta$, because $\|e^*\|_\infty \leq \bar{E}$. This absolute residual bound is needed for inactive constraints with large negative slack. Lemma 4.4 and (57) bound the repair and growth terms by $\tau/2$ each, proving (58). Every displayed constant and precision has polynomial bit length; raising a tolerance to $P + 1$ costs polynomially many bits in numerical P . The proof compares true continuous clipped responses across the boundary. It never evaluates the old branch polynomial at a point where that branch might be invalid. $\square$

For resource-only followers a sharper, entirely polyhedral recovery is available. Choose

$$\eta \leq \min \left\{ \frac{1}{4}, \tau, \frac{\tau}{6SK}, \frac{\mu(\tau/2)^{P+1}}{3S(1 + L_FK + k\Lambda)} \right\}. \quad (61)$$

Round within its rational branch cell, allowing each E_j to change by at most $S\eta$ and J by at most $k\Lambda S\eta$. Lemma 4.2 also permits any finite list of additional polynomial rounding tolerances. The true residuals at the rounded point are then $3S\eta$ and $3k\Lambda S\eta$, so lemma 4.4 gives $\|q - z(\hat{x})\|_\infty \leq \tau$. Thus the valid rounded polynomial response obeys $\|p(\hat{v}) - z(\hat{x})\|_\infty \leq \eta + \tau$. For an affine objective, set $\tau = \varepsilon/[16(1 + \|c\|_1)]$ and allow objective rounding error $\varepsilon/8$. A true optimum has polynomial lift value at most $\text{OPT} + \|c\|_1 \eta$, while the final true-to-polynomial error is at most $\|c\|_1 (\eta + \tau) \leq \varepsilon/8$. The final suboptimality is at most $5\varepsilon/16$; the rational rounded value lies between $\text{OPT} - \varepsilon/8$ and $\text{OPT} + 3\varepsilon/16$. This is the complete fixed-resource ledger, including equalities and vanishing curvature. It uses repair and polynomial growth, not (50).

4.6 Polynomial upper functions and the global theorem

For an explicitly listed upper polynomial $G(x, z) = \sum_{\alpha, \beta} a_{\alpha\beta} x^\alpha z^\beta$, put

$$\begin{aligned} L_x(G) &= \sum_{\alpha, \beta} |a_{\alpha\beta}| |\alpha|_1 2^{|\beta|_1}, \\ L_z(G) &= \sum_{\alpha, \beta} |a_{\alpha\beta}| |\beta|_1 2^{|\beta|_1}. \end{aligned} \tag{62}$$

These bound the separate infinity-norm Lipschitz constants on $[0, 1]^r \times [-1, 2]^N$, by summing absolute partial derivatives. They may vanish when their variable group is absent. The enlarged response box includes approximants with $\eta \leq 1/4$. Their encodings are polynomial in the explicit input and numerical upper degree, including the factor $2^{|\beta|_1}$.

On each candidate substitute $z = p(v)$. Let d_h bound the degrees of the univariate inverse branches, and put $d_a = \max\{1, \max_i \deg a_i\}$. Since each p_i is a branch composed with a_i , its degree is at most $d_h d_a$. The affine incentives and multiplier terms give $d_a \leq \max\{1, \deg \phi - 1\}$, with $d_a = 1$ when ϕ is constant or zero. The substituted upper polynomial therefore has degree at most $D_{\text{up}} \max\{1, d_h d_a\}$ in the fixed compressed variables. Both degree factors are polynomially bounded in the stated numerical-degree and accuracy parameters. Successive polynomial multiplication expands every listed monomial in polynomial time: only polynomially many monomials exist in fixed dimension at that degree, and coefficient growth is polynomial in numerical degree and input coefficient lengths. The number of upper polynomials and terms can grow without increasing the optimization dimension.

For the recovery in lemma 4.5, the segment between $p(v^*)$ and $z(\hat{x})$ lies in $[-1, 2]^N$, so

$$|G(\hat{x}, z(\hat{x})) - G(x^*, p(v^*))| \leq L_x(G)h_0 + L_z(G)(2\eta + \tau). \tag{63}$$

At a true response lift the corresponding error is at most $L_z(G)\eta$.

Proof of theorem 4.1. Projection decides emptiness. Continuity from lemma 4.4 and compactness of X_F ensure attainment. Let $\varepsilon = 2^{-B}$, $A_H = \max\{1, L_x(H), L_z(H)\}$, and choose $\tau \leq \varepsilon/(32A_H)$ and $h_0 \leq \varepsilon/(32A_H)$, with η as in (57). Minimize the substituted upper polynomial on every candidate set (56) and compare their exact minima. Each minimizing-point formula uses two copies of $r + s + k$ variables, so section 1.5 gives polynomial-size algebraic minima and samples. At a true optimizer the lift exists, so the winning value V satisfies $V \leq \text{OPT} + A_H \eta \leq \text{OPT} + \varepsilon/32$. Recover the winner by lemma 4.5. Equation (63) gives $|H(\hat{x}, z(\hat{x})) - V| \leq 4A_H \varepsilon/(32A_H) = \varepsilon/8$. Exact membership $\hat{x} \in X_F$ also gives $V \geq \text{OPT} - \varepsilon/8$. Approximate V rationally to error $\varepsilon/16$ to obtain the claimed value estimate and the leader guarantee. Only polynomial-degree algebraic outputs in fixed dimension are compared; no exact sum of the true inverse values is evaluated.

If $N = 0$, the follower constraints reduce to $0 \leq b(x)$ and there is no response. Minimize the polynomial upper objective on this rational polytope in fixed dimension and apply lemma 4.2. This is an LP only when the upper objective is affine. All counts, degrees, coefficient encodings and requested precisions above are polynomial in $(L_{\text{in}}, D_{\text{all}}, B)$. $\square$

For example, a convex quartic aggregate may couple many units through one total output while local second derivatives vanish at an optimum. The objective $-x_1 \sum_i z_i + d(x)$ includes tariff revenue and a polynomial intervention cost. Neither follower convexity nor the theorem requires the induced revenue to be concave. These are structural examples; the coarse constants and fixed-dimensional elimination do not establish an efficient implemented solver.

4.7 Upper constraints: approximation and exact feasibility

Let $G_j(x, z) \leq 0$, $j = 1, \dots, m$, be explicitly listed rational polynomial upper inequalities, with m unrestricted. Put $\mathcal{H}(x) = H(x, z(x))$, $\mathcal{G}_j(x) = G_j(x, z(x))$ and

$$V(t) = \min\{\mathcal{H}(x) : x \in X_F, \mathcal{G}_j(x) \leq -t \text{ for all } j\}, \quad t \geq 0, \quad (64)$$

with $V(t) = +\infty$ for an empty set. By continuity and compactness each finite minimum is attained. Leader-only affine rows should be placed directly in X and preserved exactly. Response-dependent upper equalities may be represented by opposite inequalities, but in general cannot satisfy the positive tightening assumptions below.

Margin conversion from response error to upper feasibility is standard, including in near-optimal robust bilevel models [9]. The value of perturbed inequalities and convex interpolation toward a strict point are also established [10, Section 5.6]. Here they are combined with the proved global accuracy-bit construction and its rational recovery.

Theorem 4.6 (Outer and inner guarantees). *For positive rational objective and violation tolerances ε, δ , there are algorithms with bit complexity polynomial in the input, numerical degrees, and the encodings of these tolerances, for fixed r, k, s , with the following properties.*

The outer algorithm either certifies original infeasibility, or returns rational $\hat{x} \in X_F$ with $\mathcal{G}_j(\hat{x}) \leq \delta$ for every j . If $V(0)$ is finite it returns a point and $\mathcal{H}(\hat{x}) \leq V(0) + \varepsilon$. An outer point alone does not certify original feasibility.

The inner algorithm either reports an empty inner surrogate, or returns rational $\hat{x} \in X_F$ satisfying all original upper inequalities exactly. If $V(\delta)$ is finite it returns a point with $V(0) \leq \mathcal{H}(\hat{x}) \leq V(\delta) + \varepsilon$. An empty inner surrogate implies $V(\delta) = +\infty$, but does not certify original infeasibility.

Proof. Let A be the maximum of one and all L_x, L_z constants for the objective and upper rows, and choose

$$\kappa = \min\{1/4, \varepsilon/(16A), \delta/(16A)\}, \quad \tau \leq \kappa, \quad h_0 \leq \kappa,$$

with $\eta \leq \kappa$ satisfying (57). On each candidate set write $F_j(v) = G_j(x, p(v))$ and $H_Q(v) = H(x, p(v))$. A true lift changes each criterion by at most $L_z(G_j)\eta$. At any candidate and its recovery, (63) is at most $4A\kappa \leq \min\{\varepsilon, \delta\}/4$.

For the outer algorithm impose $F_j \leq L_z(G_j)\eta$ on every candidate set and minimize H_Q exactly. Every true feasible leader has an admissible lift. Hence emptiness certifies original infeasibility, and, when $V(0)$ is finite, the winning value V_{out} is at most $V(0) + L_z(H)\eta$. Recover its algebraic minimizer using lemma 4.5. The true violation is at most $\delta/16 + \delta/4$ and objective loss at most $\varepsilon/16 + \varepsilon/4$, which prove the stated bounds.

For the inner algorithm impose $F_j \leq -\delta/2$. Every leader feasible for $V(\delta)$ has an admissible lift, since $L_z(G_j)\eta \leq \delta/16$. Any recovered candidate satisfies $\mathcal{G}_j(\hat{x}) \leq -\delta/4 < 0$. If $V(\delta)$ is finite,

its lift bounds the winning polynomial value by $V(\delta) + L_z(H)\eta$, and recovery gives the stated objective bound. All optimization still uses a fixed number of variables even though m grows. In the resource-only case one may instead use the rational-cell recovery and its additional polynomial rounding tolerances; the same conclusions follow from its smaller response-error bounds.

When $N = 0$, optimize the upper polynomials directly over $X \cap \{0 \leq b(x)\}$ with the same outer/inner margins and rational polytope recovery. This is not in general an LP. Likewise, a response-independent objective does not justify skipping response approximation when any upper constraint still depends on it. $\square$

The two algorithms give a posteriori global certificates without any regularity promise. Let rational $v_{\text{out}}, v_{\text{in}}$ approximate the exact winning surrogate values within $\omega_{\text{out}}, \omega_{\text{in}}$. For the inner recovery put $e_{\text{in}} = L_x(H)h_0 + L_z(H)(2\eta + \tau)$, using its own tolerances. If both procedures return, then

$$\underbrace{v_{\text{out}} - \omega_{\text{out}} - L_z(H)\eta_{\text{out}}}_{\text{Lower}} \leq V(0) \leq \mathcal{H}(x_{\text{in}}) \leq \underbrace{v_{\text{in}} + \omega_{\text{in}} + e_{\text{in}}}_{\text{Upper}}. \quad (65)$$

The inner point supplies the needed original feasibility. An outer objective alone is not an absolute-error estimate of $V(0)$ and can lie below it. As all tolerances tend to zero, the outer lower bounds converge to finite $V(0)$: otherwise compactness, vanishing violation, and continuity yield a feasible limit with smaller objective. Inner upper bounds also converge if $V(t) \rightarrow V(0)$ as $t \downarrow 0$. The computed finite gap is always a valid certificate when both bounds exist; a quantitative stopping guarantee needs more.

Corollary 4.7 (A supplied tightening modulus). *Suppose $V(0) < +\infty$ and supplied rationals $D_0, \sigma > 0$ and integer $\nu \geq 1$ satisfy*

$$V(t) \leq V(0) + D_0(t/\sigma)^{1/\nu}, \quad 0 < t \leq \sigma. \quad (66)$$

Then an exactly upper-feasible rational leader within additive ε of $V(0)$, and a rational absolute-error optimum estimate, are computable in polynomial time in the input, numerical degrees, numerical ν , and accuracy bits. The modulus parameters enter through their encodings; the algorithm need not verify the promise.

Proof. Use inner tightening $t = \sigma \min\{1/2, (\varepsilon/(2D_0))^{\nu}\}$ and objective tolerance $\varepsilon/2$. The modulus ensures nonemptiness and tightening loss at most $\varepsilon/2$. Its rational encoding is polynomial in numerical ν and the supplied bit lengths. Repeat with smaller constant fractions of the requested error and approximate the inner achieved objective using its surrogate/recovery bound to obtain an absolute-error estimate of $V(0)$. $\square$

Corollary 4.8 (Convex reduced constraints with a strict anchor). *Suppose each $\mathcal{G}_j$ is convex on X_F , and a supplied point $x_s \in X_F$ has $\mathcal{G}_j(x_s) \leq -\sigma$ for all j , with rational $\sigma > 0$. For $N \geq 1$, corollary 4.7 applies with*

$$\nu = P, \quad D_0 = \max\{1, L_x(H) + L_z(H)C_z\},$$

where the effective constant C_z is given in proposition C.4. One can alternatively use its simpler modulus with exponent $\nu = P + 1$ and the corresponding constant. For $N = 0$, use $\nu = 1$ and $D_0 = \max\{1, L_x(H)\}$ by direct Lipschitz interpolation. Convexity of the reduced constraints and the anchor margin are additional promises, not consequences of follower convexity.

Proof. At an original optimizer x^* , set $x_t = (1 - t/\sigma)x^* + (t/\sigma)x_s$, for $0 \leq t \leq \sigma$. It lies in X_F , has all reduced constraints at most $-t$, and moves by at most t/σ . Equation (103) and (62) bound its objective change by $D_0(t/\sigma)^{1/P}$. This proves the modulus. The algorithm uses the supplied margin; it does not silently solve an exact algebraic-sum comparison to certify the anchor. $\square$

For a concrete independent-follower subclass, assume the increasing marginals are convex, $g_i(0) = 0$, and the affine incentives are nonnegative throughout X . Then $z_i(x) = \min\{1, g_i^{-1}(\ell_i(x))\}$ is concave: the increasing inverse is concave and its constant extension above saturation preserves concavity. Thus $a_{j0} + a_j^\top x - \sum_i w_{ji} z_i(x)$ is convex for $w_{ji} \geq 0$, including minimum weighted production constraints. The incentive sign matters: clipping a negative incentive to zero can destroy concavity. This example does not assert concavity under shared resources or aggregate coupling.

Corollary 4.9 (Reserve controls with uniform headroom). *Write the leader as $(u, s) \in X_u \times S_r$, with a fixed-dimensional rational reserve polytope $S_r \subseteq [0, 1]^p$. Suppose the complete follower objective and constraints depend only on u , and at least one follower-feasible u exists. For each feasible u , assume every $G_j(u, s, z(u))$ is convex in s , and there is a supplied $s_{\text{safe}} \in S_r$ with uniform margin $G_j(u, s_{\text{safe}}, z(u)) \leq -\sigma$ for all u, j . Then corollary 4.7 applies with $\nu = 1$ and $D_0 = \max\{1, L_s(H)\}$, where the polynomial coefficient bound for reserve derivatives supplies $L_s(H)$.*

Proof. Keep the incentive component u^* of an original optimizer and interpolate its reserve toward s_{safe} . The follower response is unchanged. Convexity in the reserve gives margin t at interpolation fraction t/σ , and the objective changes by at most $L_s(H)t/\sigma$. The safe reserve also supplies original feasibility whenever the follower-feasible u set is nonempty. $\square$

For affine rows $a_{j0} + a_j^\top u + c_j^\top z - R_j s \leq 0$, uniform headroom can be certified conservatively by rational LP over the explicit polytope $X_{u,F} \times [0, 1]^N$ with $s = s_{\text{safe}}$. The matrix R need not have a prescribed sign. For an objective $h(u) + w^\top s$, use $D_0 = \max\{1, \|w\|_1\}$. Reserves are actual model controls, such as external procurement, not artificial slacks inserted into an existing hard constraint. Their dimension counts toward the fixed leader dimension. The aggregate must also be independent of them. No convexity of the reduced constraints in u is needed.

4.8 Why the feasibility qualifications are necessary

Example 4.10 (A strict point does not ensure inner convergence). For $x, z \in [0, 1]$, let

$$F_x(z) = z^2/2 + z^4/4 - z^3/6 - xz, \quad g(z) = z + z^2(z - 1/2).$$

The marginal derivative is $g'(z) = 3(z - 1/6)^2 + 11/12 > 0$, so even uniform strong curvature holds. Impose the affine upper row $z(x) - x \leq 0$ and minimize x . Since $g(0) = 0$, $g(1) = 3/2$, and $x \in [0, 1]$, the response is its unclipped inverse. Monotonicity gives

$$z(x) \leq x \iff x \leq g(x) \iff x^2(x - 1/2) \geq 0.$$

Thus the feasible set is $\{0\} \cup [1/2, 1]$ with optimum zero and strict feasible point one. Every positively tightened feasible leader has $x > 1/2$. By continuity and strict feasibility for $x > 1/2$, $V(t) \rightarrow 1/2$ as $t \downarrow 0$, so the original optimum is not approached. A strict point alone cannot justify an inner convergence or tightening-modulus claim.

Example 4.11 (No rational exactly feasible leader). The follower cost $z^3/3 - xz$ on the unit box gives $z(x) = \sqrt{x}$. The affine upper equality $z - x = 1/8$ permits precisely

$$x = 3/8 \pm \sqrt{2}/4.$$

Both roots lie in $[0, 1]$ and satisfy the unsquared equation, but both are irrational. Thus feasible rational leader output is impossible without qualifications, even with one leader and one follower.

Finally, constant incentives a_i/M^2 , cubic local costs and upper weights M , with integer $M \geq \max_i a_i > 0$, give the fixed upper value $\sum_i \sqrt{a_i}$. One exact upper inequality therefore contains sum-of-square-roots comparison. The accuracy algorithms do not resolve this exact arithmetic problem: outer constraints allow prescribed violation and exact inner feasibility uses stated margins. Numerical degree dependence is also essential by example 4.3. Neither these arithmetic limits nor the rational-output examples are claims that a convex follower cannot itself be approximated efficiently.

5 Structural and arithmetic boundaries

The preceding algorithms use more than a small leader dimension or a convex follower. This section separates four restrictions: the interaction matrix in the follower objective, the number and interaction graph of the leaders, the coupling of follower constraints, and the requested arithmetic precision. Throughout, $\text{clip}(t) = \min\{1, \max\{0, t\}\}$.

5.1 A scalar leader with a dense positive definite follower

Consider the rational problem

$$\min_{x \in X} a^\top z(x) + b^\top x, \quad z(x) = \arg \min_{z \in [0,1]^N} \left\{ \frac{1}{2} z^\top Q z + (c + Dx)^\top z \right\}, \quad Q \succ 0. \quad (67)$$

There are no upper constraints involving the response in this subsection. The response is unique and continuous: any convergent sequence of leaders has follower subsequences, and the limiting optimality inequalities identify every subsequence limit with the unique response at the limiting leader. Thus a compact leader polytope has an attained upper optimum.

Scalar-leader hardness and ternary digit gadgets have a close linear-bilevel precedent in Sugishita and Carvalho [41, Theorem 1 and Section 4]. Exponential strongly convex response paths also precede the following reduction. In particular, the full-support sign patterns in Mairal and Yu [33, Proposition 2, equation (4)] imply a Boolean-vertex box path. For their invertible square Lasso design A and target v , the normalized dual, with $t = 1/\lambda$, is

$$\min_{u \in [-1,1]^p} \frac{1}{2} u^\top A^{-1} A^{-\top} u - t(A^{-1}v)^\top u.$$

At a full-support primal sign pattern its dual point is that sign vector. Their recursion visits every full sign vector whose last coordinate is positive; an affine box change therefore gives every Boolean assignment in the first $p - 1$ coordinates. This consequence of duality does not itself establish the encoding bounds or upper-objective gap of a hardness reduction.

Lemma 5.1 (Rational certificates for box responses). *For rational data in (67), a rational upper threshold and a rational leader polytope, threshold feasibility is in NP, even when the leader dimension grows and affine upper rows are allowed.*

Proof. Guess a partition (L, F, U) of the coordinates into lower, free and upper statuses. Set $z_L = 0$, $z_U = 1$, and solve

$$z_F(x) = -Q_{FF}^{-1}(c_F + D_F x + Q_{FU} \mathbf{1}).$$

The principal matrix is positive definite; Cramer's rule gives affine rational functions of polynomial bit length. Their box bounds, the endpoint gradient signs $(Qz + c + Dx)_L \geq 0$, $(Qz + c + Dx)_U \leq 0$, and all upper rows and the threshold form a rational polyhedron in x . If nonempty, it has a rational point of polynomial bit length. Weak bounds on the free coordinates include degeneracies. A verifier reconstructs z and checks these conditions exactly; convexity makes them sufficient for global optimality. For a scalar leader the polyhedron is simply an interval, possibly a singleton. $\square$

Theorem 5.2 (Dense scalar hardness with a constant gap). *Threshold feasibility in (67) is NP-complete already for $X = [0, 1]$, a rational positive definite Q independent of the leader, and an upper objective with coefficients in $\{-2, 0, 2\}$. A polynomial reduction gives an attained optimum equal to zero in the yes case and at least two in the no case. All leaders are feasible. The follower objective can be represented as a jointly convex sum of affine squares in the leader and followers.*

Proof. Start from a 3SAT instance with $n \geq 1$ variables and m clauses, each on three distinct variables. This restriction preserves NP-hardness: remove repetitions and tautologies; pad a two-literal clause using both signs of a fresh variable, and a unit clause using all four sign pairs of two fresh variables. An empty clause can be replaced by the fixed unsatisfiable formula containing all eight clauses on three variables. Write $\ell_a(y)$ for the sum of the three continuous literal values in clause a . Put

$$\rho = \frac{1}{100 \cdot 3^n}, \quad w_i = \rho^{i-1}, \quad \eta = \rho^n, \quad \xi = \frac{\eta}{m+1}, \quad r_i = y_i - 3^i x + 2 \sum_{j < i} 3^{i-j} y_j + 1.$$

Use followers $(y, p, v) \in [0, 1]^{2n+m}$ and minimize

$$\frac{1}{2} \sum_i w_i r_i^2 + \frac{\eta}{2} \sum_i (p_i - 2y_i + 1)^2 + \frac{\xi}{2} \sum_a (v_a - 1 + \ell_a(y))^2. \quad (68)$$

The residual map in the followers is square block triangular with diagonal blocks L, I, I , where L is lower triangular with unit diagonal. Its positive weighted Gram matrix is positive definite.

Every Boolean vector $d \in \{0, 1\}^n$ occurs in the response at

$$x_d = \sum_{i=1}^n \frac{2d_i}{3^i} + \frac{1}{2 \cdot 3^n} \in (0, 1).$$

Indeed, with $t_i = 3^{i-1} x_d - 2 \sum_{j < i} 3^{i-1-j} d_j$, the residual $r_i = d_i - 3t_i + 1$ has sign $1 - 2d_i$ and magnitude between $1/(2 \cdot 3^{n-i})$ and one. The gradient of the first sum in coordinate i is $G_i = w_i r_i + \sum_{k > i} 2 \cdot 3^{k-i} w_k r_k$. Its later-coordinate contribution has magnitude at most

$$w_i \frac{6\rho}{1 - 3\rho} < \frac{w_i}{10 \cdot 3^n}, \quad (1 - 2d_i) G_i > \frac{w_i}{3^n}.$$

At any complete follower optimum, conditional minimization gives

$$p_i = \text{clip}(2y_i - 1), \quad y_i - p_i = \min(y_i, 1 - y_i), \quad v_a = \max(0, 1 - \ell_a(y)). \quad (69)$$

At $y = d$, take these conditional optima. The p -squares add derivative -2η at a zero bit and zero at a one bit. A clause residual is respectively 0, 0, 1, 2 when the clause has 0, 1, 2, 3 true literals, so the magnitude of all clause feedback at any base coordinate is at most $2m\xi < 2\eta$. The total perturbation is less than $4\eta = w_n/(25 \cdot 3^n)$, below the base margin. All box optimality signs therefore

hold, including those of the auxiliary coordinates. Positive definiteness proves that this is the unique global response.

Choose the upper objective

$$H = 2 \sum_i y_i - 2 \sum_i p_i + 2 \sum_a v_a.$$

It is nonnegative by (69), and a satisfying Boolean response gives zero. If the formula is unsatisfiable, round y at $1/2$ and take a false clause. Its literal sum is at most $D_y = \sum_i \min(y_i, 1 - y_i)$, since its variables are distinct. Consequently

$$H \geq 2D_y + 2 \max(0, 1 - D_y) \geq 2.$$

The shortfall coordinates have thus replaced all upper clause constraints by an objective penalty inside the response. No upper feasibility assumption is needed. Every coefficient and every displayed witness has polynomial bit length: powers 3^i have $O(n)$ bits and the weights have $O(n^2)$ bits. Expanding the squares yields (67) after removing a leader-only term. The unexpanded squares are jointly convex; deleting that term preserves responses but need not preserve joint convexity in (x, z) . Lemma 5.1 proves NP membership. $\square$

Corollary 5.3 (Positive objectives and multiplicative approximation). *Unless $P = NP$, no polynomial-time algorithm for the dense scalar class with strictly positive affine upper objectives can return exactly bilevel-feasible solutions with a multiplicative ratio strictly below three. Allowing polynomially encoded upper coefficients of unrestricted magnitude rules out every ratio bounded by a fixed polynomial in the total input length.*

Proof. The readout H in Theorem 5.2 is nonnegative at every exact follower response. Replacing it by $1 + H$ gives optimum one in the yes case and at least three in the no case. A ratio strictly below three would return a feasible objective below three in the yes case, while every feasible objective is at least three in the no case, deciding 3SAT.

For the general ratio, put $s = n + m$ for the source formula and replace the objective by $1 + 2^s H$. The multiplier 2^s has $s + 1$ binary bits, so the constructed input length L remains polynomial in s . The yes optimum is one and the no optimum is at least $1 + 2^{s+1}$. For any fixed polynomial ratio bound $q(L)$, we have $q(L) < 2^s$ for all sufficiently large s . The returned feasible objective then separates yes from no instances by comparison with 2^s . Exhaustive search handles the finitely many smaller source sizes, giving a polynomial 3SAT algorithm in all cases. $\square$

The additive constant is ordinary affine upper data; the exact coefficient alphabet of Theorem 5.2 is not retained. The exponential scaling also loses bounded upper-coefficient magnitudes. This elementary consequence preserves polynomial binary encoding and exact bilevel feasibility; it proves neither strong hardness nor an approximation result for relaxed follower optimality or relaxed feasibility.

5.2 Exact hardness arbitrarily close to the identity

The small weights in the proof of Theorem 5.2 do not control conditioning. A different construction does, at the cost of a much smaller upper gap. Positive diagonal network scaling is established in El Ghaoui et al. [16, equations (2.6)–(2.7)], and convex squared-residual energies with weak backward feedback are used by Zach and Estellers [44, Section 2]. The proof below supplies explicit rational scales and a relative-coordinate error bound needed for the reduction.

Theorem 5.4 (Near-identity exact hardness). *Threshold feasibility in (67) with $X = [0, 1]$ remains NP-complete when all normalized follower and upper coefficients have magnitude at most two and*

$$(49/50)^2 I \preceq Q \preceq (51/50)^2 I, \quad \kappa_2(Q) < 2.$$

For any supplied rational $0 < \zeta \leq 1$, the reduction can also ensure $\|Q - I\|_1, \|Q - I\|_\infty, \|Q - I\|_2 < \zeta$ and strict diagonal dominance. The resulting yes/no objective gap is exponentially small but has polynomial bit length. Unless $P = NP$, there is no global additive approximation algorithm polynomial in input length and accuracy bit length for this class.

Proof. Use the same clause functions. Starting with $t_1 = x$, define a feedforward ReLU computation

$$\begin{aligned} \alpha_i &= (3t_i - 1)_+, & \beta_i &= (3t_i - 2)_+, & y_i &= \alpha_i - \beta_i, \\ t_{i+1} &= 3t_i - 2y_i, & p_i &= (2y_i - 1)_+, & v_a &= (1 - \ell_a(y))_+. \end{aligned}$$

The map $t \mapsto 3t - 2 \text{clip}(3t - 1)$ sends $[0, 1]$ into itself. Order the states as $h = (\alpha_1, \beta_1, \dots, \alpha_n, \beta_n, p, v)$, with $N = 3n + m$. Substituting the recursion for each t_i gives

$$h = (Ah + b_0 + xb_1)_+,$$

where A is strictly lower triangular and its entries and those of b_0, b_1 have magnitude at most $C = 23^n$. On $[0, 1]$, $0 \leq h \leq 21$, and $0 \leq h - Ah - b_0 - xb_1 \leq 21$: the preactivations for α, β, p, v lie respectively in $[-1, 2], [-2, 1], [-1, 1], [-2, 1]$. The score

$$\Phi(h) = 2 \sum_i \alpha_i - 2 \sum_i \beta_i - 2 \sum_i p_i + 2 \sum_a v_a = \ell^\top h$$

has $\|\ell\|_1 = 2N$. The previous rounding argument gives $\Phi \geq 0$, value zero at a satisfying encoded leader, and $\Phi \geq 2$ everywhere for an unsatisfiable formula. At x_d , the ternary recursion gives $y = d$.

Set

$$M = (1 + NC)^N, \quad \theta = \frac{1}{100NCM}, \quad s_i = \frac{\theta^{i-1}}{4C}, \quad S = \text{diag}(s_i), \quad B = SAS^{-1}.$$

If the additional tolerance is requested, replace θ by $\min\{\theta, \zeta/(10C)\}$. The unit-box follower minimizes

$$\frac{1}{2} \|(I - B)u - S(b_0 + xb_1)\|_2^2. \quad (70)$$

Since $|B_{ij}| \leq C\theta^{i-j}$ for $i > j$, both its row and column norm are at most $2C\theta \leq 1/50$. Thus $Q = (I - B)^\top(I - B)$ has the asserted spectral bounds. Moreover $\|Sb_j\|_\infty \leq 1/4$, so the normalized linear-cost coefficients are less than one in magnitude. The spectral bound also bounds every entry of Q by two.

Here the small absolute size of u is not an adequate error measure. Let u be the true minimizer, $e_i = |u_i/s_i - h_i|$, $P = |A|$, and $r = (I - B)u - S(b_0 + xb_1)$. The exact KKT fixed point is

$$u = \text{clip}(Bu + S(b_0 + xb_1) + B^\top r).$$

The scaled feedforward point Sh is the same clipped fixed point without $B^\top r$; its positive coordinates are strictly below one. Coordinatewise nonexpansiveness of clipping gives

$$e \leq Pe + U|S^{-1}r|, \quad U = S^{-1}|B|^\top S = S^{-2}P^\top S^2, \quad |S^{-1}r| \leq (I + P)e + 21.$$

The nonnegative matrix $T = (I - P)^{-1} = \sum_{j=0}^{N-1} P^j$ has $\|T\|_\infty \leq M$; also $\|P\|_\infty \leq NC$ and $\|U\|_\infty \leq 2C\theta^2$. Multiplication by T therefore yields

$$\|e\|_\infty \leq 2MC(1 + NC)\theta^2 \|e\|_\infty + 4MC\theta^2, \quad \|e\|_\infty \leq 8MC\theta^2 \leq \frac{1}{16N},$$

since the first coefficient is less than $1/2$. These inequalities continue to hold after decreasing θ . Put $\delta = s_N$ and use upper objective $G(u) = \delta \ell^\top S^{-1}u$. Every coefficient has magnitude at most two, while

$$|G(u) - \delta\Phi(h)| \leq \delta/8.$$

The yes optimum is at most $\delta/8$; the no optimum is at least $15\delta/8$. Comparing with threshold δ proves hardness, and Lemma 5.1 supplies membership. All scales have polynomial bit length, including the additional input length of ζ . Finally, $\|B\|_1, \|B\|_\infty \leq \zeta/5$ gives $\|Q - I\|_{1,\infty,2} \leq 2\zeta/5 + \zeta^2/25 < \zeta$. For each row, $Q_{ii} - \sum_{j \neq i} |Q_{ij}| \geq 1 - \|Q - I\|_\infty > 0$. Accuracy $\delta/4$ separates the two cases and has polynomially many bits. $\square$

The last theorem is ordinary exact hardness. Its bounded coefficients and conditioning do not yield strong NP-hardness or an inverse-polynomial absolute gap under the same restrictions. In particular it is compatible with the following algorithm, whose dependence is on inverse accuracy.

5.3 A conditioned approximation scheme for dense box followers

Theorem 5.5 (A cost-space grid). *Fix r . For (67), let $X \subseteq [0, 1]^r$ be a nonempty rational polytope. For rational $0 < \varepsilon \leq 1$, one can return a rational leader and its exact rational follower with upper value at most $\text{OPT} + \varepsilon \|a\|_1$, in time polynomial in the rational input length, $K = \|Q\|_\infty \|Q^{-1}\|_\infty$, and $1/\varepsilon$. Here $K \leq N\kappa_2(Q)$ for $N \geq 1$. There are no response-dependent upper rows.*

The maximum-volume basis used below is the classical barycentric-spanner argument of Awerbuch and Kleinberg [3, Proposition 2.2]. Approximation of parametric convex solution paths and uniform optimizer errors have earlier treatments, including Giesen et al. [22] and Bemporad and Filippi [6, Theorem 4]. The particular purpose of the saturation step here is to remove numerical dependence on the affine-cost magnitudes; all those coefficients still contribute their binary encoding lengths. The scalar additive path bound of Giesen et al. [23, Lemma 4 and Theorem 5] is another close precedent: its coefficient involves parameter range and slope variation of the follower value. Strong convexity converts follower-objective error into response error, but alone does not remove those numerical factors.

Proof. If $N = 0$, solve the leader LP. If $r = 0$, evaluate the sole follower. If $a = 0$, solve the leader LP and evaluate its follower for the promised output. Hence assume $N, r \geq 1$ and $\|a\|_1 > 0$. Put $\mu = 1/\|Q^{-1}\|_\infty \leq \lambda_{\min}(Q)$. For row i , write

$$m_i^- = \sum_j \min(Q_{ij}, 0), \quad m_i^+ = \sum_j \max(Q_{ij}, 0), \quad R_i = m_i^+ - m_i^- = \sum_j |Q_{ij}| > 0.$$

If its linear cost $t_i \geq -m_i^-$, its response coordinate is zero: for $z_i > 0$ its gradient is at least $Q_{ii}z_i + t_i + m_i^- > 0$, contradicting box optimality. Similarly $t_i \leq -m_i^+$ forces $z_i = 1$, because at $z_i < 1$ the gradient is at most $Q_{ii}(z_i - 1) + t_i + m_i^+ < 0$. These tests include equality at thresholds.

Arrange the $2N$ hyperplanes $c_i + D_i x = -m_i^\pm$, intersect all relative cells with X , and use their closures. There are $N^{O(r)}$ such cells. On each, coordinates outside a set J are fixed at endpoints; all costs with indices in J lie in the closed slabs of width R_i . Coordinates with constant costs

inside a slab remain in J . For two leaders in this cell, the two variational inequalities and positive definiteness give

$$\|z(x) - z(x')\|_2 \leq \frac{\|D_J(x - x')\|_2}{\mu}. \quad (71)$$

Indeed the response difference vanishes outside J , so the cost difference in those coordinates contributes nothing to their summed inequalities.

Let $E = D_J/\mu$ have rank $q \leq r$. If $q = 0$, the response is constant on the cell and minimizing $b^\top x$ suffices. Otherwise choose q independent columns of E , then a maximum absolute q -row minor in those columns. Let B_0 denote the selected full rows. Every row of E is a linear combination of those rows with coefficients in $[-1, 1]$: replacing one row in the maximal minor and applying Cramer's rule proves the bound; the selected columns determine dependencies among full rows. Consequently

$$\|B_0(x - x')\|_\infty \leq h \implies \|z(x) - z(x')\|_2 \leq \sqrt{N} qh.$$

A selected coordinate $D_i x/\mu$ ranges in $[(-m_i^+ - c_i)/\mu, (-m_i^- - c_i)/\mu]$, of width $R_i/\mu \leq K$. Its offset may be large but has polynomial bit length. Divide each of these q intervals into closed intervals of length at most $h = \varepsilon/(Nr)$. For every resulting rectangle, intersect its inverse image under B_0 with the current cell, minimize $b^\top x$ there by rational LP, and evaluate the exact follower. Keep the best value. The number of LPs is $N^{O(r)}(2 + KNr/\varepsilon)^r$, and their data have polynomial bit length. An optimal leader belongs to one of these closed sets. Its LP representative has no larger direct leader cost and changes the response term by at most $\|a\|_1 \sqrt{N} rh \leq \varepsilon \|a\|_1$. Cell boundaries are included by the weak saturation tests.

For completeness an elementary exact follower oracle fits the claimed $\text{poly}(K)$ running time. At a rational leader put $t = c + Dx$ and $L_Q = \|Q\|_\infty$. Projected gradient iteration from zero,

$$z^{j+1} = \text{clip}(z^j - (Qz^j + t)/L_Q),$$

contracts Euclidean distance to the optimum by at most $1 - 1/K$. This follows from nonexpansiveness of projection and the eigenvalues of $I - Q/L_Q$. Clear all denominators in Q, t simultaneously, and bound the absolute integer entries by $H \geq 1$. In any active principal system, the denominator of each optimum coordinate in reduced form is at most $B = N!H^N$; Cramer's rule gives this bound also when free coordinates reach endpoints. After

$$j = \lceil K(2\lceil \log_2 B \rceil + \lceil \log_2 N \rceil + 4) \rceil$$

iterations every coordinate is within $1/(4B^2)$ of its optimum. Two distinct rationals of denominator at most B differ by at least $1/B^2$, so the optimum is the unique such rational in that error interval. It is found among the continued-fraction convergents of the iterate: the error is less than $1/(2q^2)$ for its denominator $q \leq B$; endpoints zero and one can be checked separately. Verify the recovered vector by exact box KKT conditions. The iteration count is polynomial in K and input length; rational iterate bit lengths grow at most linearly in that count times the input length. All recovery and verification therefore have the claimed bit complexity. Finally, symmetry gives $\|Q\|_\infty \leq \sqrt{N} \|Q\|_2$ and the same inequality for Q^{-1} , establishing $K \leq N\kappa_2(Q)$. $\square$

A grid directly in leader coordinates would not have this uniform guarantee. For $Q = 2^{-L}$ and linear cost $-xz$, the scalar response is $\text{clip}(2^L x)$, although its Hessian condition number is one. Saturation and cost normalization isolate precisely this narrow transition. Neither the number of grid cells nor the runtime above is polynomial in accuracy bit length, or fixed-parameter tractable in r .

5.4 Growing leader dimension and a bounded core

With an identity Hessian and independent boxes, each follower is $z_i(x) = \text{clip}(h_i(x))$ for an affine rational h_i . Thus affine upper optimization on a leader product box is optimization of a signed sum of clipped affine functions. Define its *leader interaction graph* by joining two leader coordinates whenever both occur in some h_i . This graph refers to cost dependence on leaders, not to constraints joining follower coordinates.

Theorem 5.6 (Growing leader dimension). *With growing leader dimension r , identity follower Hessian, independent unit boxes and no upper constraints, threshold optimization is NP-complete and $W[1]$ -hard parameterized by r . Under the exponential time hypothesis (ETH), no algorithm has runtime $f(r)\mathcal{L}^{o(r)}$, where $\mathcal{L}$ is input length. These conclusions hold with each follower depending on at most two leaders, all follower affine coefficients of magnitude at most one, and upper coefficients of magnitude at most two. The reduction has a constant absolute upper-objective gap.*

Proof. The parameterized source result is due to Froese et al. [18, Proposition 4.1, Theorem 5.3 and Corollary 5.5]. From Multicolored Clique with k colors it constructs a two-layer ReLU network in k inputs with maximum $C_0 = k + \binom{k}{2}$ in the yes case and at most $C_0 - 1$ otherwise. Its realizing points lie in a polynomially bounded box $[0, L]^k$; all affine coefficients are polynomially bounded in magnitude, and each neuron depends on at most two inputs. Output coefficients can be made ± 1 by positive homogeneity and duplication of their bounded integer magnitudes. These properties follow directly from its node and edge spikes, whose locations are integers in a greedy Sidon set of polynomial magnitude. We use the proved gap, not an assumption that continuous inputs must themselves be discrete.

Rescale the input as $t = Lx$. Write the resulting network as $\sum_i \sigma_i(h_i(x))_+$, with $\sigma_i \in \{-1, 1\}$. Choose a positive integer $R > \max_i(|h_i(0)| + \|\nabla h_i\|_1)$, polynomially bounded in the source size. Then

$$\text{clip}(h_i(x)/R) = (h_i(x))_+/R \quad (x \in [0, 1]^k).$$

Each coordinate is the response of the identity-Hessian follower with cost $-h_i(x)z_i/R$. The upper expression $C_0 - R \sum_i \sigma_i z_i(x)$ has gap zero versus at least one. Duplicate each coordinate R times to replace its coefficient $R\sigma_i$ by unit coefficients. If a constant-free linear objective is required, add C_0 fixed-one responses. All follower coefficients now have magnitude at most one; the duplication is polynomial and preserves dependence on at most two leaders. The source parameter and ETH bounds transfer with $r = k$. Lemma 5.1 proves ordinary NP membership. $\square$

This theorem transfers an established parameterized hardness result. An alternative explicit construction below records a useful continuous rounding gap and avoids reliance on the particular shape of the source spikes.

Proposition 5.7 (A mixed-radix gap). *Given a graph on $n \geq 2$ labeled vertices and $k \geq 2$, one can construct in polynomial time a nonnegative signed clipped-affine objective on $[0, 1]^k$ whose minimum is zero for a k -clique and at least two otherwise. It has an identity-Hessian realization satisfying the coefficient and two-leader restrictions of Theorem 5.6.*

Proof. Put $t_i = (n-1)x_i$, and let $d(t)$ be distance to the nearest integer label in $\{0, \dots, n-1\}$. On this interval,

$$2d(t) = \sum_{a=0}^{2n-3} (-1)^a \text{clip}(2t - a).$$

For integer $s = u + nv$, define $f(s) = 0$ if u, v are distinct adjacent vertices and $f(s) = 1$ otherwise. Its piecewise linear interpolation is

$$\varphi(s) = 1 + \sum_{a=0}^{n^2-2} (f(a+1) - f(a)) \text{clip}(s-a), \quad 0 \leq s \leq n^2-1.$$

It lies in $[0, 1]$ and is 1-Lipschitz. Take

$$H(t) = 2n \sum_i d(t_i) + 2 \sum_{i < j} \varphi(t_i + nt_j).$$

A clique labeling gives zero. Otherwise round every t_i to a label, let $D_t = \sum_i d(t_i)$, and choose a bad pair of rounded labels. For that pair $\varphi(t_i + nt_j) \geq \max(0, 1 - nD_t)$. Hence $H \geq 2nD_t + 2\max(0, 1 - nD_t) \geq 2$. The displayed formulas use $O(k^2n^2)$ clipped coordinates after duplicating the coefficient n in the first sum; upper coefficients have magnitude at most two. Every affine argument depends on at most two leaders and has polynomially bounded integer coefficients. To bound those coefficients by one, choose $R = 2n^2$ and use, for each such argument h ,

$$\text{clip}(h) = R\{\text{clip}(h/R) - \text{clip}((h-1)/R)\}.$$

On the leader box neither positive scaled argument exceeds one, so the identity follows from $\text{clip}(h) = (h)_+ - (h-1)_+$. Duplicate R copies to retain upper coefficients of magnitude at most two. This gives $O(k^2n^4)$ coordinates with follower affine coefficients bounded by one. Constant terms are realized by fixed-one responses if needed. $\square$

Theorem 5.8 (A supplied bounded leader core). *Suppose the leader domain is a product of rational intervals, the follower Hessian is positive diagonal, and all other data are as in (67), with no response-dependent upper rows. Suppose a set of at most c leader vertices is supplied whose deletion leaves components of size at most h in the leader interaction graph. For fixed c, h , exact optimization and a rational optimal leader-response pair can be computed in polynomial time.*

Proof. Normalize the leader intervals to unit intervals; fixed coordinates are removed. Normalize each positive diagonal quadratic to obtain its clipped affine response. Write the core as u , and the component leaders as v_b with $h_b \leq h$. The upper objective decomposes as $H_0(u) + \sum_b H_b(u, v_b)$, since a follower's noncore arguments all belong to one component. For fixed u , an optimum of H_b on its box occurs at a vertex of the arrangement of its clipping hyperplanes and box facets: it is affine on every closed arrangement cell. Enumerate every choice of h_b hyperplanes with independent coefficients in v_b . Solving them gives a candidate $v_b = Pu + q$, with rational affine data of polynomial bit length. There are at most $(2m_b + 2h_b)^{h_b}$ candidates, where m_b is the number of clipped terms of that component. Discard singular choices. Box facets ensure that all relevant arrangement vertices are included even when some clipping hyperplanes coincide or miss the box at particular parameters. A candidate is feasible precisely when it belongs to the component box; it need not remain a vertex away from the parameter where it was generated.

In core space arrange the hyperplanes where a candidate coordinate reaches zero or one, where a clipped term evaluated at a candidate reaches either threshold, and where a core-only term reaches a threshold. On each relative cell, candidate feasibility is constant and every feasible candidate's value is a fixed affine function of u . Add all pairwise equality hyperplanes between candidate values for the same component, and refine the cell. A minimizing candidate for each component is now fixed, and the summed objective is affine. Minimize it by LP on the closure of each resulting cell

and reconstruct the selected candidates and followers. All enumerations have polynomial size for fixed c, h ; constant core dimension also covers $c = 0$.

The closure step is sound: each selected candidate remains box-feasible and its continuous clipped value agrees with its affine formula on the closure. At a boundary, a newly feasible candidate omitted from this particular LP could be better, but that does not invalidate the returned feasible value. Completeness follows by considering the relative cell containing an optimal core point, where all its feasible candidates are considered. Thus the best constructed point is globally optimal. Rational LP recovery proves the output claim. $\square$

The vertex integrity of a graph is the minimum, over deleted vertex sets, of their size plus the largest remaining component size. Thus the supplied core in Theorem 5.8 certifies vertex integrity at most $c + h$. This is a bounded-core decomposition in the leader cost graph. Related “few global variables” decompositions for integer programming are studied by Dvořák et al. [14]; their integer constraint graphs and conclusions are different. The algorithm here is polynomial for fixed c, h , with exponents depending on these parameters. A core need not be found by an FPT algorithm: enumerating all sets of size at most c also gives a polynomial algorithm when c is fixed. If each follower depends on only one leader, the objective separates completely and each univariate clipped sum can be minimized at its breakpoints.

Theorem 5.9 (A leader path is already hard). *With a growing number of leaders, the identity-Hessian independent-box model is weakly NP-hard even when its leader interaction graph is a path and all follower affine coefficients have magnitude at most one and upper coefficients have magnitude at most two. Threshold feasibility is NP-complete. Exact continuous-state messages on a path can have exponentially many affine pieces, even for an identical local factor with a fixed finite coefficient alphabet.*

Proof. For positive Subset Sum weights $a_1, \dots, a_n$, let $W = \sum_i a_i$. Targets outside $[0, W]$, and its endpoints, can be handled directly; assume $0 < T < W$. Put $r_i = a_i/W$, $\tau = T/W$. For $-1 \leq t \leq 1$, define

$$\rho_r(t) = \min\{|t|, |t - r|\} = -t + 2 \operatorname{clip}(t) - 2 \operatorname{clip}(t - r/2) + 2 \operatorname{clip}(t - r), \quad 0 < r \leq 1.$$

On leaders $x_0, \dots, x_n \in [0, 1]$ take

$$H(x) = x_0 + |x_n - \tau| + \sum_{i=1}^n \rho_{r_i}(x_i - x_{i-1}).$$

Choose $\delta_i \in \{0, r_i\}$ nearest to the increment and telescope to get $|\sum_i \delta_i - \tau| \leq H(x)$. Conversely, cumulative sums of any chosen subset lie in the box and give that subset’s distance to τ . Therefore

$$\min H = \frac{1}{W} \min_{S \subseteq [n]} \left| \sum_{i \in S} a_i - T \right|. \quad (72)$$

The clipped expansion is

$$2x_0 - 2x_n + \tau + 2 \operatorname{clip}(x_n - \tau) + 2 \sum_i \{\operatorname{clip}(x_i - x_{i-1}) - \operatorname{clip}(x_i - x_{i-1} - r_i/2) + \operatorname{clip}(x_i - x_{i-1} - r_i)\}.$$

It uses $3n + 1$ independent identity-Hessian followers, each depending only on consecutive leaders or the terminal leader. Unary copies of x_0, x_n and a fixed-one response may replace the direct affine terms. The coefficient bounds follow, and Lemma 5.1 gives NP membership. The gap is 0 versus at

least $1/W$. Scaling the objective by W produces a constant gap but loses the bounded coefficient claim; no strong hardness is inferred.

There is also an exact message identity. Let S_n be the set of normalized subset sums, and minimize $x_0 + \sum_i \rho_{r_i}(x_i - x_{i-1})$ with $x_n = t$. The same telescope is a lower bound of $\text{dist}(t, S_n)$. For the reverse bound, follow a zero-cost subset path to its terminal sum and change only its last coordinate to t . Its last factor increases by at most that distance. The message is therefore exactly $\text{dist}(t, S_n)$. Weights $a_i = 2^{i-1}$ give $S_n = \{j/(2^n - 1) : 0 \leq j \leq 2^n - 1\}$, hence $2(2^n - 1)$ maximal affine pieces on $[0, 1]$. For a fixed alphabet use instead the identical factor $\rho_{1/2}(x_i - x_{i-1}/2)$. Its zero-cost terminal states are $S_n = \{j/2^n : 0 \leq j < 2^n\}$. If $e_i = x_i - x_{i-1}/2 - \delta_i$, $\delta_i \in \{0, 1/2\}$, then

$$t - \sum_i 2^{-(n-i)} \delta_i = 2^{-n} x_0 + \sum_i 2^{-(n-i)} e_i.$$

The absolute value is bounded by $x_0 + \sum_i |e_i|$; the same last-coordinate construction gives equality with $\text{dist}(t, S_n)$. There are $2^{n+1} - 1$ maximal affine pieces, including the last tail to one. All local coefficients now belong to one finite set (gains 1, $-1/2$, shifts 0, $-1/4$, $-1/2$, and readout coefficients ± 2). $\square$

Closure of continuous piecewise linear functions under variable elimination is classical; see Meuleau et al. [35, Lemmas 1–2 and Theorem 1] for a scheduling formulation. Closure alone gives no polynomial bound on intermediate message size. The identical-factor family above has the trivial optimum zero; its exponential message size is an output-complexity example, not another NP-hardness proof.

5.5 A path in the follower constraints

The preceding path connected *leaders* in affine follower costs. Now there is one leader, the Hessian is diagonal, and a path connects *follower* coordinates through inequalities. This distinction explains why the next example lies outside the independent-block response theorem. For rational $0 < \gamma < 1/2$, set $z_0 = 0$ and

$$P_n(\gamma) = \{z : 0 \leq z_1 \leq 1, \quad \gamma z_{j-1} \leq z_j \leq 1 - \gamma z_{j-1} \ (2 \leq j \leq n)\}. \quad (73)$$

Every row involves at most two consecutive coordinates (2VPI). Appendix D proves the needed geometric and arithmetic facts, and states with its proof the sparse-shadow construction of Gärtner et al. [21, Section 4].

Proposition 5.10 (What a sparse shadow obstructs). *For $\gamma = 1/4$, there is an explicit scalar-price LP over $P_n(\gamma)$ with 2^n distinct optimal vertices, each uniquely optimal on a price interval. Its value has 2^n affine pieces, although evaluation at a rational price requires only $O(n)$ rational arithmetic operations of polynomial bit length. The terminal-state value message has $2^n - 1$ affine pieces. An unextended quantifier-free polynomial description of the bounded epigraph in the two price/value variables requires the sum of the degrees of its nonzero polynomials to be at least 2^n .*

The explicit formulas and the boundary-factor proof are in Appendix D. This is a limitation on enumerating an entire response or on eliminating all auxiliary variables, not on solving one LP, nor on arithmetic circuits or descriptions retaining quantified states.

Theorem 5.11 (Scalar-leader hardness with an identity-Hessian path follower). *There is an NP-complete restricted family of rational bilevel feasibility problems with one leader in $[-1, 1]$, a unique follower minimizing a strictly convex diagonal quadratic over a 2VPI path polytope, one upper slab*

with two linear bounds, and one nonconvex quadratic upper row. The follower Hessian can be exactly the identity. Without upper rows, such path followers can have at least 2^n distinct constant response intervals. The feasibility hardness also holds when all local follower inequality coefficients and right-hand sides belong to fixed finite alphabets.

Proof. Take Subset Sum data $a_i > 0$, $W = \sum_i a_i$, $0 \leq T \leq W$, and put $\gamma = 1/(8W)$. Define

$$c_i = (1 - \gamma)\gamma^{2(n-i)-1} \ (i < n), \quad c_n = 0, \quad F(z) = z_n - c^\top z - z_n^2.$$

Lemma D.2 shows that $F \geq 0$ on $P_n(\gamma)$, with zeros precisely at its vertices. Each vertex has a bit description $v_j = d_j + (1 - 2d_j)\gamma v_{j-1}$, and $|v_j - d_j| \leq \gamma$. The slab

$$T - 1/4 \leq \sum_i a_i z_i \leq T + 1/4 \tag{74}$$

therefore contains a vertex if and only if the Subset Sum instance is positive: the forward error is at most $\gamma W = 1/8$, and conversely a vertex in the slab gives $|\sum_i a_i d_i - T| \leq 3/8 < 1$, forcing equality of integers.

It remains to make every required vertex an exact strongly convex response. Terminal vertex values are distinct and have denominators dividing $D = (8W)^{n-1}$. Thus their separation is at least $1/D$. Put $\Delta = D^{-2}$, $\tau = \Delta/(2n)$, and let the follower minimize

$$\frac{\tau}{2} \|z\|_2^2 - c^\top z - \lambda z_n \quad \text{over } P_n(\gamma), \quad -1 \leq \lambda \leq 1. \tag{75}$$

At vertex v choose $\lambda_v = 2v_n - 1$. Since $c^\top u = u_n - u_n^2$ at every vertex u , the loss in the linear score from v to any other vertex is $(u_n - v_n)^2 \geq \Delta$. For an arbitrary convex combination $z = \sum_u \alpha_u u$, put $a = \sum_{u \neq v} \alpha_u$. Then the score loss is at least Δa , $\|z - v\|_\infty \leq a$, and $\|z\|_2^2 - \|v\|_2^2 \geq -2na$. Consequently its follower objective exceeds that at v by at least $(\Delta - \tau n)a = \Delta a/2$, strictly if $z \neq v$. The same reasoning at price $\lambda_v + h$ loses at most $|h|a$, so v remains the unique response whenever $|h| \leq \Delta/4$ within the leader interval. There is a nontrivial one-sided interval even at an endpoint. Fixing $\gamma = 1/4$ proves the exponential constant-response claim with no upper constraints.

Impose $F(z) \leq 0$ and (74) as upper rows. The former forces an original path vertex; the preceding equivalence and exposing leaders give the reduction in both directions. Guessing its bits, computing the vertex and λ_v , and checking the slab is a polynomial-bit NP certificate for this restricted family. All coefficients have polynomial bit length, including $1/\tau$. Multiplication of the entire follower objective by $1/\tau$ preserves the response and makes its Hessian the identity.

For the fixed local alphabet, apply the padding construction of Lemma D.3: use $\gamma = 1/4$, dimension $N = n + (n - 1)r$ with $4^{-(r+1)} \leq 1/(8W)$, and bound each padding coordinate by $1/2$. Use the full N -dimensional cube's F, c , and $D = 4^{N-1}, \Delta = D^{-2}, \tau = \Delta/(2N)$. Every required free-bit pattern is an original cube vertex satisfying the padding bounds. Its strict response inequality was proved over the entire cube, so it remains valid on the padded subset. Conversely $F \leq 0$ forces an original cube vertex and its padding bits must be zero. Apply the slab only to the free coordinates; the same $1/8$ error proves the equivalence. Local inequality coefficients are now in $\{0, \pm 1, \pm 1/4\}$, and right-hand sides in $\{0, 1/2, 1\}$. $\square$

The upper quadratic row has Hessian $-2e_n e_n^\top$; it is not a convex upper constraint. The same reduction can minimize $F(z(\lambda))$ under the linear slab at threshold zero. The slab-only feasible set is nonempty: the continuous response visits zero and the all-one-bit vertex, whose weighted free-coordinate sum is at least $W - 1/8$; the intermediate value theorem covers integer targets below W , and that last vertex covers $T = W$. Its compact feasible graph gives attainment. Deleting

the quadratic row from the feasibility version therefore makes every target feasible. This proof establishes neither all-linear-upper feasibility hardness nor strong hardness. The fixed local alphabet does not bound the slab weights or the rescaled linear costs. Rank-one concave quadratic hardness more generally has the classical precedent Pardalos and Vavasis [39].

There is a narrow positive counterpart. If both sides of each affine interval transition use the *same* gain, exact feasibility projection along paths is polynomial, including negative or zero gains. Proposition D.4 in Appendix D.3 gives the full construction, its extension to trees and fixed *total* cycle rank, and same-field witness recovery. The opposite gains $+\gamma$ and $-\gamma$ in (73) do not satisfy that hypothesis. Nor does that projection permit arbitrary objectives involving all eliminated states.

5.6 Curvature, degree encoding, and arithmetic output

Proposition 5.12 (Removing positive local curvature). *Optimistic feasibility is NP-complete in restricted degree-two families with no leader and an independent unit-box follower if either zero local curvature and quadratic upper rows, or negative local curvature and linear upper rows, are permitted.*

Proof. With zero follower objective, every $z \in [0, 1]^N$ is optimal. Upper rows $z_i(1 - z_i) = 0$ and the linear literal-sum inequalities $\ell_a(z) \geq 1$ encode 3SAT. Alternatively minimize $\sum_i z_i(1 - z_i)$ in the follower. Its minimum is zero and its minimizers are exactly the Boolean vertices; then only the linear upper clause rows are needed. Boolean assignments are certificates for both restricted families. These are existential, optimistic statements; universal feasibility for every optimal response is different. $\square$

Proposition 5.13 (Sparse binary degrees and output length). *Polynomial dependence on numerical degrees cannot in general be replaced by polynomial dependence on their binary encoding, even with one power follower. For an integer $P \geq 1$, $z(x) = x^{1/P}$, $0 \leq x \leq 1$, each of the following requires $\Omega(P)$ bits in any rational leader meeting the indicated guarantee:*

1. upper objective $(z - 1/2)^2$, no upper rows, additive error $1/64$;
2. upper objective z , upper row $z \geq 1/2$, exact feasibility and additive error $1/8$.

The second example has a strict feasible anchor $x = 1$ and a convex reduced upper inequality. Sparse upper data alone can also force exponential algebraic output degree, with a trivial positive quadratic follower.

Proof. The power response minimizes $z^{P+1}/(P+1) - xz$ on the unit interval, with a sparse binary description of $O(\log(P+1))$ bits. In the first example the optimum is zero at $x = 2^{-P}$. The error guarantee forces $3/8 \leq z \leq 5/8$; hence $0 < x \leq (5/8)^P$. In the second example the optimum is $1/2$, and the guarantee forces $1/2 \leq z \leq 5/8$, giving the same upper bound on positive x . If $x = p/q > 0$ is a reduced rational, then $q \geq 1/x \geq (8/5)^P$. Its binary encoding has $\Omega(P)$ bits. The reduced row $1/2 - x^{1/P} \leq 0$ is convex, and its value at one is $-1/2$. Thus an anchor does not remove the numerical-degree dependence of Theorem 4.6. For the last assertion let the upper feasible set be $1 \leq x \leq 2$, $x^{2^t} = 2$, and minimize x . Use the independent follower $\min_{0 \leq z \leq 1} z^2$, whose response is zero. The sparse upper description has $O(t)$ bits, whereas its unique leader has degree 2^t , since $X^{2^t} - 2$ is Eisenstein at two. $\square$

Example 4.3 already gives the rational-output obstruction with an affine upper objective and no upper rows by using two different powers. Proposition 5.13 strengthens the one-follower case when polynomial upper data or a response row is allowed; it does not settle the one-power affine-objective, no-upper-row subclass.

Proposition 5.14 (An exponential common-field degree). *Independent uniformly strongly convex scalar polynomial followers can have an aggregate value of algebraic degree 2^N . Even singleton convex quadratic local sets can encode the exact Square Root Sum comparison problem through one linear aggregate row.*

Proof. For distinct primes p_i , minimize $z_i^3/3 - p_i z_i$ on $[1, p_i + 1]$. The unique response is $\sqrt{p_i}$, and the second derivative is at least two throughout the interval. The value $s = \sum_i \sqrt{p_i}$ has degree 2^N . Here is a direct field argument. Inductively, the products $\sqrt{\prod_{i \in S} p_i}$ form a basis for the field generated by the first j radicals, with independent sign-changing automorphisms. If the next radical were in that field, its image under any such automorphism would be itself or its negative, since its square is rational. It would be a common eigenvector for all sign changes. In the displayed basis the characters are distinct, so such an eigenvector is a rational multiple of one basis element. Squaring contradicts the parity of the prime valuations of p_{j+1} . The next extension therefore has degree two and all old sign changes extend with either new sign, completing the induction. The 2^N signed sums are distinct by this basis independence; they are all conjugates of s .

For arbitrary nonnegative integers a_i , the local set $Y_i = \{y_i : 0 \leq y_i \leq \max(1, a_i), y_i^2 = a_i\}$ is a compact convex singleton. Feasibility of $\sum_i y_i \leq K$ is exactly $\sum_i \sqrt{a_i} \leq K$. There is no core variable and only one aggregate. $\square$

The lower bound concerns minimal-polynomial and common-number-field output; a list of radicals can be a much shorter expression. Exact Square Root Sum comparison is not proved NP-hard by this encoding, and no such claim is used here; polynomial-time exact comparison remains an open problem [15]. It is an arithmetic obstruction to extending Appendix A’s polyhedral-leaf elimination by replacing each leaf with an arbitrary convex semialgebraic set. Accuracy-bit approximation, which does not require the entire common field, remains compatible with it.

6 Executable scalar algorithms and reproducible computations

The preceding fixed-dimensional algorithms establish bit complexity without promising an efficient general implementation of real quantifier elimination. This section gives executable specializations. We distinguish three tasks: certifying a proposed convex response path, constructing a complete nonconvex response graph, and optimizing an upper problem over that graph. In particular, an exact follower oracle at finitely many prices does not solve a continuous leader problem.

6.1 Convex quadratic paths and an aligned sweep

Consider the unique response on the unit box to

$$\min_{0 \leq z \leq 1} \frac{1}{2} z^\top Q z + (c + Cx)^\top z, \quad Q = D + U H U^\top \succ 0, \quad x \in [L, R], \quad (76)$$

where all data are rational, D is positive diagonal, H is symmetric, and the displayed decomposition is supplied. Here x is scalar. The implemented upper objective is the minimization of

$$o_x x + o_z^\top z + o_{xx} x^2 + x o_{xz}^\top z \quad \text{subject to} \quad a_j x + b_j^\top z \leq h_j. \quad (77)$$

Thus tariff maximization is represented by a negative x -response coefficient. The nonconvex interface below instead directly maximizes revenue.

For a status assignment with free coordinates F and upper coordinates B , set $z_B = 1$, set the remaining bound coordinates to zero, and solve

$$Q_{FF} z_F = -c_F - C_F x - Q_{FB} \mathbf{1}.$$

This yields a rational affine function. Free-coordinate bounds and bound-gradient signs give its exact closed validity interval. Degenerate zero gradients are allowed. The affine stationarity identities and the signs at the interval's two endpoints certify KKT throughout the interval. Positive definiteness then certifies the unique global response. This is the familiar active-set mechanism behind explicit parametric quadratic programming [7, Theorems 2 and 4]; here the certificate also includes coverage of the entire requested price interval.

No inverse of H is required. For a right-hand side v , put $S = U_F^\top D_F^{-1} U_F$ and solve

$$(I + SH)w = U_F^\top D_F^{-1} v, \quad z_F = D_F^{-1}(v - U_F H w).$$

The determinant identity $\det Q_{FF} = \det D_F \det(I + SH)$ proves invertibility, even when H is singular or indefinite. For rank k , a pattern solve uses $O(N(1 + k^2) + k^3)$ rational arithmetic operations; this count excludes validation of the supplied positive-definiteness assumption. The implementation checks that assumption, using direct exact matrix tests when the small-rank sufficient tests do not apply.

Proposition 6.1 (Certified and exhaustive convex paths). *A finite collection of rational affine pieces for (76) is a complete exact response path if every piece passes the preceding KKT test and its closed intervals cover $[L, R]$. Enumerating all 3^N status assignments always produces such a path. On any certified complete path, (77) is solved exactly by rational arithmetic.*

Proof. KKT is necessary and sufficient on a box for a convex differentiable objective. Every optimizer supplies a status assignment, including at a degenerate boundary, so exhaustive enumeration covers every price. Conversely each certified piece contains only optimizers. Substitute its affine response into each upper row and intersect intervals, retaining singleton intersections. The upper objective becomes a rational quadratic. Compare its endpoints and, when its quadratic coefficient is positive, its interior stationary point. These rational candidates cover its minimum. Taking the least value over the finite path solves the upper problem, or certifies infeasibility. $\square$

The generic implementation uses a numerical box-QP routine only to propose a status assignment at a price in an uncovered gap. It tries several numerical status tolerances, recomputes the proposed segment exactly, and verifies that the queried price belongs to it. For small N it can fall back to exhaustive search. It returns a path only after exact KKT and coverage verification; otherwise it raises a recovery or coverage error. A cap on the number of segments and unsuccessful numerical proposals are explicit failure modes. This procedure is not an unconditional polynomial-time path-construction algorithm. No numerical tolerance is used to declare global coverage.

There is a complete simpler sweep for the aligned rank-one case $Q = D + h u u^\top \succ 0$ and $C = \gamma u$, with $\gamma > 0$. Define

$$z_i(t) = \text{clip}_{[0,1]}(-(c_i + u_i t)/d_i), \quad A(t) = u^\top z(t).$$

Zero entries of u give constant coordinates. Signed nonzero entries give at most two rational breakpoints each. On a multiplier interval, $A(t) = A_0 + A_1 t$, with $A_1 = -\sum_{i \in F} u_i^2/d_i$. The relation

$$x = (t - hA(t))/\gamma, \quad t = (\gamma x + hA_0)/(1 - hA_1) \tag{78}$$

is strictly increasing: every free principal matrix is positive definite, so $1 + h \sum_{i \in F} u_i^2/d_i > 0$. This includes negative h whenever the whole Hessian is positive definite. The map tends to both infinities on the tails and therefore covers every price. Updating the affine coefficients at each grouped breakpoint gives the complete response path. Maintaining the weighted response sums in the objective and m upper rows takes $O(N \log N + N(m + 1))$ rational arithmetic operations; only the

final selected response needs reconstruction in all coordinates. This is an arithmetic count, not a constant-cost claim for arbitrarily long rational numbers. Sorting and clipped multiplier allocation are classical resource-allocation ingredients [29]; the implementation combines them with exact upper-interval optimization. Its sweep and upper solve are fused, which matters when interpreting their timings.

6.2 Complete nonconvex scalar compression

The second specialization allows an indefinite follower Hessian:

$$f_x(z) = \sum_{i=1}^N \left(\frac{1}{2} d_i z_i^2 + c_i z_i \right) - \frac{h}{2} (u^\top z)^2 + \gamma x u^\top z, \quad \ell \leq z \leq r, \quad x \in [L, R]. \quad (79)$$

Here $N \geq 1$, all data are rational, $d_i > 0$, $h \geq 0$, $\gamma \neq 0$, $\ell_i \leq r_i$, and $L \leq R$. Signed and zero u_i , fixed coordinates, negative prices, and either sign of γ are allowed. The upper problem maximizes xw , where $w = u^\top z$, subject to rational affine upper rows. Optimistic and pessimistic meanings follow Section 1, with maxima and suprema in place of minima and infima.

Let $[w_-, w_+]$ be the image of the follower box under $u^\top z$, and define

$$C(w) = \min \left\{ \sum_i \left(\frac{1}{2} d_i z_i^2 + c_i z_i \right) : \ell \leq z \leq r, \quad u^\top z = w \right\}, \quad \psi(w) = C(w) - \frac{h}{2} w^2. \quad (80)$$

The fiber problem is strictly convex even when (79) is nonconvex. Its minimizer $z(w)$ is unique. Every global follower optimum must be this minimizer on its fiber; otherwise its objective could be decreased without changing the aggregate terms. Thus the original global problem is exactly $\min_{w \in [w_-, w_+]} \{\psi(w) + \gamma x w\}$, with actual responses recovered by $z(w)$.

Lemma 6.2 (Rational fiber pieces). *If $w_- < w_+$, the interval has at most $2N - 1$ nondegenerate pieces on which $z(w)$ is rational affine and $\psi(w)$ is rational quadratic. They and their endpoints can be constructed in polynomial bit complexity. If $w_- = w_+$, there is one constant aggregate and one unique fiber response.*

Proof. For a multiplier λ , strict convexity gives

$$z_i(\lambda) = \text{clip}_{[\ell_i, r_i]}((\lambda u_i - c_i)/d_i).$$

For $u_i = 0$ this is interpreted as the constant clipped unconstrained minimizer. Ignore fixed coordinates when forming events. Sort the at most $2N$ numbers $(d_i \ell_i + c_i)/u_i$ and $(d_i r_i + c_i)/u_i$ for $u_i \neq 0$. On each intervening interval, $z = a + b\lambda$ and $w = w_0 + S\lambda$, with $S = \sum_{i \in F} u_i^2/d_i \geq 0$. If $S > 0$, invert to obtain $z(w)$. If $S = 0$, the interval maps to a single aggregate already supplied by its adjacent pieces. The two infinite tails likewise supply only endpoints. Continuity of clipping shows that the union covers $[w_-, w_+]$; the fiber KKT equations and strict convexity prove the recovered response is the unique fiber optimum, also at shared endpoints. Substituting it into (80) gives each quadratic. If all aggregates are equal, the box constraints with nonzero u_i fix those coordinates and the others minimize independently. Rational sums, products, sorting and inversions give polynomial bit bounds. Explicitly storing every length- N affine response uses at most quadratic, rather than linear, storage. $\square$

Write a piece as $\psi(w) = Aw^2 + Bw + D$ on $[a, b]$. Its complete minimizer candidates under the price tilt are:

1. its endpoints, available at every price;
2. if $A > 0$, the stationary point $w = -(B + \gamma x)/(2A)$ on the rational price interval where it lies in $[a, b]$;
3. if $A = 0$, the whole interval $[a, b]$ at $x = -B/\gamma$.

A piece with $A < 0$ contributes only endpoints. The stationary candidates have quadratic value functions in x ; endpoint candidates have affine values. These are feasible candidates, so comparison of their *original* values, not their stationarity alone, decides globality.

Theorem 6.3 (Complete scalar atlas and upper optimization). *For (79), a finite exact response description and both upper tariff problems can be computed in polynomial bit complexity. The description consists of rational affine responses on open price intervals with algebraic endpoints of degree at most two, together with all isolated responses and all flat fiber intervals at individual prices. The algorithm decides upper feasibility, computes the supremum and its attainment, and returns an attaining response when one exists. Each reported optimizer or limiting pair, together with its value, belongs to one field of degree at most two; different atlas cuts may lie in different quadratic fields. Optimistic feasibility implies attainment; pessimistic feasibility need not.*

Proof. Form all candidate domain endpoints, all flat-piece prices, and every real crossing of two candidate value polynomials in their common domain. Include L, R and discard prices outside them. There are $O(N^2)$ cuts, each rational or quadratic algebraic. On each open interval, a rational sample determines the valid candidates and every comparison sign. Keep all global winners. At every cut compare all valid candidates anew, adding a flat interval exactly when its value equals the global minimum. This proves completeness, including isolated ties and lower-dimensional price strata.

On an open interval, two winning value polynomials must agree identically. Their derivatives equal $\gamma w(x)$, both for endpoint and stationary candidates. Since $\gamma \neq 0$, they describe the same aggregate, and uniqueness of the fiber minimizer gives the same original response. Different optimal responses can therefore coexist only at the recorded individual prices.

For an open interval, substitute the affine response into the upper rows. Their intersection is an interval, possibly empty or a singleton inside the original open interval. Maximize its quadratic revenue by comparing endpoint limits and any interior stationary point with negative quadratic coefficient. Also retain an interior rational sample to recognize attainment of a constant value. Flag an endpoint as attained only if it belongs to the original open price interval; upper inequalities themselves are closed.

At a recorded price, an optimal response component is $z(w)$ on a singleton or a closed fiber interval. Optimistically, intersect each component with all upper rows and maximize the linear function xw on what remains. Pessimistically, require every upper row at both endpoints of every component, then take the least xw over all those endpoints. These endpoint tests are exact because $z(w)$ is affine on a fiber piece. Comparing the finite list of values and one-sided limits gives the supremum. Prefer an attained candidate when its value ties a limit; if none attains the best value, return the limiting price and aggregate with an explicit nonattainment flag. The optimistic graph is closed in a compact box, which separately proves its attainment assertion. The pessimistic assertion cannot be inferred from that closedness.

The all-pair construction has polynomially many branches, cuts and comparisons. All rational coefficient lengths are polynomial in the input. Quadratic root isolation and comparison, including comparisons between different quadratic fields, have polynomial bit complexity. A rational sample between distinct quadratic cuts can be found by dyadic refinement with polynomial bit length. Upper stationary points and nontrivial row boundaries on open pieces are rational; flat-piece prices

are rational; all remaining endpoint outputs lie in the field of their one quadratic price. Thus no compositum of all switch fields is needed for any reported optimizer or value. $\square$

The executable compressed implementation inserts candidates into the current lower envelope. On each existing envelope interval it splits at the new candidate's validity boundaries and its crossings with the current winner, then compares at rational interior samples. It keeps the resulting envelope and also a separate set of contact prices, including tangencies. Finally it compares *all* original candidates at retained contacts and flat prices. To justify this shorter list, consider a final global tie when the later of two tied branches was inserted. No then-existing branch could have had smaller value there. The tie was therefore at a current boundary, a new domain boundary, or a crossing with a current winner, unless the value polynomials coincided. In that last case their derivatives give the same response and there is no missing distinct contact. An isolated-domain candidate also contributes its domain endpoint. Consequently the incremental construction preserves every final global contact. Conservative bounds are $O(N^2)$ retained cuts and $O(N^3)$ interval processing steps. The implementation retains full response vectors and uses symbolic arithmetic; these counts do not predict its practical running time.

6.3 Convex-envelope values and original contacts

Extend ψ by $+\infty$ outside $[w_-, w_+]$. Its conjugate gives

$$v(x) = -\psi^*(-\gamma x) = \min_w \{\psi^{**}(w) + \gamma x w\}. \quad (81)$$

For completeness, put $m = \min_w (\psi(w) + \gamma x w)$. The affine function $m - \gamma x w$ minorizes ψ and hence lies below its closed convex envelope ψ^{**} . The rightmost minimum is at least m and, because $\psi^{**} \leq \psi$, at most m . Moreover the original minimizers are exactly

$$\{w : \psi^{**}(w) + \gamma x w = v(x), \quad \psi(w) = \psi^{**}(w)\}. \quad (82)$$

The second equality is essential for upper feasibility and tie-breaking. Convex envelopes of piecewise quadratic functions and their conjugates have established algorithms [19, 37]. Our direct candidate proof supplies the original contact sets and rational-bit guarantees needed here; it does not require a tangent formula from an external implementation.

Example 6.4 (False choices from convexification). Take $N = 1$, $d = u = 1$, $c = -1$, $h = 3$, $\gamma = 1$, $z \in [0, 1]$, and $x \in [1, 3]$. Then $\psi(z) = -z^2 - z$ has convex envelope $-2z$. The actual response is 1 for $x < 2$, 0 for $x > 2$, and $\{0, 1\}$ at $x = 2$. The convexified response at 2 is the entire interval $[0, 1]$. Consequently upper rows requiring $z = 1/2$ are infeasible for the original bilevel problem, although convexification admits revenue 1. Mixing the two original optima cannot repair this discrepancy. Without those rows, the optimistic maximum is 2, attained at $(2, 1)$, whereas the pessimistic supremum is 2 and is not attained.

Example 6.5 (Capacity at an interior jump). Let $d = (1, 1)$, $c = (-1, -1/2)$, $u = (1, 1)$, $h = 3/5$, $\gamma = 1$, $z \in [0, 1]^2$, $x \in [0, 1]$, and impose $w \leq 1$ above. The fiber formula is

$$\psi(w) = \begin{cases} w^2/5 - w, & 0 \leq w \leq 1/2, \\ -w^2/20 - 3w/4 - 1/16, & 1/2 \leq w \leq 3/2, \\ w^2/5 - 3w/2 + 1/2, & 3/2 \leq w \leq 2. \end{cases}$$

At $x_* = 17/20$, its only tilted minima are $w = 3/8$ and $13/8$, both of value $-9/320$, giving responses $(3/8, 0)$ and $(1, 5/8)$. The middle piece is strictly concave; its endpoints have value $-1/40$, so supply

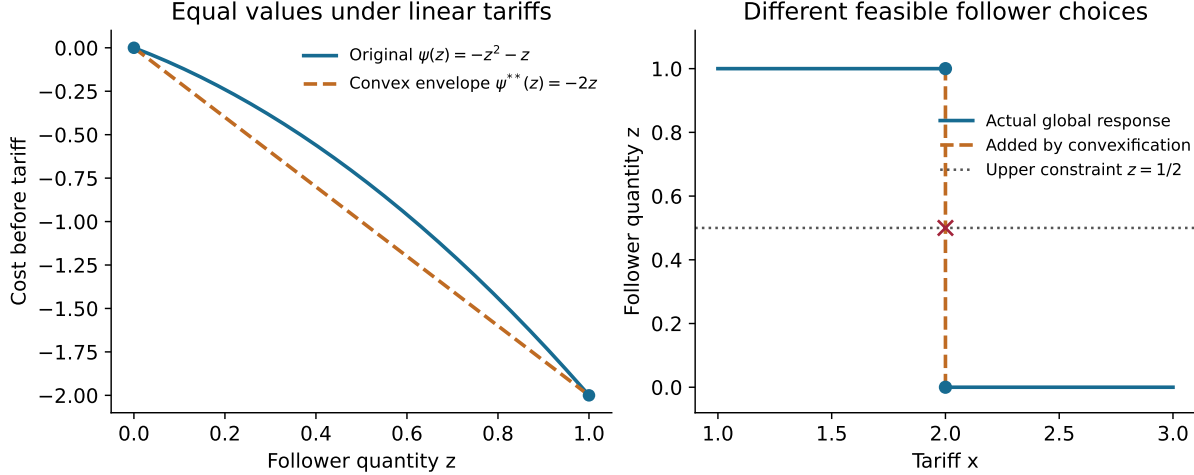


Figure 1: Example 6.4. Left: the original aggregate cost and its convex envelope. Right: original responses retain two contacts at the switch; the additional vertical segment admitted by convexification consists of false follower choices.

no additional minimizer. Comparing with the larger contact shows that for $x < x_*$ every optimal w is at least $13/8 > 1$: a putative smaller w has nonnegative excess value at x_* and gains strictly less from the negative price change. For $x > x_*$, comparison with the smaller contact forces $w \leq 3/8$; minimization on the low piece gives $w = (5/2)(1 - x)$. Its revenue decreases on $[x_*, 1]$. Only the smaller contact satisfies capacity at x_* . Thus the optimistic maximum is $51/160$, attained there, while the pessimistic supremum is the same number and is unattained. This is a failure of universal upper feasibility at a tie, not a failure to compute the nominal global follower value.

Local stationarity alone is also insufficient: on $[0, 1]$, $f(z) = z/4 - z^2/2$ has a strict one-sided local minimum at 0, of value 0, but its global minimum is at 1, of value $-1/4$. Both satisfy box KKT. The examples and the atlas construction address global selection explicitly.

6.4 An independent full-task stationary-face baseline

To compare complete bilevel tasks, we implement a second solver directly in original coordinates. It shares rational instance data with the compressed method, but no fiber allocation, envelope construction or upper-optimization code. Its declared restriction is that *every principal minor* of $Q = \text{diag}(d) - huu^\top$ is nonzero. It checks all $2^N - 1$ nonempty minors and rejects an unsupported instance before solving. This restriction excludes flat follower directions; the compressed method has no such restriction.

Proposition 6.6 (Completeness of the original-coordinate baseline). *Under the stated nonzero-principal-minor restriction, enumerating original stationary box faces, comparing their original objective values over the whole price interval, and retaining all ties yields the complete response graph of (79). Both upper tariff problems, including attainment, are then solved exactly. Enumeration is exponential in N .*

Proof. Take any global minimizer and its minimal box face. On its free coordinates F , first-order stationarity holds and the second-order necessary condition makes Q_{FF} positive semidefinite. Its nonzero determinant therefore makes it positive definite. Hence this optimizer appears in an

enumerated face with

$$z_F(x) = -Q_{FF}^{-1}(c_F + \gamma u_F x + Q_{FB} z_B).$$

Here B comprises both lower and upper fixed coordinates. The baseline solves this dense original-coordinate system, rejects non-positive-definite free faces, and intersects the free bounds and bound-gradient signs to get an exact rational validity interval. A coordinate whose lower and upper bounds coincide needs no gradient sign. Each retained candidate is feasible; every global optimum is retained. Comparing their *original* quadratic values therefore gives exactly all global optima, regardless of nonglobal KKT points.

Substitution into $\frac{1}{2}z^T Q z + (c + \gamma x u)^T z$ gives a rational quadratic in x . Form all validity endpoints and all pairwise value crossings on overlaps, including crossings above the global envelope. Exact comparison on open cells and separately at their endpoints retains all ties. The nonzero-minor condition precludes a missing continuum: any minimizer in a positive-dimensional minimal face has the unique stationary point just solved. Equal winning value polynomials on an open cell have equal derivatives $\gamma u^T z$ and hence equal aggregate responses. Strict fiber convexity then makes their entire responses equal.

The baseline independently clips each open response cell with the upper rows and compares quadratic revenue endpoints, stationary points, and interior samples, tracking whether each endpoint belongs to the cell. At a cut, optimism compares all upper-feasible global responses; pessimism requires all global responses to satisfy the rows and takes their least revenue. This finite decomposition proves the upper result, including nonattainment. There are at most 3^N faces and quadratically many pair comparisons in that number; each solve and comparison uses exact rational or quadratic algebraic arithmetic. This is an exponential reference algorithm, not a scalability claim. $\square$

An older independent oracle enumerates original stationary faces only at a requested price. It can skip singular free faces while preserving the global *value*: a stationary positive-semidefinite singular face admits a null direction along which the quadratic stays constant; moving to the boundary reduces the face until it is nonsingular or a vertex. That argument does not recover a whole flat minimizer set and does not optimize over all prices. The new baseline’s stronger checked restriction makes it a complete-response, complete-upper-task comparison. Separate tests continue to exercise the main method on singular and flat cases.

6.5 Computational protocol and findings

All fresh runs use the supplied rational data and deterministic generators in the accompanying code and data archive. The platform was Linux on an Intel Xeon w5-2565X, with Python 3.13.11, SymPy 1.14.0, NumPy 2.5.1 and SciPy 1.18.0. Each reported fresh timing is the median of three serial, fresh-process runs; all individual times are retained. The thread environment for OpenBLAS, OpenMP, MKL and NumExpr was set to one, and the SciPy MILP calls additionally forwarded `threads=1` to HiGHS. The wrapper’s warnings and solver output are retained. Processes start with the SymPy cache cleared, and the order of paired methods alternates between repetitions. This controls shared-cache ordering; the machine was shared and no hardware isolation or statistical speed guarantee is claimed. Every worker had an external 90-second limit. All 60 recorded workers completed; no timed-out algorithm run is omitted.

The new correctness tests compare both global upper outcomes on eight adversarial and ten seeded small random instances. They compare the entire response sets at every independently constructed face-atlas cut and an interior sample of every face-atlas cell. For each attained output from *both* methods, they directly check bounds, $w = u^T z$, revenue, every upper row, and original follower cost against all valid original-coordinate faces at that returned price. Pessimistic witnesses

Table 1: Fresh full-task comparisons, seconds (three-run medians). Build includes the face baseline’s input parsing and principal-minor validation. Opt. and pess. solve the two upper semantics after construction. Total includes solver input validation, construction and both upper solves; wall includes process startup, imports and input-file reading as well. The columns are independently computed medians.

N	Method	Build	Opt.	Pess.	Total	Wall
2 (jump)	Compressed	0.1322	0.0102	0.0057	0.1473	0.4348
2 (jump)	Faces	0.0034	0.0007	0.0004	0.0046	0.3018
2	Compressed	0.0597	0.0014	0.0010	0.0623	0.3541
2	Faces	0.1112	0.0005	0.0004	0.1121	0.4082
4	Compressed	0.6924	0.0973	0.0665	0.8490	1.1772
4	Faces	0.3334	0.2220	0.1637	0.7074	1.0160
6	Compressed	0.7014	0.2130	0.1479	1.0605	1.3984
6	Faces	0.6413	0.4266	0.3480	1.4041	1.7100

also undergo universal feasibility and least-revenue checks. These prices include upper row boundaries and stationary revenue points, not just follower switches. A singular principal minor is rejected by the baseline, while separate compressed tests retain flat intervals, fixed boxes and singleton aggregates.

Additional, distinct diagnostics exercise exact convex path certification, forced numerical-proposal failures, corrupted certificates and missing coverage; nonconvex original-face value comparisons on signed and fixed-coordinate instances; and incremental-envelope contacts against a full pairwise partition. The convex checks include negative rank-one coefficients with positive-definite total Hessian, nearly singular rational data, and data outside floating-point range. These finite checks support the implementation contracts and expose specified failure modes. Their counts are recorded by test family and are not combined into a claim of formal verification of the general theorems.

The same continuous leader task. The heterogeneous family draws d_i uniformly from $\{8, \dots, 16\}/8$, u_i from $\{4, \dots, 8\}/4$, and $-c_i$ from $\{4, \dots, 24\}/8$, using seed $17 + N$ in that order. It has the unit box, $x \in [0, 5]$, $h = (7/5)/(\sum_i u_i^2/d_i)$, and upper capacity $u^\top z \leq \frac{1}{2} \sum_i u_i$. Its Hessian is indefinite: diagonal congruence gives $I - hvv^\top$ with $v^\top v = \sum_i u_i^2/d_i$ and one negative eigenvalue. Both solvers construct the whole reaction graph and solve *both* tariff semantics with these same upper rows. Table 1 also includes Example 6.5. Every exact value and attainment flag agrees between the methods and across repetitions. The face algorithm is faster on the jump and $N = 4$ instances; the compressed method is faster on the tested heterogeneous $N = 2, 6$ instances. Thus even these small complete-task comparisons do not show uniform dominance. The exponential baseline was deliberately limited to $N \leq 6$, where it checks at most 63 principal minors and 729 statuses. Larger instances were not attempted with it; they are not reported as timeouts.

Symbolic cost and population structure. A profile of the unchanged scalar implementation on the $N = 12$ heterogeneous family identified repeated symbolic cancellation in sign tests as the principal cost: cancellation accumulated 5.74 of 9.88 profiled seconds across the build and one unconstrained upper solve. Profiled times include instrumentation and are not benchmark times. The paper copy first decides the sign of an already rational expression and, for two rational interval endpoints, uses their rational midpoint. The full exact algebraic fallback and the candidate, envelope

Table 2: Fresh compressed scalar runs, seconds (three-run medians). The last two upper columns concern the same capacity row. Total includes input validation, build and both upper solves; process wall times and all repetitions are in the data archive.

N	Fiber pieces	Atlas points	Build	Opt.	Pess.	Total
12	21	45	2.7076	0.4769	0.4346	3.6053
24	38	79	3.9774	0.8773	0.7808	5.6096
48	66	135	13.6697	2.7917	2.8244	19.2916
20 (two types)	3	7	0.1365	0.0413	0.0331	0.2039
200 (two types)	3	7	0.1415	0.2125	0.2828	0.6275
1000 (two types)	3	7	0.2388	1.2650	1.6078	3.1009

and contact logic are unchanged. The fallback still uses a rational sample between unrelated quadratic cuts, avoiding unnecessary output-field extensions.

The controlled ordinary measurements compare the original and paper copies on the same constrained $N = 12$ task. Original build times were 3.150, 2.929, 3.107 seconds; paper-copy times were 2.748, 2.679, 2.708 seconds. Median totals including both upper solves were 4.055 and 3.605 seconds, respectively. This modest measured improvement supports the two local changes; it does not remove the symbolic cost of heterogeneous events. Table 2 reports the remaining fresh scalar runs.

The two-type family has $N = 2m$: m copies of $(d, c, u) = (1, -1, 1)$ and m copies of $(1, -1/2, 1)$, with $h = 3/(5m)$, $x \in [0, 1]$ and capacity $w \leq m$. Its Hessian is indefinite since $h \sum_i u_i^2/d_i = 6/5$. Uniqueness of the local fiber allocation makes coordinates within each type equal. Dividing costs and aggregate by m reduces the response problem exactly to Example 6.5. Therefore the optimistic optimum is $51m/160$ at price $17/20$ and aggregate $3m/8$, while the pessimistic supremum is the same and is unattained. There are three fiber pieces and seven atlas points for every m . The $N = 1000$ run tests repeated types, whereas $N = 48$ has 66 fiber pieces; their timings measure different structural regimes.

The distinct convex tariff family uses $u_i = 1$, $h = 1/(10N)$, $d_i \in \{5, \dots, 20\}/10$, $-c_i \in \{1000, \dots, 25000\}/10000$, $x \in [0, 3]$, and a service floor $\sum_i z_i \geq N/5$. Seeds are $50001 + N$. The aligned sweep and upper solve are fused: their fresh median times were 0.014, 0.117 and 1.100 seconds for $N = 100, 1000, 10000$, respectively; input generation and validation added 0.0005, 0.0032 and 0.0336 seconds. Every returned response passed exact KKT and service-floor checks. These synthetic convex results do not measure the nonconvex atlas or a general large-scale bilevel model.

Earlier archived measurements are preserved separately. The older all-pair *compressed* implementation kept 274 atlas points for $N = 12$, compared with 45 after incremental envelope insertion; its recorded build medians were 15.882 and 3.079 seconds. It is a different comparator from the new original-coordinate baseline. The archived heterogeneous $N = 48$ timings were 18.699 seconds for construction and 2.922 for one upper solve. Archived $N = 1000$ two-type timings were 0.255 and 1.261 seconds, with only three fiber pieces and seven atlas points. The archived convex $N = 10000$ fused sweep took 1.047 seconds on the separate convex family. These historical runs used different cache and repetition protocols; their differences from fresh timings are not controlled speedup estimates. The earlier generic convex-path runs also retain their exact certificates and small dense-KKT MILP comparisons. They provide evidence for the certificate interface, not a completeness guarantee for numerical proposals on arbitrary inputs.

Table 3: Screening versus dense KKT MILP, seconds. Archived entries are single runs; fresh entries are three-run medians. Exact preprocessing includes input generation and surrogate cover; exact total also includes final rational verification. MILP preprocessing includes input generation and formulation. Tighter-tolerance follow-ups are excluded from the default-solve columns.

Source	N	Exact screening			Numerical MILP		
		Preprocess	Solve	Total	Preprocess	Solve	Total
Archived	8	0.0034	0.0984	0.1023	0.0023	0.0243	0.0265
Archived	12	0.0141	0.2962	0.3112	0.0045	0.0382	0.0427
Archived	20	0.0142	0.9332	0.9509	0.0078	0.1560	0.1638
Archived	40	0.0334	17.5689	17.6121	0.0256	0.4041	0.4297
Fresh	8	0.0046	0.1273	0.1335	0.0025	0.0180	0.0203
Fresh	20	0.0200	0.9056	0.9293	0.0084	0.1624	0.1699

Dense screening: a negative timing result. The exact screening prototype implements the scalar-cell specialization of Section 3. The identity surrogate has a diagonal response cover; the true dense Hessian is a rational symmetric perturbation with verified strict diagonal dominance. Screening fixes certified statuses, and exact original KKT recovery enumerates the ambiguous ones. Dense inversion, cover construction and certification are real computational costs: the bound $M3^t$ counts recovery LPs only.

The independent numerical comparator uses two binary variables l_i, u_i per coordinate and $g = Qz + c + Cx$, imposing

$$z_i + l_i \leq 1, \quad z_i - u_i \geq 0, \quad l_i + u_i \leq 1, \quad -M_i u_i \leq g_i \leq M_i l_i,$$

with $0 \leq z_i \leq 1$ and $M_i = 1 + |c_i| + |C_i| \max(|L|, |R|) + \sum_j |Q_{ij}|$. If both binaries vanish, $g_i = 0$; if $l_i = 1$ or $u_i = 1$, the corresponding bound and gradient sign hold. Conversely every box KKT point admits such binaries, including zero-gradient bounds. Positive definiteness makes this an exact real-arithmetic reformulation. Its SciPy–HiGHS solve is numerical, with a 30-second limit and requested relative MIP gap zero; numerical primal and dual reports are not rational optimality certificates.

The exact prototype was slower on all four archived instances, and the fresh $N = 8, 20$ repetitions preserve that finding after including preprocessing (Table 3). Their maximum ambiguity is two, with 105 and 285 completed cell/status assignments. The archived $N = 40$ case has maximum ambiguity four and 2304 completed assignments; rational recovery alone took 15.29 of 17.57 solver seconds. Small certified ambiguity therefore does not imply practical superiority over this numerical comparator.

The default $N = 8$ MILP reports optimality and zero gap but an objective $3.3703 \cdot 10^{-7}$ below the exact value $4810519997/235205877942$, with a modeled-row violation of approximately $3.5387 \cdot 10^{-7}$. This discrepancy occurs in both the archived run and all fresh repetitions. Requesting primal and MIP feasibility tolerances 10^{-9} gives objective agreement within $3.82 \cdot 10^{-17}$ and maximum row violation $1.11 \cdot 10^{-16}$. The follow-up timings and forwarded-option warnings are retained rather than substituted into the default comparison. At $N = 20$, agreement is within $1.34 \cdot 10^{-15}$. The exact returned screening solutions satisfy all rows and dense KKT equations rationally; their global guarantee rests on the screening theorem, not on checking a returned point alone.

The accompanying scripts preserve prepared rational instances, generator seeds, input hashes, software and machine metadata, individual timings, failures of the initial result-transport harness, and exact diagnostic logs. The completed experiments establish reproducible behavior for the stated

scalar subclasses. They do not implement the general polynomial quantifier-elimination algorithms, and they provide no evidence of superiority on industrial instances or on arbitrary heterogeneous follower populations.

A Polyhedral blocks coupled through a fixed core

This appendix develops a related elimination principle for a single-level problem that is linear in many small polyhedral blocks and polynomial in a fixed-dimensional core. It both supplies a broader structural comparison and makes witness recovery explicit. Its convex combinations preserve polyhedral feasibility and linear measurements; they are not combinations of globally optimal responses of a nonconvex follower.

Fix $r, k \geq 0$ and $d \geq 1$. Let $C \subset \mathbb{R}^r$ be compact and given by an explicit rational quantifier-free polynomial formula. For $b = 1, \dots, B$, let

$$P_b(x) = \{y \in \mathbb{R}^{d_b} : G_b(x)y \leq h_b(x)\}, \quad 1 \leq d_b \leq d,$$

with polynomial rational data and explicit finite rational coordinate bounds among its rows. The local sets may be empty or lower-dimensional. Let $A_b(x) \in \mathbb{Q}[x]^{k \times d_b}$, $b(x) \in \mathbb{Q}[x]^k$, $c_b(x) \in \mathbb{Q}[x]^{d_b}$ and $c_0(x) \in \mathbb{Q}[x]$. Consider

$$\begin{aligned} \min_{x, z} \quad & c_0(x) + \sum_{b=1}^B c_b(x)^\top z_b, \\ \text{subject to} \quad & x \in C, \quad z_b \in P_b(x) \quad (b = 1, \dots, B), \\ & \sum_{b=1}^B A_b(x) z_b = b(x). \end{aligned} \tag{83}$$

Let L include all explicit data and let $\delta \geq 2$ bound actual input degree. In contrast with a fixed-degree restriction, the following statement tracks the numerical degree throughout its construction.

Theorem A.1 (Fixed-core polyhedral optimization). *For fixed r, d, k , feasibility and global optimization of (83) are computable in time polynomial in (L, δ) . If feasible, an optimum is attained and an optimizer with all block coordinates and its value can be returned in a single real-algebraic field with polynomial total encoding length. Neither the number of blocks nor their row counts is fixed. A fixed number of aggregate inequalities is also permitted if finite bounds for their slacks are supplied.*

A.1 Local vertices and support directions

For block b enumerate all d_b -row subsets I . Put

$$\Delta_{bI}(x) = \det(G_b(x))_I, \quad p_{bI}(x) = \text{adj}((G_b(x))_I)(h_b(x))_I.$$

Where $\Delta_{bI} \neq 0$, the candidate is p_{bI}/Δ_{bI} . Equivalently, use the positive-denominator form

$$D_{bI} = \Delta_{bI}^2, \quad Y_{bI} = \Delta_{bI} p_{bI}, \quad y_{bI} = Y_{bI}/D_{bI}. \tag{84}$$

Its validity predicate is

$$D_{bI} > 0, \quad G_b(x)Y_{bI}(x) \leq h_b(x)D_{bI}(x). \tag{85}$$

All these tests are polynomial of degree $O(d\delta)$.

Every vertex of $P_b(x)$ has d_b independent active rows. Otherwise there is a nonzero direction orthogonal to all active normals; a small step in either direction keeps the active rows equal and preserves all finitely many strict inactive rows, contradicting extremality. A nonempty bounded polyhedron has a vertex, and every linear functional has a maximizing vertex. For completeness, a maximizing face is nonempty and compact; successively minimize the coordinates within it to produce a singleton face, which is a vertex of the original polyhedron. These arguments also apply to lower-dimensional sets and singletons. Thus (84) lists enough candidates to detect nonemptiness and maximize every linear functional.

Append the objective as a measurement. With $h = k + 1$, define

$$W_b(x) = \begin{pmatrix} A_b(x) \\ c_b(x)^\top \end{pmatrix}, \quad T_b(x) = W_b(x)P_b(x), \quad t(x, \eta) = \begin{pmatrix} b(x) \\ \eta - c_0(x) \end{pmatrix}. \quad (86)$$

For a direction $\lambda \in \mathbb{R}^h$, the support score of a valid vertex is

$$\lambda^\top W_b(x)y_{bI}(x) = n_{bI}(x, \lambda)/D_{bI}(x), \quad n_{bI} = \lambda^\top W_b Y_{bI}.$$

Collect the denominator and validity polynomials, and, for every pair of candidates in one block, the comparison polynomial

$$n_{bI}D_{bJ} - n_{bJ}D_{bI}. \quad (87)$$

The denominators of valid candidates are positive, so these signs correctly order their scores. There are polynomially many polynomials for fixed d , all in the $r + h$ coordinates (x, λ) , of degree $O(d\delta)$ and polynomial coefficient bit length.

Enumerate their realizable sign conditions, remembering identically zero polynomials and including zero signs. Each sign condition determines the valid vertices and a support-maximizing vertex for every nonempty block; choose the first maximizer in a fixed ordering. Discard a condition if some block has no valid candidate. The retained list Σ is polynomial in size and can be constructed in polynomial bit time for fixed (r, d, k) . Disconnected sign realizations cause no problem, since both validity and score order are determined by signs.

A.2 Exact membership from the support inequalities

For $\sigma \in \Sigma$, let $I_b(\sigma)$ be its selected local index and abbreviate the corresponding denominator and numerator by $D_{b\sigma}, n_{b\sigma}$. Put

$$\begin{aligned} H_\sigma(x) &= \prod_b D_{b\sigma}(x), \\ J_\sigma(x, \eta, \lambda) &= \lambda^\top t(x, \eta) H_\sigma(x) - \sum_b n_{b\sigma}(x, \lambda) \prod_{a \neq b} D_{a\sigma}(x). \end{aligned} \quad (88)$$

On that sign condition $H_\sigma > 0$ and $J_\sigma \leq 0$ states the support inequality

$$\lambda^\top t(x, \eta) \leq \sum_b \max_{y \in P_b(x)} \lambda^\top W_b(x)y.$$

Writing $S_\sigma(x, \lambda)$ for the sign conjunction, define

$$\Phi(x, \eta, \lambda) = \bigvee_{\sigma \in \Sigma} [S_\sigma(x, \lambda) \wedge J_\sigma(x, \eta, \lambda) \leq 0]. \quad (89)$$

Lemma A.2 (Support membership). *For $x \in C$, there exist blocks satisfying (83) with objective exactly η if and only if*

$$\forall \lambda \in \mathbb{R}^h \quad \Phi(x, \eta, \lambda). \quad (90)$$

Proof. If a block is empty, no sign condition realized at (x, λ) is retained, for any λ . Thus (90) is false. Otherwise the Minkowski sum $T(x) = \sum_b T_b(x) = \{\sum_b t_b : t_b \in T_b(x)\}$ is compact and convex. Its support function is the sum of the support functions of its summands: a maximizing choice can be made independently in every block. The formula (89) therefore gives its exact support inequality in every direction.

A point t in a compact convex set T satisfies all its support inequalities. Conversely, if $t \notin T$, choose a nearest point $p \in T$ and put $\lambda = t - p \neq 0$. The derivative of squared distance along the segment from p to any $u \in T$ gives $(t - p)^\top(u - p) \leq 0$. Hence $\max_{u \in T} \lambda^\top u \leq \lambda^\top p < \lambda^\top t$, violating a support inequality. Thus (90) is equivalent to $t(x, \eta) \in T(x)$. By (86), this membership states precisely existence of blocks satisfying all linking equations and attaining the specified objective value. $\square$

The degrees in (88) are $O((B + 1)d\delta)$, which need not be constant. Their ambient dimension $r + h + 1$ is fixed, however, so their expanded monomial count and bit size are polynomial in (L, δ) by the product argument in lemma 2.3. Define

$$E(x, \eta) = (x \in C) \wedge \forall \lambda \Phi(x, \eta, \lambda).$$

This fixed-variable formula describes exact attainable core–value pairs. The full feasible set of (83) is closed and lies in the product of compact C and the finite block boxes. Therefore the upper minimum is attained whenever feasible. Apply quantifier elimination and joint sampling to

$$E(x, \eta) \wedge \neg \exists x', \eta' [E(x', \eta') \wedge \eta' < \eta]. \quad (91)$$

This gives an optimal (x^*, η^*) in one field $K = \mathbb{Q}(\alpha)$ of polynomial degree and encoding length. It remains to recover all block variables. Giving each of them a new quantified coordinate would destroy the fixed-dimensional argument.

A.3 Constructive recovery from a few support tuples

The same support enumeration already supplies a polynomial-size set of full block tuples sufficient for recovery. For each $\sigma \in \Sigma$ test its selected local denominators at x^* and discard the tuple if any is zero. Evaluate the remaining formulas and retain the tuple if every selected local vertex is feasible at x^* . Denote the retained tuples by $y^1, \dots, y^M$, and set

$$a^j = \sum_b W_b(x^*) y_b^j \in K^h \quad (j = 1, \dots, M).$$

There is no need to require that the original sign condition σ be realizable at x^* with some direction: feasibility of the tuple suffices for this filtering step. It never adds a point outside $T(x^*)$.

Lemma A.3 (A small common convex combination recovers every block). *The retained aggregate points satisfy $\text{conv}\{a^1, \dots, a^M\} = T(x^*)$. For every target $t \in K^h \cap T(x^*)$ one can find at most $h + 1$ retained tuples and nonnegative weights in K that sum to one and whose weighted aggregate is t , in polynomial bit time in the original input and target encoding. Using the same weights in every block gives a feasible original tuple with aggregate t .*

Proof. Every retained tuple is blockwise feasible, so its aggregate belongs to $T(x^*)$ and the indicated convex hull is contained in $T(x^*)$. For every real direction λ , the actual sign condition at (x^*, λ) is retained in Σ : no block is empty at an optimal core. Its selected vertices give a feasible tuple that attains the support of $T(x^*)$ in direction λ . This tuple survives the filter. Hence the convex hull has the same support function as $T(x^*)$. The separation argument in lemma A.2 proves equality.

The usual Carathéodory reduction gives a convex representation of a target in a finite convex hull with affinely independent support. To see this directly, start with a representation with all supported weights positive. If its lifted columns $(a^j, 1)$ are dependent, choose a nonzero dependence u . Its components sum to zero, so it has a positive component. Replace the weights θ_j by $\theta_j - su_j$, with $s = \min_{u_j > 0} \theta_j / u_j$. This preserves the target and the sum of weights, keeps weights nonnegative, and removes at least one support point. Repeat until the lifted columns are independent. Their number is at most $h + 1$.

For construction, enumerate all subsets of at most $h + 1$ retained points. On each affinely independent subset solve

$$\sum_j \theta_j a^j = t, \quad \sum_j \theta_j = 1.$$

Choose independent rows to solve this fixed-size system, then check all remaining equations and the nonnegativity of its unique weights. If a suitable subset exists over the reals, its independent linear system has its unique solution in K , since every coefficient and the target belong to K . Thus the enumeration finds it. There are at most $\sum_{q=1}^{h+1} \binom{M}{q}$ subsets, polynomial for fixed h . All arithmetic, zero tests, and comparisons take place in the already sampled ordered field K , whose degree and coefficient encodings are polynomial. Each system has fixed order, so determinant expressions and rational evaluation give polynomial coefficient lengths for the weights; summing the resulting fixed number of terms preserves the same bound for every recovered block coordinate.

Finally, put $z_b = \sum_j \theta_j y_b^j$ for every block. Convexity of $P_b(x^*)$ makes z_b feasible, and linearity of $W_b(x^*)$ gives $\sum_b W_b(x^*) z_b = t$. The same weights must be used in all blocks: this is how the prescribed total measurements are preserved. $\square$

Applying lemma A.3 to $t = t(x^*, \eta^*)$ recovers the original optimizer in the field K , completing the main claim of theorem A.1. With no blocks, the image sum is $\{0\}$ and the empty tuple is its sole support tuple; all the formulas use empty sums and products with their usual meanings. Blocks of dimension zero can be removed after testing their scalar rows. A fixed number of aggregate inequalities can be made equalities using bounded scalar slack blocks. Supplied slack bounds ensure that the result stays in the same compact model and do not change the fixed structural dimensions.

This argument gives polynomial dependence on the numerical input degree as well as input length: local determinants and pair comparisons have degree $O(d\delta)$; the global products have degree $O((B + 1)d\delta)$; sign enumeration, elimination and sampling are all in fixed dimension; and reconstruction adds only fixed-size algebraic linear systems. Under polynomially bounded degree, this is polynomial time in L .

The common support-direction construction has a direct predecessor in Gritzmann and Sturmfels [24, Algorithm 2.3.6, Theorem 2.3.7, and Corollary 2.3.10]. For fixed-dimensional input polytopes, they enumerate a shared arrangement of support directions and compatible maximizing summand vertices, with polynomial binary complexity. Here the sign enumeration is uniform over the polynomially varying core; exact optimization and the subsequent reconstruction recover all original blocks in one common field. These additional steps are proved above. The support characterization, finite-dimensional convex-hull reduction, and real algebraic algorithms are established tools. An alternative recovery would solve a linear program over the sampled common field; Adler and Beling [1] analyze that route with explicit dependence on the common extension degree. The constructive

recovery above uses the support tuples already generated and needs no large-dimensional algebraic LP solve. It depends on the affine block measurements, including the objective coordinate. It would not justify convexifying a nonconvex follower reaction set or mixing its tied responses to create additional follower choices.

B Uniform rational approximation of polynomial inverses

This appendix proves the approximation interface used in section 4. Polynomial inverse approximation and algebraic-series computation have established algorithms [17, 43]. The purpose here is to give the complete uniform error and rational encoding bounds, including flat points of a strictly increasing marginal. The analytic ingredients are the classical holomorphic inverse, Cauchy estimates, and Rouché theorem [38].

Lemma B.1 (Effective increment and inverse bounds). *Let $g \in \mathbb{Q}[t]$ have degree $D \geq 1$, be strictly increasing on $[0, 1]$, and satisfy $g(0) = 0$, $g(1) = 1$. For $[a, a + h] \subseteq [0, 1]$,*

$$g(a + h) - g(a) \geq \frac{1}{2} \left(\frac{h}{2D} \right)^D. \quad (92)$$

Consequently the clipped inverse changes by less than e whenever its argument changes by less than $\frac{1}{2}(e/(2D))^D$, for $0 < e \leq 1$. For an unnormalized marginal with endpoint difference $G > 0$, multiply this argument tolerance by G .

Proof. For $h > 0$ put $v = g(a + h) - g(a)$. Interpolate $g - g(a)$ at $a + jh/D$, $j = 0, \dots, D$. All node values lie in $[0, v]$. On $[0, 1]$ every factor in the numerator of a Lagrange basis polynomial has absolute value at most one, and its denominator has magnitude $(h/D)^D j!(D - j)!$. Thus

$$|g(t) - g(a)| \leq v(D/h)^D \sum_{j=0}^D \frac{1}{j!(D - j)!} \leq v(2D/h)^D.$$

Applying this at $t = 0, 1$ gives $1 \leq 2v(2D/h)^D$. The case $h = 0$ is immediate. An inverse displacement of at least e contains a response interval of length e , so requires at least the displayed argument displacement. If clipping occurs, the argument interval only gets longer. Positive affine normalization proves the last assertion. $\square$

Theorem B.2 (Rational inverse approximation). *For a densely encoded rational polynomial g strictly increasing on $[0, 1]$, and a positive rational tolerance η , one can construct rational breakpoints and rational polynomial branches approximating its clipped inverse uniformly to error η . Each branch is valid on its closed interval, with constant branches outside $[g(0), g(1)]$. Construction time, number of branches, degrees, and rational encoding lengths are polynomial in input bits, numerical degree, and $\max\{0, \log(1/\eta)\}$. The encoding of η is part of the input; equivalently one can request $\eta = 2^{-B}$ at cost polynomial in B . No positive lower bound on g' is required.*

Proof. Normalize to $g(0) = 0, g(1) = 1$ by a rational affine change of target. Replace η by $\min\{\eta, 1/2\}$, and put

$$e = \eta/4, \quad \delta_0 = \frac{1}{4}(e/(2D))^D, \quad \rho = \text{largest dyadic number at most } \min\{\delta_0/[4(D + 1)], 1/16\}. \quad (93)$$

Their bit lengths are polynomial in the stated parameters. An interval of target length at most δ_0 has inverse oscillation less than e , by lemma B.1.

Critical values, including nonreal ones. The real parts of all finite complex critical values are the real s satisfying

$$\exists u, v : \quad \Re g'(u + iv) = \Im g'(u + iv) = 0, \quad s = \Re g(u + iv). \quad (94)$$

The formula has three real variables and polynomials of degree at most D with polynomial-size dense expansions. Fixed-dimensional elimination and univariate isolation from section 1.5 compute this finite set, of cardinality at most $D - 1$, in polynomial bit time. For $D = 1$ it is empty. Repeated critical points or coincident real parts cause no problem. Keeping only real critical values would be insufficient for the analytic argument below.

Retain the real parts in $[-1, 2]$, bracket each in a rational interval of width at most ρ , and pad that interval by ρ at both ends. Add $[-\rho, \rho]$ and $[1 - \rho, 1 + \rho]$, intersect the union with $[0, 1]$, and merge overlapping intervals. Denote the result by B_0 . The sum of untrimmed lengths is at most $3(D - 1)\rho + 4\rho \leq \delta_0$. Each merged component therefore has length at most δ_0 . On it use a constant rational approximation, within e , to the inverse at its rational midpoint. Monotone bisection in response space computes that constant in polynomial time. Its error on the entire component is less than $2e$. This also treats endpoints and interior derivative zeros.

For a complementary gap $[a, b]$, define

$$d(t) = \min\{t - a + \rho, b - t + \rho\}.$$

Every critical real part is at distance at least $d(t)$ from t . For retained parts, padding places them at least ρ behind the exposed endpoint of their merged component. Omitted parts lie outside $[-1, 2]$, at distance greater than one, whereas $d(t) \leq 1/2 + \rho < 1$.

Rational analytic panels. Split each gap at $m = (a + b)/2$. Starting at $x = a$, set

$$y = \min\{m, x + (x - a + \rho)/32\}$$

and take $[x, y]$ as the next panel; reflect this construction on the right half. Before the final panel, $x - a + \rho$ grows by $33/32$. Thus each gap has $O(1 + \log(1/\rho))$ panels, all with polynomial rational bit length. There are at most $D + 1$ gaps. At a panel midpoint τ , put $d = d(\tau)$; the panel half-width is at most $d/64$.

Writing $g(z) = \sum a_j z^j$, the rational number $L_g = \max\{1, \sum j|a_j|\}$ bounds g' in absolute value on $[0, 1]$. Bisection of the inverse bracket to width $d/(64L_g)$ provides a rational midpoint $z_0 \in (0, 1)$ such that $|g(z_0) - \tau| \leq d/64$. Set $t_0 = g(z_0)$ and $R = d/4$. Every complex critical value has distance at least $63d/64$ from t_0 . Hence a neighborhood of the closed radius- R disk contains no critical value. Every panel target satisfies

$$|t - t_0| \leq d/32 \leq R/8. \quad (95)$$

For completeness, the inverse branch exists on that entire disk, not merely on its real diameter. A polynomial is proper: the modulus of its leading term forces $|g(z)| \rightarrow \infty$ as $|z| \rightarrow \infty$. Away from critical values each preimage is simple, and the local holomorphic inverse theorem gives disjoint inverse neighborhoods for its finitely many roots. Properness ensures that these neighborhoods exhaust preimages after shrinking the target neighborhood; otherwise a sequence of additional preimages has a bounded convergent subsequence, contradicting local uniqueness. Thus the restriction is a finite covering. Path lifting over the simply connected disk gives one inverse on each connected component, with no monodromy. It is holomorphic by local inversion. Select the component through z_0 . Along the real segment to every panel target this branch equals the real increasing inverse by uniqueness and continuity. In particular $g'(z_0) \neq 0$.

Taylor approximation and rational arithmetic. On the disk $|t| < 2$. The elementary polynomial root bound gives

$$|Z(t)| \leq M_g := 2 + \frac{2 + \sum_{j < D} |a_j|}{|a_D|}.$$

Indeed, a root of $a_D z^D + \sum_{j < D} \tilde{a}_j z^j$ has modulus at most $1 + \max_{j < D} |\tilde{a}_j/a_D|$, and the constant coefficient changes by less than two. Cauchy's estimate for $Z(t_0 + v) = z_0 + \sum_{n \geq 1} c_n v^n$ yields $|c_n| \leq M_g R^{-n}$. With $q \geq \lceil \log_2(4M_g/\eta) \rceil$, the tail on the panel is at most $M_g 2^{-q} \leq \eta/4$, using the weaker ratio $1/2$ in (95).

All coefficients are rational, since the expansion center is a *rational response* rather than an approximate irrational inverse. Write

$$g(z_0 + u) - t_0 = Q^{-1} \sum_{h=1}^D A_h u^h, \quad Q \in \mathbb{Z}_{>0}, \quad A_h \in \mathbb{Z}, \quad A_1 \neq 0.$$

Formal substitution gives

$$\begin{aligned} c_1 &= Q/A_1, \\ c_n &= -\frac{1}{A_1} \sum_{h=2}^{\min(D,n)} A_h [v^n] \left(\sum_{j=1}^{n-1} c_j v^j \right)^h. \end{aligned} \tag{96}$$

A factor containing c_n cannot contribute to degree n in a power $h \geq 2$, which proves the triangular recurrence. Inductively

$$c_n = Q^n N_n / A_1^{2n-1}, \quad N_n \in \mathbb{Z}. \tag{97}$$

A product of h earlier coefficients with indices summing to n has denominator A_1^{2n-h} after factoring Q^n . The additional division by A_1 and conversion to the displayed denominator multiplies its integer numerator by A_1^{h-2} . Signs are irrelevant.

Translation of g to z_0 has polynomial encoding; so do Q, A_h . The denominator in (97) and Cauchy's magnitude bound give polynomial numerator and denominator bit lengths, because $q, \log M_g$, and $\log(1/R)$ are polynomially bounded. The same holds for intermediate truncated convolutions: each term has at most q earlier factors, and at most 2^{q-1} compositions contribute to any fixed degree before the polynomially many sums over h . Using these common denominators keeps all arithmetic polynomial in bit cost. Truncated polynomial multiplication uses polynomially many operations, without enumerating the compositions. Ordinary-power expansion of the Taylor polynomials preserves the bounds.

The constant pieces on B_0 , Taylor pieces on the gaps, and exact clipped constants cover the line. Adjacent branches each satisfy the error bound at their shared endpoint; they need not agree exactly. A half-open convention defines a single approximating function if needed. Undoing the target normalization proves the theorem. $\square$

Proposition B.3 (Sharper positive-coefficient construction). *If $g(z) = \sum_{j=1}^P a_j z^j$ with $a_j \geq 0$ and $g(1) > 0$, the approximation in theorem B.2 can use $O(P^2 \log(1/\eta))$ nonconstant branches, each of degree $O(\log(1/\eta))$, independently of coefficient conditioning in these two counts. Coefficient bit lengths still depend polynomially on the input encoding. Numerical, rather than sparsely encoded binary, P is the complexity parameter.*

Proof. Normalize $g(1) = 1$. Termwise,

$$z^P \leq g(z) \leq z, \quad 0 \leq g'(z) \leq P, \quad g(z) \leq z g'(z) \leq P g(z) \quad (0 < z \leq 1).$$

Take $m = \lceil \log_2(1/\eta) \rceil \geq 1$ after reducing η to at most $1/2$, and $K_0 = Pm$. Below 2^{-K_0} the constant zero has error at most 2^{-m} ; above one use the exact constant one.

For real $0 < z_0 \leq 1$, put $t_0 = g(z_0)$, $r_z = z_0/(4P)$ and $R_t = t_0/(8P)$. Nonnegative coefficients give, for complex $|u| \leq r_z$,

$$|g'(z_0 + u) - g'(z_0)| \leq g'(z_0)((1 + 1/(4P))^{P-1} - 1) < g'(z_0)/3.$$

Here $e^{1/4} < 4/3$ follows by termwise comparison with a geometric series. Integrating along the segment gives

$$|g(z_0 + u) - t_0 - g'(z_0)u| \leq g'(z_0)|u|/3.$$

On $|u| = r_z$, the shift $|t - t_0| \leq R_t$ is at most $g'(z_0)r_z/2$. The nonlinear remainder plus shift is therefore at most $5/6$ of the linear comparison magnitude. Rouché gives exactly one root in the response disk; the derivative bound makes it simple. Local inverses agree by uniqueness, and the strict boundary margin extends them to a neighborhood of the closed target disk. The resulting branch satisfies $|Z(t)| < 2$ and agrees with the positive real inverse on real targets in $[0, 1]$.

Split each dyadic interval $[A, 2A]$, $A = 2^{-(k+1)}$, $k = 0, \dots, K_0 - 1$, into $32P$ equal rational panels. At midpoint τ the panel half-width is at most $\tau/(64P)$. Bisection to response width $\tau/(64P^2)$ produces rational z_0 with $|t_0 - \tau| \leq \tau/(64P)$ and $t_0 \geq 63\tau/64$. Thus on its panel

$$|t - t_0|/R_t \leq 16/63 < 1/2.$$

The bisection takes $K_0 + O(\log P)$ steps. Use the exact recurrence (96). Cauchy's estimate is now $|c_n| \leq 2R_t^{-n}$, so truncating at $q = m + 3$ gives tail $2^{1-q} \leq \eta/4$. The denominator and intermediate-bit proof above applies unchanged. There are $32P^2m$ panels and the claimed degree. This argument uses coefficient nonnegativity essentially in the derivative-disk estimate; it is not used for signed marginals. $\square$

C Explicit bounds for resource-coupled followers

The estimates below give polynomial-bit constants for the accuracy algorithms. Their numerical values need not be small. They instantiate classical polyhedral error bounds [26]; no optimal Hoffman constant is needed.

Lemma C.1 (Exact resource projection). *For a rational matrix $C \in \mathbb{Q}^{k \times N}$, with fixed k , the set*

$$X_F = \{x \in X : \exists z \in [0, 1]^N, Cz \leq b(x)\}$$

has a polynomial-size rational inequality description when X is a rational polytope and b is rational affine. That description and its feasibility can be computed in polynomial bit time.

Proof. The right-hand sides permitting a follower form the closed polyhedron $C[0, 1]^N + \mathbb{R}_+^k$. Separation from this set shows that membership is equivalent to

$$\lambda^\top b(x) \geq \sum_{i=1}^N \min\{0, \lambda^\top C_i\} \quad \text{for all } \lambda \geq 0. \quad (98)$$

Indeed a finite lower support functional on the orthant sum must be nonnegative, and its box minimum is the displayed sum. On every cone of the central arrangement $\lambda^\top C_i = 0$ in the nonnegative orthant this sum is linear. Such cones are pointed, so it suffices to check their extreme rays. Each ray

has $k - 1$ independent active hyperplanes from the column and coordinate hyperplanes. Enumerate all such subsets, compute their one-dimensional nullspaces, and retain each nonnegative direction. The resulting $O((N + k)^{k-1})$ tests include all necessary rays; extra nonnegative directions give valid inequalities. Cramer's rule supplies polynomial rational bit lengths. Coordinate hyperplanes cover rank deficiency and zero columns, including lower-dimensional cones. For $k = 1$, the empty subset gives the ray $+1$; for $k = 0$, $X_F = X$. Substitute $b(x)$ in these finitely many inequalities and use rational LP. $\square$

Lemma C.2 (Uniform multipliers and feasibility repair). *Assume $N \geq 1$ and put*

$$\begin{aligned} \mathbf{A} &= d_0[C; I; -I], \quad M = \max\{1, \max_{ij} |A_{ij}|\}, \\ V &= N!(1 + NM^2)^N, \quad K = d_0 N^3 MV, \end{aligned} \tag{99}$$

where the positive integer d_0 clears all denominators of C . These constants have polynomial bit length. For every $x \in X_F$ and $q \in [0, 1]^N$ with $Cq - b(x) \leq \delta \mathbf{1}$, $\delta \geq 0$, there is a feasible y with

$$\|q - y\|_\infty \leq K\delta. \tag{100}$$

If a differentiable convex follower objective has gradient bounded by $\bar{G}$ in infinity norm on the boxes, each optimum has resource KKT multipliers in $[0, \Lambda]^k$, where $\Lambda = \max\{1, K\bar{G}\}$. No strict feasibility is required.

Proof. A vector v in a cone of rows of the integer matrix $\mathbf{A}$ has a nonnegative representation using linearly independent rows $\mathbf{A}_J$, with $|J| \leq N$, by the conic support reduction used in lemma 2.2. Its coefficients are

$$\alpha = (\mathbf{A}_J \mathbf{A}_J^\top)^{-1} \mathbf{A}_J v.$$

The Gram determinant is a positive integer. Entries have magnitude at most NM^2 and every cofactor has magnitude at most V by the permutation expansion, including the zero-order cofactor one. Hence

$$\|\alpha\|_1 \leq N^3 MV \|v\|_2.$$

For $\|v\|_\infty \leq \bar{G}$, the same formula bounds each individual coefficient by $N^2 MV \bar{G} \leq N^3 MV \bar{G}$. At an optimum apply this to $-\nabla F_x(z^*)$, using only active normals. Undoing their scale multiplies coefficients by d_0 , so resource multipliers are at most $K\bar{G}$. The active-normal formula is valid for every polyhedron, including degenerate faces. Convexity makes the resulting KKT conditions sufficient.

For repair, let y be the Euclidean projection of q onto the nonempty follower polytope and apply the representation to $v = q - y$. For active scaled rows, $\mathbf{A}_J(q - y)$ is the violation at q . Resource violations are at most $d_0\delta$ and box violations are nonpositive. Thus

$$\|v\|_2^2 = \alpha^\top \mathbf{A}_J(q - y) \leq d_0\delta \|\alpha\|_1 \leq K\delta \|v\|_2.$$

For $v \neq 0$ divide by $\|v\|_2$; for $v = 0$ the result is immediate. The bit bounds follow from $\log N! = O(N \log N)$ and the explicit integer powers. Neither projection nor the correct active set is computed by the accuracy algorithm. $\square$

Lemma C.3 (Polynomial growth without strong curvature). *Let $g_i(0) = 0$, $g_i(1) = G_i > 0$ be strictly increasing rational polynomials of degrees at most $P \geq 1$, and $f'_i = g_i$. For the convex aggregate model (47), at every fixed follower-feasible leader and all $y, z \in [0, 1]^N$,*

$$F_x(y) - F_x(z) - \nabla F_x(z)^\top (y - z) \geq \mu \|y - z\|_\infty^{P+1}, \quad \mu = \frac{\min_i G_i}{4(4P)^P}. \tag{101}$$

If every coefficient of each g_i is nonnegative, one may instead use the sharper $\mu = \min_i G_i / (P + 1)$.

Proof. For one marginal of actual degree d , let $h = |y_i - z_i|$. In either direction the Bregman integral contains an interval of length $h/2$ where the marginal difference from its base value is at least $G_i(h/(4d))^d/2$, by lemma B.1 on a response interval of length $h/2$. Its integral is therefore at least $G_i h^{d+1}/[4(4d)^d] \geq G_i h^{P+1}/[4(4P)^P]$. For nonnegative coefficients, the Bregman divergence of $t^{j+1}/(j+1)$ is at least $|y_i - z_i|^{j+1}/(j+1)$: in the forward direction integrate $(z_i + t)^j - z_i^j \geq t^j$; in the reverse direction integrate $z_i^j - (z_i - t)^j \geq t^j$. Both follow from the binomial theorem with nonnegative arguments. Multiply by the coefficients and use $h \leq 1$ and $j \leq P$. Sum over coordinates; the affine terms cancel, and the aggregate has nonnegative Bregman divergence by convexity. The coordinate of largest displacement gives the asserted infinity-norm bound. $\square$

Polynomial solution-map regularity has a direct predecessor in Jeyakumar, Lasserre, Li, and Pham [27, Theorem 2.3]. For a compact polynomial follower feasible set independent of the leader, their theorem gives a one-sided inclusion of the nearby solution set $Y(x)$ in a Hölder neighborhood of $Y(\bar{x})$ at a fixed reference leader $\bar{x}$. Its exponent bound depends on polynomial degree, follower dimension, and constraint count. For the structured unique-response class here, the following proposition gives a uniform pairwise exponent $1/P$, independent of follower count and aggregate degree, with a computable rational constant of polynomial encoding length. It also allows affine resource right-hand sides that vary with the leader.

Proposition C.4 (Leader-response modulus and a sharper exponent). *For (47) with $N \geq 1$, effective rational constants with polynomial encoding give, for $x, x' \in X_F$ and $\Delta = \|x - x'\|_\infty$,*

$$\|z(x) - z(x')\|_\infty \leq KB_b\Delta + [2(T_x + L_F KB_b)\Delta/\mu]^{1/(P+1)}. \quad (102)$$

Here B_b bounds the Lipschitz constant of b and $|F_x(z) - F_{x'}(z)| \leq T_x\Delta$ uniformly on the box. If $\phi = 0$, replace $2(T_x + L_F KB_b)$ by $2L_F KB_b + B_\ell$, where $B_\ell = \sum_i \|\nabla \ell_i\|_1$. In fact there is an effective rational constant C_z of polynomial encoding such that

$$\|z(x) - z(x')\|_\infty \leq C_z \Delta^{1/P}. \quad (103)$$

The exponent $1/P$ is best possible for this class.

Proof. For the first bound, repair $z(x)$ to y feasible at x' , and $z(x')$ to y' feasible at x , each within $KB_b\Delta$. Optimality and the two Lipschitz bounds give

$$\begin{aligned} F_{x'}(y) &\leq F_x(z(x)) + T_x\Delta + L_F KB_b\Delta \\ &\leq F_x(y') + T_x\Delta + L_F KB_b\Delta \\ &\leq F_{x'}(z(x')) + 2(T_x + L_F KB_b)\Delta. \end{aligned}$$

Apply lemma C.3 at the constrained optimum $z(x')$ and add the repair distance. For $\phi = 0$, retaining the cancellation of the leader-linear terms bounds the gap instead by

$$2L_F KB_b\Delta + |(\ell(x) - \ell(x'))^\top (z(x) - z(x'))| \leq (2L_F KB_b + B_\ell)\Delta.$$

These estimates need only coefficient sums in the fixed aggregate variables and in the local marginals.

We prove the sharper assertion separately. Let $A_x, A_z \geq 0$ be rational bounds such that the full follower gradient obeys

$$\|\nabla F_x(z) - \nabla F_{x'}(z)\|_\infty \leq A_x\Delta, \quad \|\nabla F_x(z) - \nabla F_x(z')\|_\infty \leq A_z \|z - z'\|_\infty.$$

They have polynomial encoding and are computable without expanding in N variables: local derivative coefficient sums, bounds on the aggregate Hessian and its mixed leader derivatives, and the row and column sums of U suffice. Set $D_b = KB_b$.

First suppose $z = z(x)$ and $z' = z(x')$ have the same active resource rows and the same lower/free/upper box pattern. Take an independent basis of their common active rows from $[C; I; -I]$. The row-space solution d of their active right-hand-side difference satisfies

$$\|d\|_\infty \leq D_b \Delta, \quad e - d \text{ is tangent to all active rows,} \quad e = z - z'. \quad (104)$$

Indeed for the scaled integer basis A_J the solution is $A_J^\top (A_J A_J^\top)^{-1}$ times the scaled right-hand-side difference. The latter has infinity norm at most $d_0 B_b \Delta$, and the cofactor bounds from lemma C.2 give $\|d\|_\infty \leq d_0 N^2 M V B_b \Delta \leq D_b \Delta$. Dependent active rows follow from the same row space; their differences are consistent because e solves the complete system.

Both optimal gradients lie in the common active row span by the polyhedral KKT conditions, so their difference is orthogonal to $e - d$. Splitting the gradient difference through $\nabla F_{x'}(z')$ therefore gives

$$\begin{aligned} (\nabla F_x(z) - \nabla F_x(z'))^\top e &= (\nabla F_x(z) - \nabla F_{x'}(z'))^\top d \\ &\quad + (\nabla F_{x'}(z') - \nabla F_x(z'))^\top e. \end{aligned}$$

The first gradient difference has infinity norm at most $A_z \|e\|_\infty + A_x \Delta$, and the second at most $A_x \Delta$. Adding the two Bregman inequalities at fixed leader x now gives

$$2\mu \|e\|_\infty^{P+1} \leq (\nabla F_x(z) - \nabla F_x(z'))^\top e \leq N(A_z \|e\|_\infty + A_x \Delta) D_b \Delta + N A_x \Delta \|e\|_\infty.$$

If $\|e\|_\infty \geq D_b \Delta$ and it is positive, division yields

$$\|e\|_\infty^P \leq \frac{N(A_z D_b + 2A_x)}{2\mu} \Delta.$$

Otherwise $\|e\|_\infty \leq D_b \Delta$. As $\Delta \leq 1$, both cases give the same-pattern bound with rational constant

$$C_0 = D_b + \max\{1, N(A_z D_b + 2A_x)/(2\mu)\}.$$

Continuity extends this bound to limits of any common-pattern subset.

It remains to control pattern changes along the segment $x(t) = (1-t)x + tx' \in X_F$, $0 \leq t \leq 1$. This argument bounds their number; the algorithm does not construct their high-dimensional description. A pattern has three choices per box coordinate and two per resource row. For each pattern, its graph is described in (t, z, λ) by box bound equalities or strict interior inequalities, active resource equalities or strict slacks, $0 \leq \lambda \leq \Lambda$, zero multipliers on inactive resources, and the corresponding gradient signs or equations for the box KKT conditions. These conditions are necessary and sufficient by convexity, and use at most $3N + 4k + 2$ polynomial conditions, of degree at most $d = \max\{P, \deg \phi, 2\}$. Real coefficients involving the two fixed endpoint leaders are harmless for component counting.

Convert each weak inequality $p \geq 0$ to $p - u^2 = 0$, each strict inequality $p > 0$ to $pu^2 - 1 = 0$, and retain equations. Sum their squares. The resulting real hypersurface has degree at most $D_* = 2(d+2)$ in at most $n_* = 8(N+k+1)$ variables. Projection maps it onto the pattern graph. The classical hypersurface component bound [36] is at most $D_*(2D_* - 1)^{n_*-1}$. Continuous projection cannot increase the number of connected components, so this also bounds the number of interval or point components of the pattern's set of parameter values t . It applies to strict inequalities because of the reciprocal-square variables, without a boundedness assumption on those variables.

Consequently, over all patterns, the component count is at most the explicit integer

$$\Gamma = 3^N 2^k D_* (2D_* - 1)^{n*}.$$

The endpoints of those components partition $[0, 1]$ into at most $2\Gamma + 1$ successive subintervals. On the interior of each choose a pattern that occurs there; absence of component endpoints implies that it occurs throughout that interior. Apply the same-pattern bound and continuity at endpoints. Summing the bounds gives

$$\|z(x) - z(x')\|_\infty \leq C_0 \sum_j [\Delta(t_j - t_{j-1})]^{1/P} \leq C_0 (2\Gamma + 1) \Delta^{1/P}.$$

Thus $C_z = C_0(2\Gamma + 1)$ is a rational sufficient constant. Its logarithm is polynomial in the input and numerical degrees, despite its potentially large numerical value. Finally the independent marginal $g(z) = z^P$ with incentive x has response $x^{1/P}$ near zero. No larger uniform Hölder exponent holds even for this one-variable subclass. $\square$

D Path geometry and exact projection

This appendix supplies the statements used in Section 5.5. The path shadow is an established construction. The telescoping identity gives a second exposure formula tailored to the quadratic follower, and the final projection result applies to the narrower class of equal-gain strips.

D.1 The sparse Klee–Minty shadow and its messages

Every coordinate of $P_n(\gamma)$ lies in $[0, 1]$. Since $\gamma < 1/2$, the two bounds on a coordinate cannot be simultaneously tight. A vertex has n independent tight rows, hence exactly one bound per coordinate. Conversely, every endpoint selection is nonsingular triangular. The vertices are therefore

$$v_j(d) = d_j + (1 - 2d_j)\gamma v_{j-1}(d), \quad v_0 = 0, \quad d \in \{0, 1\}^n. \quad (105)$$

Two vertices are adjacent when their bit vectors differ in one coordinate. Their terminal values are distinct: the two final choices lie in the disjoint intervals $[0, \gamma]$ and $[1 - \gamma, 1]$; within either interval the terminal map is injective in the previous coordinate, so induction applies.

Lemma D.1 (Sparse shadow, after Gärtner et al.). *Let $\gamma = 1/4$, $\hat{c}_j = \gamma^{3(n-j)}$ for $j < n$, and $\hat{c}_n = 0$. Every vertex $v(d)$ uniquely maximizes $\hat{c}^\top z + \lambda z_n$ over $P_n(\gamma)$ at the price*

$$\lambda(d) = - \sum_{j=0}^{n-1} \left(\prod_{k=j+1}^n (1 - 2d_k) \right) \gamma^{2(n-j)} \in (-1/15, 1/15).$$

Proof. We reproduce the edge calculation in Gärtner et al. [21, Definition 11 and Lemma 12] to fix the exact version used here. Put $p_i^j = \prod_{k=i}^j (1 - 2d_k)$, with an empty product equal to one. If bit ℓ is flipped, the difference in coordinate $j \geq \ell$ is $p_{\ell+1}^j \gamma^{j-\ell} q_\ell$, where $q_\ell = (1 - 2d_\ell)(1 - 2\gamma v_{\ell-1})$; earlier coordinates do not change. Divide the objective difference along this edge by q_ℓ . The contributions with $j \geq \ell$ cancel between $\hat{c}$ and $\lambda(d)e_n$, leaving

$$- \sum_{j=0}^{\ell-1} p_{j+1}^n p_{\ell+1}^n \gamma^{3n-\ell-2j}.$$

The lowest-power coefficient, at $j = \ell - 1$, is $-(1 - 2d_\ell)$. All other coefficients have magnitude one and their powers increase by two; their total magnitude relative to the leading term is less than $\gamma^2/(1 - \gamma^2) < 1$. Multiplying by q_ℓ gives a strictly negative edge difference. The feasible direction cone at a simple polytope vertex is generated by its incident edge directions (solve in its independent active-row coordinates); hence negativity on every edge implies unique global maximality. Finally $|\lambda(d)| \leq \sum_{h=1}^n \gamma^{2h} < \gamma^2/(1 - \gamma^2) = 1/15$. $\square$

Proof of Proposition 5.10. Let $V_n(\lambda) = \max_{z \in P_n} \hat{c}^\top z + \lambda z_n$. Each of its 2^n vertex lines is uniquely maximal at the price in Lemma D.1, and remains so on an open neighborhood. A maximum of affine functions cannot leave and later return to the same strictly active line, since that line's dominance set is an interval. Thus all 2^n lines give distinct maximal affine pieces. Pointwise evaluation does not require their construction. Eliminate the last coordinate, and continue backwards using

$$a_n = \lambda, \quad a_j = \hat{c}_j - \gamma|a_{j+1}| \quad (j = n-1, \dots, 1), \quad V_n(\lambda) = \sum_{j=1}^n \max(0, a_j).$$

A nonnegative coefficient selects $z_j = 1 - \gamma z_{j-1}$, contributing its constant and subtracting γa_j from the previous coefficient. A negative coefficient selects $z_j = \gamma z_{j-1}$, contributing no constant and adding γa_j . At zero either choice is valid. This proves the recurrence. It uses $O(n)$ rational operations, with bit lengths growing polynomially in n and the input price length. The terminal-state message $M_n(t) = \max\{\hat{c}^\top z : z \in P_n, z_n = t\}$ is the upper boundary of the planar projection $(z_n, \hat{c}^\top z)$. Every projected vertex is uniquely exposed by a direction with coefficient one on the second coordinate, so all belong to this upper boundary. Their terminal coordinates are distinct, and no three consecutive projected vertices are collinear (otherwise the middle one could not be uniquely exposed). The extreme terminal values zero and one occur. Hence M_n has $2^n - 1$ maximal affine intervals.

Consider the bounded epigraph

$$E_n = \{(\lambda, v) : -1/15 \leq \lambda \leq 1/15, \quad V_n(\lambda) \leq v \leq 1\}.$$

Here $V_n < 1$, since $\sum_j \hat{c}_j < 1/63$ and $|\lambda| \leq 1/15$. Suppose a quantifier-free Boolean formula in these two variables uses the nonzero polynomials $P_1, \dots, P_s$ to describe E_n . At every point on an open linear piece of its lower boundary, some polynomial must vanish: if all were nonzero, their signs and thus the formula's truth would be constant in a neighborhood. The product $\prod_i P_i$ therefore vanishes on each of 2^n distinct open line segments. Its restriction to each supporting line is a univariate polynomial vanishing on an interval, so that line's defining linear polynomial is a factor. Distinct line factors are coprime; their product divides $\prod_i P_i$. Thus $\sum_i \deg P_i \geq 2^n$. Identically zero predicates can be removed from the formula in advance. This lower bound explicitly excludes auxiliary variables and quantified or circuit representations. $\square$

D.2 A vertex polynomial, a slab, and padding

Lemma D.2 (Telescoping and parabolic exposure). *For $P_n(\gamma)$, let*

$$L(z) = z_n - (1 - \gamma) \sum_{i < n} \gamma^{2(n-i)-1} z_i, \quad F(z) = L(z) - z_n^2.$$

Then

$$F(z) = \sum_{j=1}^n \gamma^{2(n-j)} (z_j - \gamma z_{j-1}) (1 - \gamma z_{j-1} - z_j) \geq 0. \quad (106)$$

Its zeros in $P_n(\gamma)$ are exactly the vertices. Every vertex with terminal value t uniquely maximizes $2tz_n - L(z)$. The Hessian of F is $-2e_n e_n^\top$.

Proof. The product in the j -th term expands to $z_j - z_j^2 - \gamma z_{j-1} + \gamma^2 z_{j-1}^2$. The weighted quadratic terms telescope to $-z_n^2$; the interior linear coefficient is $\gamma^{2(n-j)} - \gamma^{2(n-j)-1} = -(1 - \gamma)\gamma^{2(n-j)-1}$. This proves the identity. Both product factors are nonnegative on the polytope, and neither pair can simultaneously vanish. Equality therefore selects one endpoint per coordinate, precisely (105). For any $z \in P_n$,

$$2tz_n - L(z) \leq 2tz_n - z_n^2 = t^2 - (z_n - t)^2 \leq t^2.$$

Equality forces $F = 0$ and $z_n = t$, hence the unique vertex with that terminal value. The Hessian follows directly from the definition. $\square$

Lemma D.3 (Subset Sum slab and fixed local coefficients). *For positive integers a_i , $W = \sum_i a_i$, and $0 \leq T \leq W$, the system $z \in P_n(1/(8W))$, $F(z) \leq 0$, and $T - 1/4 \leq \sum_i a_i z_i \leq T + 1/4$ is feasible precisely for a positive Subset Sum instance. The same equivalence holds with $\gamma = 1/4$, additional unary padding bounds, and polynomially increased dimension. Its feasible set without $F \leq 0$ is nonempty and compact.*

Proof. For the unpadding system, a vertex's coordinate differs from its bit by at most γ . A satisfying bit pattern therefore gives slab error at most $\gamma W = 1/8$. Conversely, $F \leq 0$ forces a vertex, and its integer subset sum differs from T by at most $1/4 + 1/8 = 3/8 < 1$, so it equals T . For fixed $\gamma = 1/4$, take the least integer $r \geq 0$ such that $4^{-(r+1)} \leq 1/(8W)$. Set $N = n + (n-1)r$, with free positions $j_i = 1 + (i-1)(r+1)$, and require $z_j \leq 1/2$ at every other position. At an original cube vertex, an upper endpoint is at least $3/4$. Thus all padding coordinates take their lower endpoints. Between successive free positions there are r multiplications by $1/4$, and

$$|z_{j_i} - d_{j_i}| \leq 4^{-(r+1)} \quad (i > 1), \quad z_1 = d_1.$$

Every free bit pattern extends by zero padding bits to an original cube vertex satisfying all unary bounds. Apply the slab only to $\sum_i a_i z_{j_i}$. The same $1/8$ and $3/8$ estimates prove the equivalence. Here $r = O(\log W)$, so dimension and bit lengths are polynomial. The local rows use coefficients $0, \pm 1, \pm 1/4$ and right-hand sides $0, 1/2, 1$; only the aggregate and vertex polynomial retain variable binary data. For nonemptiness without the vertex row, the zero vertex has sum zero, and the vertex with every free bit one has sum at least $W - 1/8$. Their segment reaches every integer $T < W$; for $T = W$ the latter vertex already meets the slab. Compactness follows from the unit box. $\square$

D.3 Equal-gain strips: a polynomial projection

Proposition D.4 (Paths, trees, and fixed total cycle rank). *Fix the dimension of a parameter t and the number of retained state variables. Suppose scalar states have finite bounds, with rational polynomial coefficients in t , and each directed edge $u \rightarrow v$ imposes*

$$a_e(t)x_u + b_e(t) \leq x_v \leq a_e(t)x_u + c_e(t). \quad (107)$$

If the underlying graph is a forest, one can eliminate all nonretained states with polynomial description size, numerical degree and bit complexity. The same holds for fixed total cycle rank. Signs of gains, including zero, are unrestricted. Additional objectives and constraints may depend on the parameters and retained states only. For fixed remaining dimension, exact feasibility, finite-infimum computation and attainment decisions consequently have polynomial bit complexity under dense polynomial encoding. Witnesses can be recovered in the same real number field as the retained solution.

Proof. First consider a path with nonzero gains a_i , indexed so that its edges are $i - 1 \rightarrow i$. On a realizable sign condition of the input gains put $A_0 = 1$, $A_i = \prod_{h=1}^i a_h$, and $y_i = x_i/A_i$. The node bounds become $L_i \leq y_i \leq U_i$, with endpoints interchanged when $A_i < 0$. The transition becomes $\alpha_i \leq y_i - y_{i-1} \leq \beta_i$, obtained from $b_i/A_i, c_i/A_i$ with the same sign rule. Require each interval to be consistent. Interpret these as directed difference constraints, with edge length β_i from $i - 1$ to i and $-\alpha_i$ in the reverse direction. Every two-edge cycle has nonnegative length. Removing backtracks shows that the unique simple path between two vertices is a shortest path. Denote its length by $d(j, i)$, including $d(i, i) = 0$. Thus feasibility is exactly

$$L_i \leq U_j + d(j, i) \quad \text{for every pair } i, j. \quad (108)$$

Necessity follows by summing along a path. For sufficiency, set

$$y_i = \min_j \{U_j + d(j, i)\}. \quad (109)$$

The all-pairs inequalities give the lower bounds, the term $j = i$ gives the upper bounds, and the shortest-path triangle inequality gives every directed edge inequality. To retain a specified state, add its prescribed value as both a lower and an upper bound, while keeping its original bounds. This makes (108) an exact projection onto the parameters and retained values, rather than only a feasibility test.

If a gain is zero, its original row imposes only $b_e \leq x_v \leq c_e$. Delete the edge and add these bounds to its original head v ; its original direction matters here. Restart the products in each remaining component. Keep multiple lower and upper candidates at a node without selecting a largest or smallest one. Require every lower candidate to be at most every propagated upper candidate; in (109), minimize over all upper candidates. This covers isolated nodes, consecutive zero gains, inconsistent bounds and lower-dimensional parameter sign conditions.

For an arbitrarily oriented tree with nonzero gains, choose a root and factors A_v satisfying $A_v = a_e A_u$ on each original edge $u \rightarrow v$. Traversing against an edge uses its reciprocal gain. The normalization $y_v = x_v/A_v$ again makes each strip a bounded difference interval. The unique-simple-path proof, all-pairs conditions, retained-value bounds and minimum recovery all remain valid. Zero edges are removed before this normalization as above. With $O(N)$ bound candidates there are $O(N^2)$ all-pairs conditions. Removing c nonforest edges from a graph retains at most $2c$ further endpoint states. Project the forest with these endpoints retained and then reinsert the original removed rows. For fixed *total* cycle rank c , the remaining dimension stays fixed. A bound on the cycle rank of each of many blocks would not suffice.

It remains to check the symbolic and bit bounds. Realizable sign conditions of the input gains can be enumerated in polynomial time in fixed parameter dimension, retaining zeros. Within each, the normalization is a rational function with a denominator dividing a product of input nonzero gains; tree reciprocals cause no iterative algebraic extensions. Prefix products, path sums and all-pairs conditions use only polynomially many such factors. One can use a common product of all nonzero gains, with additional powers if needed, to clear denominators of known sign, or multiply by its square and retain its nonvanishing condition. The degrees are $O(N\delta)$ up to a fixed polynomial factor when input degree is δ , and the number of monomials is polynomial in this degree because the parameter dimension is fixed. Exact multiplication and addition therefore yield polynomial coefficient bit length and total description size. The fixed number of retained state variables only adds a fixed number of polynomial coordinates. The resulting union of projected sign conditions is a fixed-dimensional semialgebraic formula, to which the operations of Section 1.5 apply. Finally, (109) selects one of finitely many rational functions of the retained solution; comparisons and rational arithmetic stay in its common field. Multiplication by A_i recovers each original state.

Finite state intervals alone do not make the parameter domain compact. The conclusion therefore distinguishes a finite infimum from an attained minimum; an optimizer is returned when attained, and unboundedness can also be detected. Under a closed compact feasible-domain assumption a continuous objective attains its minimum. An objective or additional row involving all eliminated states is outside the projection claim unless independently expressible in the retained variables. The common gain on both sides of (107) is essential. $\square$

References

- [1] Ilan Adler and Peter A. Beling. Polynomial algorithms for linear programming over the algebraic numbers. *Algorithmica*, 12:436–457, 1994. doi: 10.1007/BF01188714.
- [2] Daniel Arnström and Daniel Axehill. A unifying complexity certification framework for active-set methods for convex quadratic programming. *IEEE Transactions on Automatic Control*, 67(6):2758–2770, 2022. doi: 10.1109/TAC.2021.3090749.
- [3] Baruch Awerbuch and Robert D. Kleinberg. Adaptive routing with end-to-end feedback: Distributed learning and geometric approaches. In *Proceedings of the 36th Annual ACM Symposium on Theory of Computing*, pages 45–53, 2004. URL <https://www.cs.cornell.edu/~rdk/papers/OLSP.pdf>.
- [4] Saugata Basu, Richard Pollack, and Marie-Françoise Roy. On the combinatorial and algebraic complexity of quantifier elimination. *Journal of the ACM*, 43(6):1002–1045, 1996. doi: 10.1145/235809.235813.
- [5] Yasmine Beck, Ivana Ljubić, and Martin Schmidt. A survey on bilevel optimization under uncertainty. *European Journal of Operational Research*, 311(2):401–426, 2023. doi: 10.1016/j.ejor.2023.01.008.
- [6] Alberto Bemporad and Carlo Filippi. Suboptimal explicit MPC via approximate multi-parametric quadratic programming. In *Proceedings of the 40th IEEE Conference on Decision and Control*, volume 5, pages 4851–4856, 2001. doi: 10.1109/.2001.980975. URL <https://cse.lab.imtlucca.it/~bemporad/publications/papers/cdc01-sub-mpqp.pdf>.
- [7] Alberto Bemporad, Manfred Morari, Vivek Dua, and Efstratios N. Pistikopoulos. The explicit linear quadratic regulator for constrained systems. *Automatica*, 38(1):3–20, 2002. doi: 10.1016/S0005-1098(01)00174-1.
- [8] Mathieu Besançon, Miguel F. Anjos, and Luce Brotcorne. Complexity of near-optimal robust versions of multilevel optimization problems. *Optimization Letters*, 15:2597–2610, 2021. doi: 10.1007/s11590-021-01754-9.
- [9] Mathieu Besançon, Miguel F. Anjos, and Luce Brotcorne. Robust bilevel optimization for near-optimal lower-level solutions. *Journal of Global Optimization*, 90:813–842, 2024. doi: 10.1007/s10898-024-01422-z.
- [10] Stephen Boyd and Lieven Vandenbergh. *Convex Optimization*. Cambridge University Press, 2004. URL <https://web.stanford.edu/~boyd/cvxbook/>.
- [11] Christoph Buchheim. Bilevel linear optimization belongs to NP and admits polynomial-size KKT-based reformulations. *Operations Research Letters*, 51(6):618–622, 2023. doi: 10.1016/j.orl.2023.10.006.

- [12] Cássio Fraga Dantas and Rémi Gribonval. Stable safe screening and structured dictionaries for faster ℓ_1 regularization. *IEEE Transactions on Signal Processing*, 67(14):3756–3769, 2019. doi: 10.1109/TSP.2019.2919404.
- [13] Xiaotie Deng. Complexity issues in bilevel linear programming. In *Multilevel Optimization: Algorithms and Applications*, pages 149–164. Springer, 1998. doi: 10.1007/978-1-4613-0307-7_6.
- [14] Pavel Dvořák, Eduard Eiben, Robert Ganian, Dušan Knop, and Sebastian Ordyniak. The complexity landscape of decompositional parameters for ILP: Programs with few global variables and constraints. *Artificial Intelligence*, 300:103561, 2021. doi: 10.1016/j.artint.2021.103561.
- [15] Friedrich Eisenbrand, Matthieu Haeberle, and Neta Singer. An improved bound on sums of square roots via the subspace theorem. In *40th International Symposium on Computational Geometry*, volume 293 of *Leibniz International Proceedings in Informatics*, pages 54:1–54:8, 2024. doi: 10.4230/LIPIcs.SoCG.2024.54.
- [16] Laurent El Ghaoui, Fangda Gu, Bertrand Travacca, Armin Askari, and Alicia Tsai. Implicit deep learning. *SIAM Journal on Mathematics of Data Science*, 3(3):930–958, 2021. doi: 10.1137/20M1358517.
- [17] Rida T. Farouki. Convergent inversion approximations for polynomials in Bernstein form. *Computer Aided Geometric Design*, 17(2):179–196, 2000. doi: 10.1016/S0167-8396(99)00046-1.
- [18] Vincent Froese, Moritz Grillo, Christoph Hertrich, and Moritz Stargalla. Parameterized hardness of zonotope containment and neural network verification. arXiv:2509.22849v3, 2026. URL <https://arxiv.org/abs/2509.22849v3>. Version 3, September 3, 2026.
- [19] Bryan Gardiner and Yves Lucet. Convex hull algorithms for piecewise linear-quadratic functions in computational convex analysis. *Set-Valued and Variational Analysis*, 18:467–482, 2010. doi: 10.1007/s11228-010-0157-5.
- [20] Michael R. Garey, David S. Johnson, and Larry Stockmeyer. Some simplified NP-complete graph problems. *Theoretical Computer Science*, 1(3):237–267, 1976. doi: 10.1016/0304-3975(76)90059-1.
- [21] Bernd Gärtner, Christian Helbling, Yoshiki Ota, and Takeru Takahashi. Large shadows from sparse inequalities. arXiv:1308.2495, 2013. URL <https://arxiv.org/abs/1308.2495>.
- [22] Joachim Giesen, Martin Jaggi, and Sören Laue. Approximating parameterized convex optimization problems. *ACM Transactions on Algorithms*, 9(1):10:1–10:17, 2012. doi: 10.1145/2390176.2390186. URL <https://www.m8j.net/math/approxPaths.pdf>.
- [23] Joachim Giesen, Jens Müller, Sören Laue, and Sascha Swiercy. Approximating concavely parameterized optimization problems. In *Advances in Neural Information Processing Systems*, volume 25, 2012. URL https://papers.neurips.cc/paper_files/paper/2012/file/bdb106a0560c4e46ccc488ef010af787-Paper.pdf.
- [24] Peter Gritzmann and Bernd Sturmfels. Minkowski addition of polytopes: Computational complexity and applications to Gröbner bases. *SIAM Journal on Discrete Mathematics*, 6(2): 246–269, 1993. doi: 10.1137/0406019.

- [25] Dorit S. Hochbaum and J. George Shanthikumar. Convex separable optimization is not much harder than linear optimization. *Journal of the ACM*, 37(4):843–862, 1990. doi: 10.1145/96559.96597.
- [26] Alan J. Hoffman. On approximate solutions of systems of linear inequalities. *Journal of Research of the National Bureau of Standards*, 49(4):263–265, 1952. doi: 10.6028/jres.049.027.
- [27] V. Jeyakumar, J. B. Lasserre, G. Li, and T. S. Phạm. Convergent semidefinite programming relaxations for global bilevel polynomial optimization problems. *SIAM Journal on Optimization*, 26(1):753–780, 2016. doi: 10.1137/15M1017922.
- [28] Sergey S. Ketkov and Oleg A. Prokopyev. On the complexity of bilevel linear and quadratic programs in fixed dimensions, 2026. URL <https://arxiv.org/abs/2511.15592v2>. arXiv:2511.15592, version 2, June 10.
- [29] Krzysztof C. Kiwiel. Breakpoint searching algorithms for the continuous quadratic knapsack problem. *Mathematical Programming*, 112(2):473–491, 2008. doi: 10.1007/s10107-006-0050-z.
- [30] Thomas Kleinert, Martine Labbé, Ivana Ljubić, and Martin Schmidt. A survey on mixed-integer programming techniques in bilevel optimization. *EURO Journal on Computational Optimization*, 9:100007, 2021. doi: 10.1016/j.ejco.2021.100007.
- [31] Jun Liu, Zheng Zhao, Jie Wang, and Jieping Ye. Safe screening with variational inequalities and its application to Lasso. In *Proceedings of the 31st International Conference on Machine Learning*, volume 32(2) of *Proceedings of Machine Learning Research*, pages 289–297, 2014. URL <https://proceedings.mlr.press/v32/liuc14.html>.
- [32] Yi-Hsin Liu and Thomas H. Spencer. Solving a bilevel linear program when the inner decision maker controls few variables. *European Journal of Operational Research*, 81(3):644–651, 1995. doi: 10.1016/0377-2217(94)00005-W.
- [33] Julien Mairal and Bin Yu. Complexity analysis of the Lasso regularization path. In *Proceedings of the 29th International Conference on Machine Learning*, 2012. URL <https://arxiv.org/abs/1205.0079>.
- [34] Nimrod Megiddo and Arie Tamir. Linear time algorithms for some separable quadratic programming problems. *Operations Research Letters*, 13(4):203–211, 1993. doi: 10.1016/0167-6377(93)90041-E.
- [35] Nicolas Meuleau, Robert A. Morris, and Neil Yorke-Smith. A variable elimination approach for optimal scheduling with linear preferences. In *CP/ICAPS Joint Workshop on Constraint Satisfaction Techniques for Planning and Scheduling Problems*, 2008. URL <https://homepage.tudelft.nl/0p6y8/papers/n58.pdf>.
- [36] John Milnor. On the betti numbers of real varieties. *Proceedings of the American Mathematical Society*, 15(2):275–280, 1964. doi: 10.1090/S0002-9939-1964-0161339-9.
- [37] Nicholas Moehle, Jack Gindi, Stephen Boyd, and Mykel J. Kochenderfer. Portfolio construction as linearly constrained separable optimization. *Optimization and Engineering*, 24:1667–1687, 2023. doi: 10.1007/s11081-022-09748-x.

- [38] National Institute of Standards and Technology. DLMF, section 1.10: Functions of a complex variable. Digital Library of Mathematical Functions, n.d. URL <https://dlmf.nist.gov/1.10>. Accessed September 7, 2026.
- [39] Panos M. Pardalos and Stephen A. Vavasis. Quadratic programming with one negative eigenvalue is NP-hard. *Journal of Global Optimization*, 1:15–22, 1991. doi: 10.1007/BF00120662.
- [40] Michael Patriksson and Christoffer Strömberg. Algorithms for the continuous nonlinear resource allocation problem—new implementations and numerical studies. *European Journal of Operational Research*, 243(3):703–722, 2015. doi: 10.1016/j.ejor.2015.01.029.
- [41] Nagisa Sugishita and Margarida Carvalho. Complexity of bilevel linear programming with a single upper-level variable, 2026. URL <https://arxiv.org/abs/2510.21126v2>. arXiv:2510.21126, version 2, March 20.
- [42] Antoine Vigneron. Geometric optimization and sums of algebraic functions. *ACM Transactions on Algorithms*, 10(1), 2014. doi: 10.1145/2532647.
- [43] P. G. Walsh. A polynomial-time complexity bound for the computation of the singular part of a Puiseux expansion of an algebraic function. *Mathematics of Computation*, 69(231):1167–1182, 2000. doi: 10.1090/S0025-5718-00-01246-1.
- [44] Christopher Zach and Virginia Estellers. Contrastive learning for lifted networks. In *British Machine Vision Conference*, 2019. URL <https://arxiv.org/abs/1905.02507>.